

DISS. ETH NO. 23918

Optimization Methods to Manage Uncertainty and Risk in Power Systems Operation

A thesis submitted to attain the degree of

DOCTOR OF SCIENCES of ETH ZURICH

(Dr. sc. ETH Zurich)

presented by

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2016

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ISBN: 978-3-906327-65-5

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For a copy visit: <http://www.eeh.ee.ethz.ch>

Cover illustration by Hannah Hewett
Based on Ansel Adams “Transmission lines in Mojave Desert, 1941”
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Printed in Switzerland by the ETH Druckzentrum

To my family.

Du kan hvis du vil.

Preface

This thesis summarizes work done over the course of four years at the Power Systems Laboratory of ETH Zurich. It has been a wonderful time, and I would like to extend my sincere gratitude to everyone I have interacted with, both on a professional and a personal level.

First and foremost, I would like to thank my advisor Prof. Göran Andersson for providing me with the opportunity to pursue this PhD. I am very grateful for the trust, freedom and generosity with which I have been allowed to explore the world of research, both in my office at ETH and through various trips.

I am thankful to Prof. Louis Wehenkel from University of Liege for accepting to be my co-examiner and taking the time to review this thesis. From the very beginning of my PhD, Louis' work as well as our interactions at various conferences and workshops have been an inspiration to me. I would also like to thank Prof. Gabriela Hug, both for accepting to be my co-examiner and for having me around at PSL after Göran retired. It has been a pleasure to work with Gabriela and her group, and I am looking forward to staying in touch.

My next thanks goes to Thilo Krause and Frauke Oldewurtel for their importance both as mentors, office mates and partners on the UMBRELLA project. Thilo helped me set up my first optimal power flow code. From him, I learnt that academic writing is about conveying a story and that there is no contradiction between being a first-class poet and a researcher. Frauke deserves credit for introducing me to chance-constraints (anyone reading beyond this preface will quickly understand how important this has been), and for bothering to be a perfectionist, including proof-reading my papers over and over.

When I first started the PhD, there were many people in our group who were generous in sharing their experience with me. In particular, I would like to thank Spyros Chatzivasileiadis for his help and our many discussions, and for being an inspiration both personally and professionally. I am also thankful to Maria Vrakopoulou for introducing me to several important concepts, such as the scenario approach, and to Bart Van Parys for showing me that distributionally robust optimization is not necessarily hard.

I would like to extend my thanks to all members of the UMBRELLA consortium, both research and industry partners. In particular, I would like to thank our close collaborators at TU Delft and TU Graz, as well as all members of the project management team. I greatly appreciated all feedback and all discussions with the transmission system operators, which were pivotal in making our work practically relevant.

Some of my most intense and fun collaborations were carried out during and after my stay at Center for Non-Linear Studies at Los Alamos National Laboratory. I would like to thank Scott Backhaus and Michael Chertkov for hosting me, and including me in the many discussions and projects that have both inspired and profoundly changed my way of thinking about “grid science”. I would like to thank my Los Alamos colleagues and friends for all good days and nights, including Marc, Andrey, Theodor, Harsha, Kaarthik, Miles, Yury and Amanda. Special thanks goes to Anatoly Zlotnik for our collaboration on gas-electric systems - and for letting me explore the many trails around Los Alamos on his mountain bike. I am deeply grateful to Sidhant Misra for his inputs on optimization and coding, which contributed so significantly to the completion of this thesis, and for spending time talking to me about everything from boundary conditions to Belgian beers.

Throughout my PhD, I have been fortunate to supervise a number of master and semester thesis projects. I would like to thank Jaka Strumbej, Bing Li, Georgios Chatzis, Haoyuan Qu, Eren Cam, Jeremias Schmidli, Tobias Mohrhauer, Andrew Morrison and Aldo Tobler for their contributions and for choosing to work with me. I would also like to thank my collaborators who co-supervised some of these projects.

The experience of the PhD would have been completely different without the enthusiasm of the crew at the Power Systems Laboratory, be it in discussing improbable concepts for renewable generation or joining for the monthly lønningspils. Thanks to Marina, Maria, Frauke,

Spyros, Matthias B., Olivier, Matthias G., Andreas, Johanna, Philipp, Hubert, Roger, Monika, Osvaldo, Stephan, Markus, Vaggelis V., Thilo, Emil, Marcus, Olli, Louis, Jo, Adrian, Dmitry, Conor, Uros, Vaggelis K., Stavros, Andrew, Jack, Nadezdha, Tomas, Petros, Thierry, Chenye and our visitors Martin and Sven Christian for many good discussions, collaborations, ski trips, conferences and excursions. A special thanks to Marina Gonzalez Vaya for being my Frauen-Power partner over all these years, and to Johanna Mathieu for her precious advice on all aspects of (academic) life.

There is more to life than work, and my thanks also goes to all my friends in Zurich, Oslo and beyond for their friendship and support. In particular, I would like to thank my flat mates Hannah Hewett and Simone Kubik for being so patient and caring, and for sharing our common home. I would also like to thank Sara Kijewski for being my sister in Zurich, and for opening her home to me when needed.

Last, but not the least, I would like to thank my family. My mother, grandmas and aunts provide ample evidence that being a woman is no excuse for not reaching far. The determination of my younger brother Petter in pursuing his dreams is inspiring, and I am proud to share the Alnæs Roald identity with him. Finally, I am endlessly grateful for the love and support from my parents Helge and Monika, as well as their respect for my freedom to choose my own ways in life. Without them, I would never have been where I am today.

Thank you!

Line A. Roald

Zurich, December 2016

Abstract

Electricity from renewable energy sources is essential for a sustainable energy future. One inherent property of renewable generation is however that it is partially unpredictable, with uncertainty arising from forecast errors and real-time fluctuations. As the share of renewable generation grows, this uncertainty challenges existing practices for electric grid operation.

In the past, the operation of the electric grid was deemed secure if the system could withstand any credible contingency, typically defined as the failure of any single component. With increasing levels of forecast uncertainty, the system experiences larger deviations from the planned operating point. It therefore becomes more important to account for possible adverse impacts of those deviations to ensure that the system will remain secure in real time operation. This thesis is concerned with the development of optimal power flow methods that help system operators manage uncertainty and mitigate risk in day-to-day operational planning.

In the first part of the thesis, we focus on chance-constrained optimal power flow, which limits the probability of constraint violation. We propose to formulate the problem using separate chance constraints, which are reformulated using an analytical reformulation based on either full or partial knowledge about the uncertainty distribution. The approach allow us to model policy-based control actions in reaction to uncertainty realizations, which contribute to managing and mitigating possible adverse impacts. The method is applied to selected cases in power system operations, including investigations of different reserve activation policies and corrective control by HVDC connections and phase shifting transformers. These studies show that a more flexible system decreases

the cost of handling uncertainty. We also develop and demonstrate a solution algorithm that allow us to solve the problem for large systems.

While the above mentioned applications are based on the linear DC power flow equation, we also suggest an extension based on the non-linear AC power flow equations. We model the system using the full AC equations for the nominal operating point, but apply a linearization around this point to represent the impact of uncertainty. Due to the linear dependence on the uncertain variables, we can apply the analytical chance constraint reformulation, which allows for a tractable optimization problem. We further suggest different solution approaches, including an iterative approach which leverages scalability of existing solvers to tackle large problems, and compare the suggested analytical reformulation method with two sample-based methods. We find that the proposed method provides a good performance at low computational cost.

In the second part of the thesis, we focus on risk-based optimal power flow, where risk functions are used to model how the size of constraint violations influence risk. First, we suggest a method to define risk functions for post-contingency overloads based on system properties such as cost and availability of remedial actions, and provide a comparison with the N-1 criterion to argue for the choice of risk limits. We further combine the proposed risk functions with a chance constrained optimal power flow to limit the probability of high risk events.

Second, we introduce the weighted chance constraints, which measure and limit the expected risk of constraint violations, as defined by the probability distribution of the forecast errors and the chosen risk function. It is shown that the weighted chance constraint remains convex for convex risk functions and general control policies. We investigate how the choice of a risk function influences the number and size of constraint violations, and apply the method to a situation where tertiary control is activated during large wind deviations. Further, we investigate the optimal use of active power control from wind turbines, including reserve provision and enforcement of output caps.

Kurzfassung

Elektrizität aus erneuerbaren Energieträgern ist essentiell für eine nachhaltige Zukunft. Die produzierte Leistung aus erneuerbaren Energiequellen hängt allerdings stark von den momentan herrschenden Wetterverhältnissen, wie zum Beispiel der vorhandenen Sonneneinstrahlung und Windstärke, ab. Da diese aufgrund von kurzfristigen Schwankungen und Prognosefehlern nicht perfekt vorhergesagt werden können, ist die Elektrizitätsproduktion aus erneuerbaren Energien mit Unsicherheiten behaftet. Mit dem steigenden Anteil erneuerbarer Elektrizitätsproduktion, wird die Unsicherheit zur Herausforderung für den Betrieb des elektrischen Energiesystems.

Traditionell wurde der Netzbetrieb als N-1 sicher eingestuft, wenn das System jede wahrscheinlich auftretende Störung, typischerweise definiert als der Ausfall einer einzelnen Komponente, überstehen konnte. Da der zukünftige Systemzustand zunehmend durch Unsicherheiten beeinflusst wird, wird es wichtiger, dass nicht nur Komponentenausfälle, sondern auch mögliche Entwicklungen in der Einspeisung aus erneuerbaren Energien in der Betriebsplanung mitberücksichtigt werden, um einen sicheren Netzbetrieb zu garantieren. Diese Dissertation befasst sich mit der Entwicklung stochastischer Optimierungsmethoden für Lastflussberechnungen. Die Methoden erlauben es dem Netzbetreiber, die im Netz vorhandene Unsicherheit zu erkennen und das damit verbundene Risiko zu minimieren.

Im ersten Teil der Dissertation liegt der Fokus auf dem Thema optimaler Lastfluss mit probabilistischen Nebenbedingungen, mit dem Ziel die Wahrscheinlichkeit von Überlastsituationen im Netz zu limitieren. Um ein lösbares Optimierungsproblem zu erhalten, werden die probabilistischen Nebenbedingungen analytisch in deterministische Nebenbedin-

gungen überführt. Diese analytische Umformulierung kann für Wahrscheinlichkeitsverteilungen angewandt werden, deren Parameter bekannt oder teilweise bekannt sind. Um die Unsicherheit flexibler handhaben zu können und den Einfluss von eventuellen negativen Entwicklungen zu minimieren, werden korrektive Betriebsmassnahmen als Funktion der realisierten Unsicherheiten modelliert. In unterschiedlichen Fallstudien werden verschiedene Massnahmen betrachtet, wie die Aktivierung von Reserven oder Änderungen im Betriebspunkt von Hochspannungs-Gleichstrom-Übertragungsleitungen bzw. Phasenschiebertransformatoren. Die Studien zeigen, dass die zusätzliche Flexibilität durch korrektive Massnahmen die Kosten des Systembetriebs unter Unsicherheit verringern. Über dieses Modell hinaus wird auch ein Algorithmus formuliert, der stochastische Optimierung auf grosse Systeme anwendbar macht.

Die oben genannte Formulierung beruht auf einer linearen Approximation des Lastflusses (sogenannter DC-Lastfluss). Um eine genauere Beschreibung des Systems zu ermöglichen, wird eine Erweiterung auf die nicht-linearen Gleichungen für einen “vollen” AC-Lastfluss vorgeschlagen. Bei dieser Methode wird der vorhergesagte Systemzustand durch die kompletten AC Gleichungen modelliert, wobei der Einfluss der Unsicherheit durch eine Linearisierung um den prognostizierten Betriebspunkt analysiert wird. Die resultierenden Gleichungen hängen linear von den unsicheren Variablen ab, was es erlaubt, die oben genannte analytische Umformulierung für die probabilistischen Nebenbedingungen zu benutzen. Unterschiedliche Lösungsalgorithmen werden vorgeschlagen und diskutiert, u. a. eine iterative Methode, die in schon vorhandene Algorithmen für optimalen AC-Lastfluss integriert werden kann. Die Eigenschaften der analytischen Umformulierung werden weiter mit zwei szenariobasierten Umformulierungsmethoden verglichen. Die Resultate zeigen, dass die vorgeschlagene analytische Methode gute Resultate mit wenig Rechenaufwand erreicht.

Der zweite Teil der Dissertation befasst sich mit risiko-basierten Methoden für den optimalen Lastfluss. Das Risiko von Überlasten wird durch Risiko-Funktionen modelliert, die nicht nur die Häufigkeit, sondern auch die Grösse einer Überlast berücksichtigen. Wir fokussieren zuerst auf das Risikolevel des Systems nach einem Ausfall, und definieren die Parameter der Risiko-Funktion in Bezug auf die Kosten einer Lastflussreduktion, unter Berücksichtigung der möglichen korrektiven Massnahmen. Ein Vergleich mit dem traditionellen N-1 Kriterium wird genutzt um angemessene Risiko-Grenzen zu setzen. Wir formu-

lieren einen optimalen Lastfluss mit risikobasierten Nebenbedingungen für den Systemzustand nach jedem Ausfall, und integrieren diese Formulierung mit einer stochastischen Nebenbedingung, die Unsicherheiten in den Einspeisungen berücksichtigt und die Wahrscheinlichkeit für gefährliche Systemzustände begrenzt.

In einer weiteren risikobasierten Formulierung führen wir gewichtete Wahrscheinlichkeitsbedingungen ein, die das erwartete Risiko von Überlasten messen und limitieren. Das erwartete Risiko wird durch die Wahrscheinlichkeitsverteilung der unsicheren Variablen und der gewählten Risiko-Funktion evaluiert. Es wird gezeigt, dass die gewichteten Wahrscheinlichkeitsbedingungen konvex sind für konvexe Risikofunktionen und allgemeine korrektive Betriebsmassnahmen. Wir untersuchen wie die Wahl einer Risikofunktion die Häufigkeit und die Grösse von Überlasten beeinflusst, und betrachten eine Situation in der Tertiärreserven bei grossen Abweichungen vom geplanten Betriebspunkt aktiviert werden. Wir untersuchen darüber hinaus wie Kontrollmechanismen von Windturbinen optimal eingesetzt werden können, insbesondere in Bezug auf Reserven und Einspeisegrenzen.

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Nomenclature

The notation is given in the following order: General notation, sub- and superscripts, Roman letters, Greek letters and abbreviations.

General notation

$\mathbf{X}$	Matrices are denoted in bold
$\mathbf{X}_{(i,j)}$	Entry at the i th row and j th column of matrix $\mathbf{X}$
$\mathbf{X}_{(i,\cdot)}$	i th row vector of matrix $\mathbf{X}$
$\mathbf{X}_{(\cdot,j)}$	j th column vector of matrix $\mathbf{X}$
x_i	Lower case letters are used as indices for generators, buses, contingencies, HVDC and PSTs
x_{ij}	Double lower case letters are used as indices for transmission lines (denoting a connection from bus i to j)
$\tilde{x}$	Variables influenced by the uncertainty realization
$ \mathcal{X} $	Number of elements in set $\mathcal{X}$

Sub- and superscripts

x_D	Variables related to demand
x_{DC}	Variables related to HVDC lines
x_G	Variables related to generators
x_I	Variables related to current magnitude
x_L	Variables related to lines

x_{PQ}	Variables related to PQ buses
x_{PV}	Variables related to PV buses
x_R	Variables related to generation redispatch
x_V	Variables related to voltage magnitude
x_W	Variables related to wind or uncertainty sources
$x_{\theta V}$	Variables related to the reference bus
x_{ϑ}	Variables related to PSTs
x^{max}	Upper bound on a variable
x^{min}	Upper bound on a variable
x^T	Transposed variable
x^+	Variables related to up-reserves
x^-	Variables related to down-reserves

Roman letters

a_n	Slope of the n^{th} piece of the severity function
b_n	Constant for the n^{th} piece of the severity function
$\mathbf{b}_{\vartheta}$	Matrix connecting PSTs to lines
$\mathbf{B}_{\vartheta}$	Matrix connecting PSTs to buses
$\mathbf{B}_{Bus}$	Bus susceptance matrix
$\mathbf{B}_F$	Line susceptance matrix
c	Linear cost coefficients for active power generation
$c_{ij}^{+,k}$	Cost of remedial measure for line ij following outage k
c_R^+	Cost of positive redispatch
c_R^-	Cost of negative redispatch
c^+	Linear cost coefficients for up-reserves
c^-	Linear cost coefficients for down-reserves
c_0	Constant cost coefficients for active power generation
c_1	Linear cost coefficients for active power generation
c_2	Quadratic cost coefficients for active power generation
$\cos \phi$	Power factor

$\mathcal{C}$	Set of distributions with a given mean and covariance
$\mathcal{C}_{DC}$	Connectivity matrix of the HVDC connections
$\mathcal{C}$	Set of wind power plant with output caps
d	Active power consumption of demands
d_{peak}	System peak load
$\mathcal{D}$	Uncertainty set for the realizations of uncertain variables
$\mathcal{G}$	Set of generators
$\mathcal{H}$	Set of HVDC connections
i_{ij}	Current on line ij
$\mathbf{I}$	Identity matrix
$\mathcal{K}$	Set of contingencies
l	Number of lines
$\mathcal{L}$	Set of lines
LF_{ij}^{kl}	Line outage distribution factor for the line outages
m	Number of nodes, generators, loads and uncertainty sources
$\mathbf{M}$	Matrix of power transfer distribution factors
N_X	Number of decision variables
N_S	Number of samples for the scenario approach
$\mathcal{N}$	Set of nodes
p_{ij}	Active power flow on line ij
$p_{ij}^{\delta_{kl}}$	Active power flow on line ij after activation of corrective control
$p_{ij}^{kl,\omega}$	Active power flow on line ij after outage kl and fluctuation ω
p_{DC}	Power set-point of the HVDC lines
p_G	Active power generation at generators
p_R^+	Active power redispatch in positive direction
p_R^-	Active power redispatch in negative direction
$P(y)$	Probability distribution of y
$P(y \Omega)$	Probability distribution of given Ω

$\mathbb{P}$	Probability distribution
$\mathcal{P}$	Set of probability distributions
$\mathcal{P}$	Probability of outage
$PTDF$	Power transfer distribution factor
q_G	Reactive power generation at generators
q_D	Reactive power consumption of demands
q_U	Expected reactive power generation from uncertain sources
r^+	Generator up-reserve capacity
r^-	Generator down-reserve capacity
$\bar{r}^+$	Limit on available generator up-reserve capacity
$\bar{r}^-$	Limit on available generator down-reserve capacity
R^+	Total up-reserve requirement
R^-	Total down-reserve requirement
$\mathcal{R}$	Set of wind power plant with active power control
$\overline{\mathcal{R}}$	Upper bound on risk
$\mathcal{R}^{line}$	Risk related to a transmission line
$\mathcal{R}^{out}$	Risk related to an outage
$\mathcal{R}^{spec}$	Risk of a specific outage and transmission line
$\mathcal{R}^{tot}$	Total system risk
$\mathcal{S}$	Set of symmetric, unimodal distributions
$\mathcal{S}$	Set of PSTs
$\mathcal{S}_{(ij k)}$	Severity of post-contingency flow on line ij , given outage k
t_{ν, σ_T}	Cumulative distribution function of the Student's t distribution with ν degrees of freedom and scale parameter σ_T .
u	Expected active power generation from uncertain sources
u_f	Forecasted active power generation from uncertain sources
U	Set of unimodal distributions
$\mathcal{U}$	Set of uncertain power injections
v_j	Voltage magnitude at bus j
w	Uncontrolled wind power generation
$\hat{w}$	Controlled wind power generation

$\mathcal{W}$	Set of wind power plants
WCC	Weighted chance constraint
y	Overload of transmission line or generator
$\mathbf{0}$	Matrix with entries equal to zero
$\mathbf{1}_{a,b}$	Matrix with entries equal to one of dimension $a \times b$

Greek letters

α	Participation factors of generators in system balancing
α_G	Participation factors of generators in system balancing
α_1	Participation factors of generators in global system balancing with Policy I
α_2	Participation factors of generators in global system balancing with Policy II
α_3	Participation factors of generators in local system balancing with Policy III
α	Participation factors of generators in local balancing
α_{DC}	Parameter for HVDC corrective control
α_{ϑ}	Parameter for PST corrective control
β	Confidence level for the scenario approach
γ	Power ratio of the wind power fluctuations
Γ_I	Sensitivity factor for current magnitudes
Γ_P	Sensitivity factor for active power generation
Γ_Q	Sensitivity factor for reactive power generation
Γ_V	Sensitivity factor for voltage magnitudes
δ	Left hand side of the chance constraint
δ_n	Normalized left hand side of the chance constraint
δ_{DC}^{kl}	Post-contingency corrective control by HVDC after outage of line kl
$\bar{\delta}_{DC}^{kl}$	Limits on post-contingency corrective control by HVDC
δ_{ϑ}^{kl}	Post-contingency corrective control by PST

	after outage of line kl
$\bar{\delta}_{\vartheta}^{kl}$	Limits on post-contingency corrective control by PST
δi	Deviations in current magnitudes
δp_{ij}^{ω}	Deviations in power flows due to ω
δp	Deviations in active power injections
δq	Deviations in reactive power injections
δv	Deviations in voltage magnitudes
δu	Fluctuations of uncertainty sources
Δp_{ij}^{+k}	Maximum overload that can be relieved with remedial measures after outage ij (for positive power flows)
Δp_{ij}^{-k}	Maximum overload that can be relieved with remedial measures after outage ij (for negative power flows)
$\Delta p_{g \rightarrow h}$	Maximum generation shift from generator g to h
ϵ	Violation probability (for separate chance constraints), or risk limit (for weighted chance constraints)
ϵ_{emp}	Empirical violation probability
ϵ_J	Joint violation probability
η	Difference in uncertainty margin between iterations
$\hat{\eta}$	Stopping criterion for iterative approach
ϑ	Set-point of the PST
θ_j	Voltage angle at bus j
κ	Iteration count
λ	Uncertainty margin
μ	Mean forecast error
μ_y	Mean overload
μ_{δ}	Mean of the left hand side of the chance constraint
ν	Degrees of freedom for the Student's t distribution
$\xi(\Omega)$	Piecewise affine reserve activation policy
ξ^{+}	Change in generation for large, positive deviations
ξ^{-}	Change in generation for large, negative deviations
π	Corrective control from a generic power flow controller

ρ	Correlation coefficient
ρ_C	Parameter for cascading risk
σ_T	Scale parameter of the Student's t distribution
σ_W	Standard deviations of uncertainty sources
σ_Ω	Standard deviations of total power deviation
σ_δ	Standard deviation of the chance constraint
Σ_W	Covariance matrix of uncertainty sources
v	Mean wind power output after control
Φ	Cumulative distribution function of the standard normal distribution
Ψ	Generalized generation distribution factors
ω	Fluctuation (per uncertainty source)
$\hat{\omega}$	Fluctuation after control
Ω	Total fluctuation
$\hat{\Omega}$	Total fluctuation after control
$\hat{\Omega}_{1-\epsilon_G}$	$1 - \epsilon_G$ quantile of Ω

Abbreviations

AC	Alternating Current
AC CC-OPF	Chance-Constrained AC Optimal Power Flow
AC OPF	AC Optimal Power Flow
ACE	Area Control Error
AGC	Automatic Generation Control
APG	Austrian Power Grid
AVR	Automatic Voltage Regulator
CC-OPF	Chance-Constrained Optimal Power Flow
CC-SCOPF	Chance-Constrained Optimal Power Flow with Security Constraints
<i>CDF</i>	Cumulative Distribution Function
<i>CoS</i>	Cost of Security

<i>CoU</i>	Cost of Uncertainty
CVaR	Conditional-Value-at-Risk
DACF	Day-Ahead Congestion Forecast
DC	Direct Current (in context of HVDC)
DC	DC Approximation (in context of power flow equations or OPF)
EENS	Expected Energy Not Served
ENTSO-E	European Network of Transmission System Operators for Electricity
EV	Expected Value
FACTS	Flexible AC Transmission Systems
GDFs	Generation Distribution Factors
HVDC	High Voltage Direct Current
IGCC	International Grid Control Cooperation
LMP	Locational Marginal Pricing
LODF	Line Outage Distribution Factor
OPF	Optimal Power Flow
PST	Phase Shifting Transformer
PTDFs	Power Transfer Distribution Factors
PV	Photovoltaics
RB-CC-SCOPF	Chance-Constrained Optimal Power Flow with Risk-Based Security Constraints
RB-SCOPF	Optimal Power Flow with Risk-Based Security Constraints
SCOPF	Security Constrained Optimal Power Flow
SOC	Second-Order Cone
SOCF	Second-Order Cone Program
TSO	Transmission System Operator
WCC	Weighted Chance Constraints
WCC-OPF	Optimal Power Flow with Weighted Chance Constraints

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Chapter 1

Introduction

Over the past decade, electric power systems in many parts of the world have experienced significant changes. Market liberalization and increasing shares of renewable generation imply that power is not necessarily produced close to where it is consumed, but rather where production is cheap, the wind is blowing or the sun is shining. This trend increases the need for transmission capacity, makes power flow patterns more variable, and forces the Transmission System Operator (TSO) to operate the system closer to the limits. At the same time, fluctuations in renewable in-feed and short-term trading lead to larger deviations from the planned generation schedules, and make the future state of the system less predictable. The combination of a highly loaded system and significant uncertainty increases operational risk.

While grid expansion is often difficult due to long regulatory processes and public opposition, there exist new possibilities to cope with high loads and frequently changing operating conditions. Installation of equipment such as Phase Shifting Transformer (PST) and High Voltage Direct Current (HVDC) connections, implementation of demand-side response schemes and stricter grid-connection requirements for renewable generation contribute to improved controllability in the grid. Simultaneously, increased situational awareness, probabilistic forecasting methods and advances within optimization and control allow for better decision making and extended ability to adjust to the real time situation.

To understand and manage this more complex situation, many TSOs

are currently redesigning existing operational processes. The aim of this thesis is to make a contribution towards this development, and help in addressing the above challenges and opportunities. The thesis focuses on methods to handle uncertainty and model risk within the Optimal Power Flow (OPF) problem, an optimization problem that plays an integral role in power systems operation and operational planning. The main idea is that, by utilizing additional available information about forecast uncertainty, operational risk and availability of corrective control actions, transmission capacity can be used more efficiently, while maintaining security and robustness against disturbances.

1.1 Background

The time frame considered in this thesis is *power system operation and operational planning*, from a few days until a few minutes ahead of real time. Over this time horizon, the system itself can be considered as given, and we do not consider, e.g., installation of new equipment or maintenance scheduling. The main degrees of freedom include the generation dispatch, allocation of reserves and the system configuration. The planned schedules for dispatchable generation are obtained from the electricity market, which is typically cleared in either of the two following ways.

In *central-dispatch markets*, electricity markets are cleared by the TSO under consideration of the physical limits of the transmission grid. This is the predominant market type in the United States (US), with two thirds of electricity demand being served by such markets [1]. The market is typically cleared in several stages, including unit commitment to determine which generation units are online, day-ahead market clearing to obtain hourly generation schedules, and real-time markets to make adjustments in intra-day operation. Each stage typically involves solving some version of the OPF problem. The OPF is an optimization problem which typically minimizes the total cost of electricity while enforcing physical constraints such as limits on transmission capacity. The OPF problems used for market clearing purposes in the US are often based on the DC approximation of the power flow problem. The resulting optimization problem is linear and has well defined Lagrangian multipliers, which are used to determine electricity prices, the so-called Locational Marginal Prices (LMPs).

In *self-dispatch markets*, electricity markets are cleared without consid-

eration of the transmission grid, or with some limited information such as import/export constraints. This is the most common method for market clearing in Europe. In this set-up, the TSO has no direct influence on the outcome of the market clearing, but are provided with the planned generation schedules in the day-ahead. The TSO uses available control actions to make the market-based schedule comply with the physical grid limitations. Common control actions include transmission switching and set-point changes for PST or HVDC, generation redispatch or curtailment of renewable energy. The TSO is typically required to take the actions that have lowest cost and intervene the least with the market outcome. The TSO in a self-dispatch market hence also benefit from solving an OPF to determine the least cost, optimal changes to the system for the given a market outcome.

The OPF problem is thus an important tool in the operational planning for any power system, independent of whether the markets follow a central- or self-dispatch set-up.

Incorporating Uncertainty

Currently, the OPF problem is typically solved as a deterministic optimization problem, based on a point forecast for the load and renewable generation. The system is considered secure if it remains within the operational limits during normal operation and during *outage of any single component*. This principle is referred to as the $N - 1$ criterion [2], and is reflected in the OPF through additional constraints, leading to a Security Constrained Optimal Power Flow (SCOPF).

In recent years, uncertainty in power system operational planning has been increasing. On the one hand, higher shares of generation from renewable energy sources increase uncertainty of power production and net load. On the other hand, the liberalization of electricity markets has lead to increased intra-day electricity trading, and frequent changes in the generation schedules between the day-ahead planning and real time. In Europe, TSOs increasingly face a situation in which operational measures for system security are exhausted in real time [3]. Accounting for *forecast uncertainty* when planning system operation is therefore becoming more important, and forces a change in power system operational procedures. However, the main task of the TSO remains the same: To provide the customers with reliable supply of electricity, and achieve this in a cost efficient manner. The OPF problem can hence be recast as a stochastic optimization problem.

Several stochastic versions of the OPF have been proposed, including robust and worst-case methods [4, 5, 6, 7], two- and multi-stage stochastic programming based on samples [8, 9, 10, 11] or stochastic approximation techniques [12], and chance-constrained formulations [13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23]. While the different approaches vary in how they model uncertainty, which is typically a trade-off between accuracy and computational complexity, all share the common property that they prepare the system for a range of uncertainty realizations. Ensuring secure operations for all possible forecast errors will, in most situations, provide very conservative and thus costly solutions. An important aspect when solving an OPF under uncertainty is therefore to find an appropriate trade-off between security and cost, which is typically dictated by the choice of the uncertainty set (robust optimization), the considered scenarios (stochastic programming) or the acceptable violation probability (chance constraints).

Assessing Risk

The traditional N-1 criterion requires that no contingency should cause overloads or other damage that would lead the outage of further components. It can thus be seen as a criterion meant to secure the system against cascading. The benefits of the N-1 security criterion are its conceptual simplicity and its ability to provide a clear answer on whether or not the system is considered secure. However, while the N-1 criterion implicitly accounts for outage probabilities by considering a list of *credible contingencies*, it lacks explicit modelling of outage probabilities or assessment of the actual risk related to an N-1 violation. To obtain a more accurate description of the risk, we can model it as the product between the probability and the severity of an event,

$$\text{Risk} = \text{Probability} \cdot \text{Severity} \quad (1.1)$$

To model the probability of events, we distinguish between disturbances in form of outages and deviations.

- A *component outage* is a binary event, with a probability of occurrence. The probability of outages can be estimated in different ways, e.g. based on historical data, and can vary depending on several external factors, such as the weather forecast.
- *Forecast uncertainty* is better represented as continuous random variables with corresponding probability distributions, which can

be obtained, e.g., through probabilistic forecasts or by analysis of historical data.

For severity modeling, there exist two main approaches:

- On the one hand, severity can be modeled through overall reliability parameters like Expected Energy Not Served (EENS). These parameters incorporate the effect of cascading events and best reflects the impact on the customers in the system. However, computing the risk requires extensive calculations (i.e., Monte-Carlo simulations), and these types of risk measures are typically used to analyze the risk for a given operating condition [24, 25], as opposed to inclusion in an optimization problem.
- On the other hand, severity can be modeled in terms of violation of technical limits, e.g., dependent on the power flow of a line or on the voltage magnitude [26, 27, 28, 19]. Such models typically consider the situation after an N-1 outage, and do not simulate how a potential cascade would develop further. Thus, these risk measures do not reflect the full risk of cascading events, but are relatively easy to compute. Further, technical violations are easily understood and influenced by the system operator.

The choice of the severity measure depends on the application. For implementation within risk-based OPF, severity measures of the second type are used due to their relatively simpler computational evaluation. For a more comprehensive risk assessment, e.g. for a-posteriori evaluation of an OPF solution, Monte-Carlo simulations can give a more accurate picture of the true system risk.

Managing Impacts of Disturbances

Power flow control devices such as HVDC and PSTs allow for more flexible use of transmission assets, by enabling redirection of power flows from congested to less loaded transmission lines. Other types of control, such as demand response, control of renewable generation or generation redispatch further increase system flexibility. In power system terminology, system security is ensured either *preventively* or *correctively*:

- A *preventively* secure system state refers to a situation in which the power system remains secure after any credible contingency

(typically defined as an N-1 situation) without any additional control action.

- A *correctively* secure state refers to a situation where additional, post-disturbance controls might be required.

Corrective control to handle contingencies is known to reduce the cost of enforcing N-1 security constraints [29]. However, corrective control actions can also be used to handle uncertainty, e.g. by changing the set-points of HVDC and PSTs in reaction to forecast deviations. More generally, this type of corrective control can be understood as recourse actions taken in reaction to a certain realization of the uncertain parameters.

When the system operator has many possibilities to react, the impact of disturbances is less severe and the constraints related to contingencies and uncertainty become less binding. Assuming that the corrective control reactions work as planned, we are then able to maintain a similar security level at lower operational cost. This makes corrective control very attractive to TSOs, and the use of corrective actions has increased over the past years [30]. While corrective control is applied routinely in real-time system operation, the real-time set-points of, e.g. HVDC and PSTs are typically chosen in an ad-hoc fashion. This is partially due to the difficulty of planning corrective control actions in response to forecast uncertainty, as it requires the consideration of a large number possible uncertainty scenarios in addition to consideration of contingencies. However, the need to ensure that corrective control will be sufficient in real-time calls for efficient ways of modelling both the possible system states and the corresponding corrective control actions.

1.2 Contributions

This thesis attempts to address the challenges and opportunities described above. On the one hand, we propose methods to model uncertainty and risk within OPF problems. On the other hand, we apply these models to specific problems with different kinds of power flow control, and investigate the effect on system security and operational cost. Below, the contributions are divided into the development of chance-constrained OPF formulations, risk-based OPF formulations and the application of those to selected problems in power systems operation.

Chance-Constrained Optimal Power Flow

We develop a Chance-Constrained Optimal Power Flow (CC-OPF) formulation which accounts for uncertainty and ensures that the probability of constraint violation remains low.

- We formulate the CC-OPF based on a DC power flow approximation and separate chance constraints, where each constraint is enforced with a given probability. We present analytical reformulations of the chance constraints for both known and partially known distributions.
- Based on the analytical reformulation, we characterize the impact of uncertainty in terms of a tightening of the transmission and generation constraints. This tightening can be interpreted as a security margin against uncertainty, i.e., an “uncertainty margin”.
- We develop a sequential SOCP algorithm to solve the CC-OPF for large systems with security constraints, and demonstrate scalability by solving the problem for large test instances.
- Finally, we extend the modelling to AC power flows and propose an approximate AC CC-OPF formulation. The model uses the full AC power flow to represent the nominal operating point, but linearizes the system around this point to model the impact of uncertainty. The linearization allows for an analytic chance constraint representation, and a computationally tractable representation of the optimization problem. We compare different solution approaches and show scalability to large systems. We further contrast the analytical chance constraint reformulation with two sample-based reformulations based on (i) a Monte Carlo simulation and (ii) the scenario approach for joint chance-constrained problems.

Risk-Based Optimal Power Flow

We develop models for the risk of constraint violations, incorporating both the impact of uncertainty, as well as the probability and severity of contingencies.

- For the security constraints, we develop a risk model that explicitly accounts for the probability of outages and the cost of removing post-contingency overloads. A chance-constrained formulation of the risk-based OPF is used to limit the probability of high risk events.
- We further develop a risk-based alternative to the standard chance constraints, the so called Weighted Chance Constraints (WCC) which limits the expected risk of constraint violations. Instead of measuring only the probability of violation, we use a weighting function to assign a higher penalty to large violations. Depending on the choice of weighting function, we can measure different types of risk.
- We show that the Optimal Power Flow with Weighted Chance Constraints (WCC-OPF) remains convex for general types of control, and use this property to propose an efficient solution algorithm based on outer-approximation with cutting planes.

Managing Uncertainty and Risk

We apply the developed methods to selected cases in power system operations. Throughout several case studies, we show the effectiveness of the proposed approaches in reducing constraint violations and risk:

- We investigate how the choice of ϵ influences cost, and compare this with the cost of enforcing security constraints.
- We run in-sample and out-of-sample Monte Carlo simulations to test the performance of the analytical chance constraint reformulations. We observe that assuming normally distributed power flows can provide a good approximation even when the power injections are not normally distributed, because the power flows are weighted sums of a large number of uncertain power injections.
- We observe that the WCC limit large violations more effectively than the standard chance constraints, but allow for small violations to occur more frequently.

Further, we investigate how more flexible controls can reduce the cost of managing uncertainty:

- We investigate an integrated approach to balancing and congestion management in systems with significant uncertainty. The reserves are activated not only based on the total power mismatch, but also the location of the fluctuations in the system. We observe that a more flexible activation based on location reduces cost, particularly in congested systems.
- We propose a framework for corrective control in response to uncertainty. The framework is based on an optimization over affine policies, and guarantees (with a desired probability) that a secure solution can be found in real time. We show that corrective control can reduce cost, while maintaining a similar level of security, and that the method is applicable to large systems.
- We review the grid-code requirements for active power control with wind power plants, including reserve provision and enforcing caps on the power output. We investigate how to make optimal use of these capabilities, and observe that both types of control can contribute to cost reductions.

1.3 Thesis Organization

The thesis is divided into two main parts. The first part discusses the formulations and applications for CC-OPF, whereas the second part describes extensions to risk-based OPF.

Part I: Chance-Constrained Optimal Power Flow

Chapter 2 describes the power system model with uncertainty, and formulates the Chance-Constrained Optimal Power Flow with Security Constraints (CC-SCOPF). Further, it discusses different analytical reformulations of the chance constraints, including both exact reformulations for both known and partially known distributions. In two case studies, we investigate the impact of uncertainty on the cost of operation and assess the performance of the different reformulations.

Chapter 3 investigate the impact of different balancing policies in systems under uncertainty. In particular, we compare the impact of activating reserves based on the location of a specific fluctuation, as opposed to activation based only on the total power mismatch.

Chapter 4 proposes a framework for modelling corrective control actions to handle uncertainty. It is proposed to use affine control policies to model the control actions, as this allows for a continuous representation of the uncertain variables and ensures feasibility in real time. The chapter further describes a sequential Second-Order Cone Program (SOCP) algorithm to solve the CC-SCOPF for large systems.

Chapter 5 extends the chance-constrained model to AC Optimal Power Flow (AC OPF). We solve the problem using the full AC power flow equations for the forecasted operating point, but linearize the impact of uncertainty. The resulting chance constraints are reformulated using the analytical approach from Chapter 2. We investigate different solution approaches, and show that the problem can be implemented with existing AC OPF algorithms and scale to large systems. We further contrast the analytical reformulation two sample-based approaches based on a Monte Carlo simulation, and the scenario approach for joint chance constraints.

Part II: Risk-Based Optimal Power Flow

Chapter 6 discusses the Chance-Constrained Optimal Power Flow with Risk-Based Security Constraints (RB-CC-SCOPF). The security constraints on line flows are formulated as risk based constraints, and account for both the outage probabilities and the severity of the resulting operating point. The severity is modelled through a piecewise affine function, which reflects the cost and availability of remedial actions as well as the risk of cascading events. To account for the impact of forecast uncertainty, we use chance constraints to prescribe a limit on the probability of high risk events.

Chapter 7 introduces the concept of the weighted chance constraints (WCC), discusses different weight functions and establishes the connection to a standard chance constraint. We further prove that the WCC remains convex for convex weight functions and very general generation control policies. In the case study, we investigate the impact of the choice of the weight function, and demonstrate the benefit of more general control policies.

Chapter 8 extends the WCC-OPF to incorporate active power control with wind power plants. We review the requirements for grid connection of wind power plants, and choose the control capabilities most relevant for active power control. These capabilities allow the wind power plants

to provide both up- and down-reserves, or to limit their output to a given value. We extend the WCC-OPF to incorporate this kind of control, and propose an outer-approximation cutting-planes algorithm to solve the problem efficiently.

Chapter 9 summarizes the contents of the thesis, gives an overview of the most important observations and concludes. It further provides an outlook of directions of future work.

1.4 List of Publications

The work presented in this thesis has been reported by the following publications:

Journal Publications

1. L. Roald, S. Misra, T. Krause and G. Andersson, "Corrective Control to Handle Forecast Uncertainty: A Chance Constrained Optimal Power Flow", *IEEE Transactions on Power Systems*, in press [22]
2. L. Roald, M. Vrakopoulou, F. Oldewurtel and G. Andersson, "Risk-Based Optimal Power Flow with Probabilistic Guarantees", *International Journal of Electrical Power and Energy Systems*, vol. 56, pp. 66 - 74, 2015 [31]

Conference Publications

1. J. Schmidli, L. Roald, S. Chatzivasileiadis and G. Andersson, "Stochastic AC Optimal Power Flow with Approximate Constraints", *IEEE PES General Meeting*, Boston, 2016 [32]
2. L. Roald, S. Misra, M. Chertkov, S. Backhaus and G. Andersson, "Optimal Power Flow with Wind Power Control and Limited Expected Risk of Overloads", *Power Systems Computation Conference (PSCC)*, Genova, Italy, 2016 [33]
3. L. Roald, T. Krause and G. Andersson, "Integrated Balancing and Congestion Management under Forecast Uncertainty", *IEEE EnergyCon*, Leuven, Belgium, 2016 [21]

4. L. Roald, S. Misra, M. Chertkov and G. Andersson, "Optimal Power Flow with Weighted Chance Constraints and General Policies for Generation Control", *IEEE Conference on Decision and Control (CDC)*, Osaka, Japan, 2015 [20]
5. H. Qu, L. Roald and G. Andersson, "Uncertainty Margins for Probabilistic AC Security Assessment", *IEEE PowerTech, Eindhoven*, Netherlands, 2015 [34]
6. L. Roald, M. Vrakopoulou, F. Oldewurtel and G. Andersson, "Risk-Constrained Optimal Power Flow with Probabilistic Guarantees", *Power Systems Computation Conference (PSCC)*, Wroclaw, Poland, 2014 [19]
7. L. Roald, F. Oldewurtel, T. Krause and G. Andersson, "Analytical reformulation of security constrained optimal power flow with probabilistic constraints", *IEEE PowerTech*, Grenoble, France, 2013 [17]

Other Publications and Reports

1. L. Roald, F. Oldewurtel, B. Van Parys and G. Andersson, "Security Constrained Optimal Power Flow with Distributionally Robust Chance Constraints", Technical Report, arxiv: <http://arxiv.org/pdf/1508.06061.pdf> [23]
2. UMBRELLA FP7 Project, "Deliverable 4.3: Methods for Optimisation of Power Transits", www.e-umbrella.eu, December 2014 (co-author and coordinator) [35]
3. UMBRELLA FP7 Project, "Deliverable 4.2: Risk Assessment Methods", www.e-umbrella.eu, December 2013 (co-author and coordinator) [36]
4. UMBRELLA FP7 Project, "Deliverable 4.1: Risk-based Assessment Concepts for System Security - State-of-the-Art Review and Concept Extensions", www.e-umbrella.eu, December 2012 (co-author and coordinator) [3]

The following relevant publications have been published in the course of this PhD, but their contents are not included in the thesis:

Journal Publications

1. A. Zlotnik, L. Roald, S. Backhaus, M. Chertkov and G. Andersson, "Coordinated Operational Planning for Integrated Electric Power and Natural Gas Infrastructures", *IEEE Trans. on Power Systems* (in press) [37]

Conference Publications

1. A. Zlotnik, L. Roald, S. Backhaus, M. Chertkov and G. Andersson, "Control Policies for Operational Coordination of Electric Power and Natural Gas Transmission Systems", *IEEE American Control Conference (ACC)*, Boston, United States, 2016 [38]
2. K. Sundar, H. Nagarajan, M. Lubin, L. Roald, S. Misra, R. Bent, D. Bienstock, "Unit Commitment with N-1 Security and Wind Uncertainty", *Power System Computation Conference (PSCC)*, Genova, Italy, 2016 [39]
3. G. Chatzis, I. Avrimotis-Falireas, L. Roald, F. Abbaspourtorbati, M. Zima and G. Andersson, "Joint Scheduling of Frequency Control Reserves and Energy Dispatch for Small Islanded Power Systems", *Power System Computation Conference (PSCC)*, Genoa, Italy, 2016 [40]
4. G. Morales et al, "Innovative tools for the future coordinated and stable operation of the pan-European electricity transmission system", *CIGRE PARIS 2016*, Paris, France, 2016 [41]
5. F. Oldewurtel, L. Roald, G. Andersson and C. Tomlin, "Adaptively constrained stochastic model predictive control applied to security constrained optimal power flow", *IEEE American Control Conference (ACC)*, Chicago, United States, 2015 [42]
6. L. Roald, B. Li, F. Oldewurtel, T. Krause and G. Andersson, "An optimal power flow (OPF) formulation including risk of cascading events", *XIII SEPOPE*, Foz du Iguazu Brazil, 2014 [43]
7. M. Vrakopoulou et al., "A Unified Analysis of Security-Constrained OPF Formulations Considering Uncertainty, Risk, and Controllability in Single and Multi-area Systems", *IREP Symposium on Bulk Power System Dynamics and Control - IX Optimization, Security and Control*, Crete, Greece, 2013 [44]

8. Z. Liu et al, “Challenges, Experiences and Possible Solutions in Transmission System Operation with Large Wind Integration”, *11th International Workshop on Large-Scale Integration of Wind Power into Power Systems*, Lisbon, Portugal, 2012 [45]
9. Z. Liu et al, “Innovative operational security tools for the development of a stable pan-European grid”, 2012 CIGRE Canada Conference, Montreal, Canada, 2012 [46]

Other Publications and Reports

1. UMBRELLA FP7 Project, “Deliverable 1.3: Final Report”, www.e-umbrella.eu, December 2015 (co-author) [47]
2. UMBRELLA FP7 Project, “Deliverable 6.3: Recommendations to ENTSO-E Regarding TSO RSCI Rules”, www.e-umbrella.eu, December 2015 (co-author) [48]

Part I

Chance-Constrained Optimal Power Flow

Chapter 2

Analytical Reformulations of Chance-Constrained Optimal Power Flow

The growing amount of fluctuating renewable in-feeds and market liberalization increases uncertainty in power system operation. To capture the influence of uncertainty in operational planning, we model the forecast errors of the uncertain in-feeds as random variables and formulate a security constrained optimal power flow using chance constraints. The chance constraints prescribe an upper bound on the probability of violations of technical constraints, such as generation and transmission limits, but require a tractable reformulation. To address this problem, we propose different analytical reformulations of the chance constraints, based on a given set of assumptions concerning the forecast error distributions. We discuss reformulations which assume a specific distribution such as the Gaussian and the Student's t distribution, as well as distributionally robust reformulations based on partial information about the distribution. In two case studies, we investigate the impact of uncertainty on system operation. We further assess the impact of the chosen violation probability and compare the performance of the different reformulations.

2.1 Motivation and Related Work

A fundamental tool in power system analysis is the *optimal power flow* (OPF) [49]. Several tasks central to power system operation, such as unit commitment, reserve procurement, market clearing and security assessment rely on the solution of an OPF problem. Typically, the main goal of the OPF is to minimize operational cost, while ensuring secure operation that respects technical limits of the power system. In current operational schemes, the system is considered secure if it remains within the operational limits during normal operation and during *outage of any single component*. This principle is referred to as the $N - 1$ criterion [2], and is reflected in the OPF through additional constraints, leading to a *security constrained optimal power flow* (SCOPF).

While the $N - 1$ criterion secures the system against individual outages, it does not account for *forecast uncertainty*. Forecast uncertainty arises from unforeseen fluctuations in the power injections, such as inaccurate predictions of load or renewable in-feeds, or short-term electricity trading. While load profiles are relatively predictable, higher shares of electricity production from renewable sources and liberalization of energy markets (particularly in Europe) have increased the forecast uncertainty by orders of magnitude [45]. In current operational planning, uncertainty is usually ignored and uncertain quantities are typically replaced by a forecast value. While this approach has provided good solutions in the past, the increased levels of uncertainty lead to frequent $N-1$ violations in real-time operation. To mitigate these problems, it is proposed to explicitly account for uncertainty during operational planning, in particular while solving the OPF. Towards this objective, a wide variety of approaches and methods to account for uncertainty within the OPF have recently been proposed in literature. These include, among others, robust and worst-case methods [4, 5, 6, 7], two- and multi-stage stochastic programming based on samples [8, 9, 10, 11] or stochastic approximation techniques [12], and chance-constrained formulations [13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23].

Chance Constraints

In this thesis, we choose to work with chance constraints, which ensure that the system constraints will be satisfied with a desired probability. Discussions with TSO partners within the UMBRELLA project

[50] showed that choosing an acceptable violation probability ϵ (or conversely a security level $1 - \epsilon$) was perceived as a relatively intuitive way of defining the security level. Chance constraints also align well with several practices applied in industry, such as the probabilistic reserve dimensioning applied in ENTSO-E [51, 52] or the definitions of reliability margins for flow-based market coupling in parts of Europe [53].

Here, we formulate a chance-constrained optimal power flow with security constraints (CC-SCOPF) using *separate* chance constraints, such that *each* constraint is assigned an maximum acceptable violation probability ϵ . Separate chance constraints allow the operator to define different violation probabilities for different types of constraints, reflecting the particular importance of certain elements or possibly less restrictive conditions for, e.g., N-1 constraints. In comparison, *joint* chance constraints (see e.g. [14]) ensures that the probability of constraint violation of *any* constraint remains below ϵ . While this provides more comprehensive guarantees for the overall system security, it is harder to analyze how the choice of the violation probability affects the system on a component level.

Chance Constraint Reformulations

While the CC-SCOPF allow us to account for uncertainty in a comprehensive way, it is generally hard to reformulate chance constraints as tractable constraints. Two main reformulation approaches include either sampling based or analytical reformulation. In [14], a CC-SCOPF with joint chance constraints is reformulated using the scenario approach [54]. The formulation was extended to include market clearing with co-optimization of energy and reserves in [15], where a different sampling based reformulation from [55] was used. Both sampling based reformulations require no knowledge about the underlying distribution, except for availability of a given number of samples (which increases with the problem size).

In contrast, the CC-SCOPF shown here is based on separate chance constraints which admit an analytical reformulation [17, 23]. Based on the information available to the TSO, more or less tight reformulations can be applied.

If the forecast errors are assumed to follow a multivariate Gaussian distribution, it is possible to obtain a *tight* reformulation which meets the acceptable violation probability with equality [17, 18]. Other tight

reformulations can be obtained for forecast error distributions in the elliptical family, such as the more heavy tailed Student's t ., or in special cases where the cumulative distribution function can be estimated directly, based on, e.g., historical data as in [56]. Assuming a normal distribution might also be appropriate when the chance constraint depends on a large number of forecast errors, even though the forecast errors are not normally distributed. In this case, we can justify the assumption using arguments similar to the central limit theorem [57].

For cases where little is known about the forecast error distributions, we can use *distributionally robust* reformulations based on optimal probability inequalities. These reformulations require less stringent assumptions on the uncertainty distributions, as they only assume some general properties such as known mean and covariance. This approach is well-known in operations research and control theory, see e.g. [58, 59, 60]. They have also been discussed in the context of power systems. In [61], a distributionally robust reformulation of the OPF problem based on the Chebyshev inequality (assuming known mean and covariance, but unknown distribution) was discussed along with a scenario based approach. A different kind of distributionally robust CC-OPF was presented in [62], where a reformulation based on the Gaussian distribution was robustified against ambiguity in the (unknown) mean and covariance. In [63], distributional robustness for both aspects of the problem is considered by assuming that both the problem parameters (mean, covariance) and the type of distribution are unknown.

Here, we take the former approach and assume that the mean and covariance is known, but only partial information about the overall distribution is available. By extending the assumed knowledge about the distribution to include general properties like unimodality and symmetry, we can use tighter reformulations than the Chebyshev reformulation from [61]. We include a comparison and discussion of the applicability of different reformulations, and assess their relative performance in a case studies. Generally, we observe that a lower acceptable violation probability and a more conservative chance constraint reformulation increases system security, but also the operational cost. It is therefore important to find a good trade-off between the security level we want and the cost we are willing to pay.

The analytical reformulations presented here all share a number of attractive properties. First, the reformulated chance constraints are all in the form of an Second-Order Cone (SOC), admitting the formulation of

tractable OPF problems that are scalable to large systems. Second, the approach is scalable to a large number of random variables, as the number of random variables does not influence the number of constraints, and has limited impact on problem complexity. Third, the solution is more transparent than a sample based solution since it is possible to trace the influence of each random variable through the analytical relations. Finally, the solution based on the analytical reformulation is deterministic, i.e., the OPF will always find the same optimal solution with the same optimal cost. While this might seem trivial, it is not the case for all reformulation methods. For example, the OPF solution based on the scenario approach is actually random, since it depends on a set of randomly sampled uncertainty realizations. Thus, we typically obtain different solutions with different costs if we redraw the samples.

Organization

The remainder of this chapter is organized as follows. First, we present our model of the power system operation under uncertainty, which we build on in subsequent chapters. Based on the described model, we formulate the CC-SCOPF, incorporating both traditional N-1 constraints to secure the system against uncertainty and chance constraints to secure the system against a range of uncertainty realizations. We then present the analytical reformulations of the chance constraints, including two tight and three distributionally robust reformulations. We discuss the applicability of each reformulation and provide a general comparison, before showing the full reformulated problem.

The chapter finishes with two case studies. In the first, we investigate the impact of the uncertainty and the chosen violation probability on the OPF problem and its solution. In the second case study, we compare the different exact and distributionally robust reformulations of the chance constraints. We investigate both the empirical violation probability, as well as the impact of the formulations on cost.

The presented material is based mainly on [17] and [23].

2.2 Modelling Forecast Uncertainty

Forecast errors arise from power in-feeds that deviate from their forecasted values. The deviations in demand or production at any given

node can be due to various sources, such as load fluctuations, forecast errors for wind or PV or intra-day electricity trading. The characteristics of forecast errors ω differ between systems, and depend on the generation mix, load characteristics and how the market for short-term electricity trading is organized. Further, the forecast errors also depend on parameters such as the time of the day, or the forecast horizon. For example, the activation of frequency containment reserves in the continental European grid can be modelled as a Student t-distribution [51], while the day-ahead distribution of wind forecast depends on the forecasted wind power, and might be non-symmetric with heavy tails [64].

Load consumption and renewable energy production can take on any value within a certain range, and are naturally described using continuous probability distributions. Hence, we model the uncertainty sources $\tilde{u} \in \mathbb{R}^m$ as the sum of the forecasted production of active power $u_f \in \mathbb{R}^m$ and a random fluctuation δu :

$$\tilde{u} = u_f + \delta u . \quad (2.1)$$

Since the characteristics of the uncertainty depend on the system and the forecast horizon, the full distribution of δu is generally not known. However, we assume that at least some partial information about the distribution is available. In particular, we assume that the mean μ and covariance matrix $\Sigma_W \in \mathbb{R}^{m \times m}$ of the forecast errors exist, and can be estimated either based on historical data or through forecasting methods. Note that we allow for non-zero mean, since forecasts are not necessarily based on the expectation of $\tilde{u}$, but rather on the most probable realization (which are not the same, e.g., for skewed distributions). Further, we do not assume independence of the forecast errors, but allow off-diagonal elements representing non-zero correlation in Σ_W .

With the above information, we rewrite (2.1) in terms of the expected power production $u = u_f + \mu$ and a zero mean fluctuating component $\omega \in \mathbb{R}^m$:

$$\tilde{u} = u_f + \mu + \omega \quad (2.2)$$

$$= u + \omega . \quad (2.3)$$

In the subsequent chapters, we will mostly use (2.3) to refer to uncertainty, but will also comment on the impact of a forecast error bias μ , referring to (2.2).

For many power system applications, such as balancing, the *total* mismatch arising from the fluctuations is of interest. We define the total power mismatch $\Omega \in \mathbb{R}$ based on the zero mean fluctuating component,

$$\Omega = \sum_{i \in \mathcal{U}} \omega_i, \quad \text{with } \sigma_\Omega = \sqrt{\mathbf{1}_{1,m} \Sigma_W \mathbf{1}_{1,m}^T}, \quad (2.4)$$

where $\sigma_\Omega \in \mathbb{R}$ is the standard deviation of Ω and $\mathbf{1}_{1,m} \in \mathbb{R}^{1 \times m}$ is a row vector of ones with dimension m .

2.3 Power System Modelling

We consider a power system where $\mathcal{N}$, $\mathcal{L}$ denote the set of nodes and lines. The number of nodes and lines are given by $|\mathcal{N}| = m$ and $|\mathcal{L}| = l$. The set of nodes with uncertain demand or production of energy is given by $\mathcal{U} \subseteq \mathcal{N}$. The set of conventional generators is denoted by $\mathcal{G} \subseteq \mathcal{N}$, and are assumed to be controllable within their limits. To simplify notation, we assume that there is one conventional generator $p_{G,i}$, one composite uncertainty source u_i and one demand d_i per node, such that $|\mathcal{G}| = |\mathcal{U}| = |\mathcal{N}| = m$. Nodes without generation or load can be handled by setting the respective entries to zero, and nodes with multiple entries can be handled through a summation. We consider the outage of any single line or generator for the N-1 security constraints, and use a DC approximation to model the power flows.

2.3.1 Power Balance and Generation Control

Secure power system operation requires balance between consumed and produced power at all times. Here, we split the modelling of the power balance into the case with and without deviations ω from the schedule. For the base case (with $\omega = 0$ and no outages), we enforce the total power balance constraint,

$$\sum_{i \in \mathcal{N}} (p_{G,i} - d_i + u_i) = 0. \quad (2.5)$$

When forecast errors occur, the total power mismatch Ω must be balanced by an adjustment in the controllable generation. The amount of requested up- or down-regulation is proportional to the total mismatch Ω , but with opposite sign. This reflects the actions of the Automatic

Generation Control (AGC), which establishes power balance within tens of second to a few minutes [65]. We model the reserve activation through an affine control policy as in [14]:

$$\tilde{p}_G(\Omega) = p_G - \alpha\Omega. \quad (2.6)$$

Here, p_G , $\tilde{p}_G(\Omega) \in \mathbb{R}^m$ represent the scheduled and actual generation set-point, respectively, and the vector of participation factors $\alpha \in \mathbb{R}^m$ describes the contribution of each generator towards balancing the system. The elements α must sum to one to ensure that any given fluctuation Ω is balanced by the same amount $\sum_{i \in \mathcal{G}} (\alpha_i \Omega) = \Omega$.

The participation factors α can be chosen in different ways. Here, we assume that each generator contributes according to its maximum nominal output, similar to [29]. When all generators operate, the balancing contribution of each generator i is given by

$$\alpha_i = \frac{p_{G,i}^{max}}{\sum_{j \in \mathcal{G}} p_{G,j}^{max}} \quad \forall i \in \mathcal{G}. \quad (2.7)$$

This participation vector is valid during normal operating condition and in situations with line outages. During the outage of generator k , the compensation vector of the remaining generators i is given by

$$\alpha_i^k = \frac{p_{G,i}^{max}}{\sum_{j \in \mathcal{G} \setminus k} p_{G,j}^{max}} \quad \forall i \in \mathcal{G} \setminus k, \quad (2.8)$$

while the participation factor of the outaged generator k is set to zero $\alpha_k^k = 0$. The vectors α , $\alpha^k \in \mathbb{R}^m$ thus describe the compensation of any power mismatch in the system, for any operating situation without or with outages $k \in 0, \mathcal{L}, \mathcal{K}$. Note that by definition we have $\sum_{i \in \mathcal{G}} \alpha_i^k = 1$.

2.3.2 Power Flows

The power flows p_{ij} on each line are computed according to the linear DC approximation [65]. For the nominal operating condition without outages, the power flows can be expressed as

$$p_{ij} = \mathbf{M}_{(ij, \cdot)}(p_G - \alpha\Omega + u + \omega - d), \quad \forall ij \in \mathcal{L}. \quad (2.9)$$

The matrix $\mathbf{M} \in \mathbb{R}^{l \times m}$ is the matrix of Power Transfer Distribution Factors (PTDFs). It relates the line flows to the nodal power injections, which are expressed as the sum of generation $p - \alpha\Omega$, wind power

production $u + \omega$ and demand $-d$. The matrix $\mathbf{M}$ is defined as

$$\mathbf{M} = \mathbf{B}_F \begin{bmatrix} (\tilde{\mathbf{B}}_{Bus})^{-1} & \mathbf{0} \\ \mathbf{0} & 0 \end{bmatrix} \quad (2.10)$$

where $\mathbf{B}_F$ is the line susceptance matrix and $\tilde{\mathbf{B}}_{Bus}$ the bus susceptance matrix (without the column and row corresponding to the slack bus) [15]. The zero row and column correspond to the row and column of the slack bus.

Power flows with generation outages

When a generator is outaged, the system topology and hence the matrix $\mathbf{M}$ remain the same, whereas the power injections change. The new power flows are given by

$$p_{ij}^k = \mathbf{M}_{(ij, \cdot)}(p_G^k - \alpha^k(\Omega + p_{G,k}) + u + \omega - d), \quad \forall ij \in \mathcal{L}, \forall k \in \mathcal{G}, \quad (2.11)$$

where the vector of generation p_G^k is equal to p_G , but with the k^{th} entry (corresponding to the outaged generator) set to zero.

Power flows with transmission line outages

When a transmission line fails, the system topology and hence the matrix $\mathbf{M}^k$ changes, whereas the power injections remain the same. The new power flows are given by

$$p_{ij}^k = \mathbf{M}_{(ij, \cdot)}^k(p_G - \alpha\Omega + u + \omega - d), \quad \forall ij \in \mathcal{L}, \forall k \in \mathcal{L}, \quad (2.12)$$

where matrix $\mathbf{M}^k$ is defined based on the susceptance matrices $\mathbf{B}_{Bus}^k, \mathbf{B}_F^k$ of the system without the outaged line k .

2.4 Chance-Constrained Optimal Power Flow

Based on the power system model above, we formulate the CC-SCOPF. We use traditional N-1 security constraints to enforce security against outages, and chance constraints to account for the impact of uncertainty. We formulate the problem as a single period problem, where the time-step corresponding to the period can be, e.g., one hour or 15 min.

Objective Function

The objective is to minimize the total, nominal generation cost,

$$\min_{p_G} \sum_{i \in \mathcal{G}} (c_i p_{G,i}), \quad (2.13)$$

where $c_i \in \mathbb{R}^m$ represents the bids of the generators for providing energy.

Power Balance and Generation Constraints

The power balance and generator constraints are given by

$$\sum_{i \in \mathcal{N}} (p_{G,i} - d_i + u_i) = 0, \quad (2.14)$$

$$\mathbb{P} [p_{G,i} - \alpha_i \Omega \leq p_{G,i}^{max}] \geq 1 - \epsilon, \quad \forall i \in \mathcal{G} \quad (2.15)$$

$$\mathbb{P} [p_{G,i} - \alpha_i \Omega \geq p_{G,i}^{min}] \geq 1 - \epsilon, \quad \forall i \in \mathcal{G} \quad (2.16)$$

$$\mathbb{P} [p_{G,i} - \alpha_i^k (\Omega + p_{G,k}) \leq p_{G,i}^{max}] \geq 1 - \epsilon, \quad \forall i \in \mathcal{G}, \quad \forall k \in \mathcal{G} \quad (2.17)$$

$$\mathbb{P} [p_{G,i} - \alpha_i^k (\Omega + p_{G,k}) \geq p_{G,i}^{min}] \geq 1 - \epsilon, \quad \forall i \in \mathcal{G}, \quad \forall k \in \mathcal{G} \quad (2.18)$$

Here, (2.14) defines the power balance. Eq. (2.15) - (2.18) ensure that the generation remains within the technical generation limits p_G^{max} , p_G^{min} during normal operation, as well as during generator outages. Since the reserve activation depends on the random variable Ω , we cannot enforce (2.15) - (2.18) as standard, deterministic constraints. Instead, we formulate these constraints as chance constraints, and require each constraint to hold with probability $1 - \epsilon$. We refer to ϵ as the violation probability and to $1 - \epsilon$ as the security level. The value of ϵ is prescribed as an input parameter to the optimization, and can be chosen separately for each constraint.

Note that the chance constraints (2.15), (2.16) leave a non-zero probability ϵ of violating the generator limits. It is however physically impossible for the generator to increase their output above or below their limits. Therefore, a constraint violation should not be understood as an actual violation of the generator limit, but rather as the point at which the AGC has exhausted its ability to balance the system. Despite an increase in the power mismatch Ω , a generator which is already operating at full output cannot increase its generation further. Handling the situation would require manual intervention by the operator.

Power Flow Constraints

The power flow constraints are given by

$$\mathbb{P} [\mathbf{M}_{(ij, \cdot)}(p_G - \alpha\Omega + u + \omega - d) \leq p_{ij}^{max}] \geq 1 - \epsilon, \quad \forall_{ij \in \mathcal{L}} \quad (2.19)$$

$$\mathbb{P} [\mathbf{M}_{(ij, \cdot)}(p_G - \alpha\Omega + u + \omega - d) \geq -p_{ij}^{max}] \geq 1 - \epsilon, \quad \forall_{ij \in \mathcal{L}} \quad (2.20)$$

$$\mathbb{P} [\mathbf{M}_{(ij, \cdot)}(p_G^k - \alpha^k(\Omega + p_{G,k}) + u + \omega - d) \leq p_{ij}^{max}] \geq 1 - \epsilon, \quad \forall_{ij \in \mathcal{L}, k \in \mathcal{G}} \quad (2.21)$$

$$\mathbb{P} [\mathbf{M}_{(ij, \cdot)}(p_G^k - \alpha^k(\Omega + p_{G,k}) + u + \omega - d) \geq -p_{ij}^{max}] \geq 1 - \epsilon, \quad \forall_{ij \in \mathcal{L}, k \in \mathcal{G}} \quad (2.22)$$

$$\mathbb{P} [\mathbf{M}_{(ij, \cdot)}^k(p_G - \alpha\Omega + u + \omega - d) \leq p_{ij}^{max}] \geq 1 - \epsilon, \quad \forall_{ij \in \mathcal{L}, k \in \mathcal{L}} \quad (2.23)$$

$$\mathbb{P} [\mathbf{M}_{(ij, \cdot)}^k(p_G - \alpha\Omega + u + \omega - d) \geq -p_{ij}^{max}] \geq 1 - \epsilon, \quad \forall_{ij \in \mathcal{L}, k \in \mathcal{L}} \quad (2.24)$$

The constraints (2.19), (2.20) limit the power flows to stay within $\pm p_{ij}^{max}$ during normal operation¹. Similarly, the power flows are limited during generator outages by (2.21), (2.22) and during transmission line outages by (2.23), (2.24). These constraints depend on both the total fluctuation Ω and the individual deviations ω , and are enforced as chance constraint with security level $1 - \epsilon$.

As for the generator constraints, the chance constraints on the line flows leave a non-zero violation probability ϵ . In contrast to the generator constraints, the line constraints do however physically allow for overloads. A violation of the line flow constraints during normal operation (2.19), (2.20) could either be tolerated (given it is not too large or long lasting), or the operator could initiate additional remedial actions, such as generation redispatch or switching, to relieve the overload.

For the N-1 security constraints on generators (2.17), (2.18), and transmission lines (2.21) - (2.24), the physical interpretation of the violation probability is different. While N-1 violations increase the operating risk, a violation does not have an adverse impact on the system equipment unless the actual contingency takes place. Therefore, if the operator observes an N-1 violation in real time, he usually has sufficient time to perform calculations and initiate remedial actions to bring the system

¹We assume symmetric limits for the power flow constraints, which is reasonable in thermally constrained systems. Note that the formulation could easily be extended to account for asymmetric upper and lower bounds, where p_{ij}^{max} and p_{ij}^{min} are separately defined.

back to N-1 security. Hence, N-1 constraints can usually be treated as soft constraints, even for generators.

2.5 Analytical Reformulation of Chance Constraints

To obtain a solution of the CC-SCOPF problem, the chance constraints (2.15) - (2.24) must be reformulated to deterministic and tractable constraints. The constraints (2.15)-(2.24) are all univariate or single chance constraints of the general form

$$\mathbb{P}[a(p_G) + b(p_G)\delta u \leq c] \geq 1 - \epsilon . \quad (2.25)$$

where $a(p_G) \in \mathbb{R}$ and $b(p_G) \in \mathbb{R}^{1 \times m}$ are functions of the decision variables p_G and c is a constant. The term $a(p_G)$ represents the generation output or the line flows without forecast errors and c represents the generation or line flow limit. The vector $b(p_G)$ expresses the influence of the forecast errors δu on the respective constraint. Regardless of the exact expressions for $b(p_G)$, and for any dimension or distribution of the random vector δu , the left hand side of the constraint (2.25) is a scalar random variable $\delta = a(p_G) + b(p_G)\delta u$ with mean $\mu_\delta(p_G)$ and variance $\sigma_\delta(p_G)$ given by

$$\mu_\delta(p_G) = a(p_G) + b(p_G)\mu , \quad (2.26)$$

$$\sigma_\delta(p_G) = \sqrt{b(p_G) \Sigma_W b(p_G)^T} = \| b(p_G) \Sigma_W^{1/2} \|_2 . \quad (2.27)$$

The constraint (2.25) can hence be equivalently represented as

$$\mathbb{P} \left[\frac{\delta - \mu_\delta(p_G)}{\sigma_\delta(p_G)} < \frac{c - \mu_\delta(p_G)}{\sigma_\delta(p_G)} \right] = \mathbb{P} \left[\delta_n < \frac{c - \mu_\delta(p_G)}{\sigma_\delta(p_G)} \right] \geq 1 - \epsilon \quad (2.28)$$

where the scaled random variable $\delta_n := (\delta - \mu_\delta(p_G)) / \sigma_\delta(p_G)$ has zero mean and unit variance by construction.

In the following, we present five different analytical reformulations for (2.28), where each reformulation depends on a different set of assumptions regarding the distribution of δ_n . To unify the analysis, we consider the general situation in which the distribution $\mathbb{P}$ of δ_n is known to belong to a set of distributions $\mathbb{P} \in \mathcal{P}$. The set $\mathcal{P}$ is constructed such that it contains all distributions which share the assumed properties. We will

first show the reformulation steps for a general choice of the distribution set $\mathcal{P}$, before providing details for the specific reformulations we propose for the CC-SCOPF.

Given the set $\mathcal{P}$, we can rewrite constraint (2.28) as

$$\mathbb{P}[\delta_n < (c - \mu_\delta(p_G)) / \sigma_\delta(p_G)] \geq 1 - \epsilon \quad \forall \mathbb{P} \in \mathcal{P}. \quad (2.29)$$

Defining the function $f_{\mathcal{P}}(k)$ as the "worst-case" probability distribution in the set $\mathcal{P}$,

$$f_{\mathcal{P}}(k) := \inf_{\mathbb{P} \in \mathcal{P}} \mathbb{P}[\delta_n < k] , \quad (2.30)$$

we can replace (2.29) by

$$f_{\mathcal{P}}((c - \mu_\delta(p_G)) / \sigma_\delta(p_G)) \geq 1 - \epsilon . \quad (2.31)$$

As the function $f_{\mathcal{P}}$ is increasing, it has a well defined generalized inverse

$$f_{\mathcal{P}}^{-1}(\lambda) = \inf \{k \mid f_{\mathcal{P}}(k) \geq \lambda\} . \quad (2.32)$$

Hence, we have that

$$(c - \mu_\delta(P_G)) / \sigma_\delta(P_G) \geq f_{\mathcal{P}}^{-1}(1 - \epsilon) \quad (2.33)$$

After rearranging the terms and inserting the definitions for μ_δ , σ_δ from (2.26), (2.27), we obtain

$$a(p_G) \leq c - b(p_G)\mu - f_{\mathcal{P}}^{-1}(1 - \epsilon) \parallel b(p_G) \Sigma_W^{1/2} \parallel_2 \quad (2.34)$$

The set $\mathcal{P}$ can be chosen differently for different systems and even for individual constraints. The choice of reformulation depends on various aspects, such as the dominant source of uncertainty (e.g., load, renewables or short-term trading), the time frame of the forecast (e.g., day-ahead planning or close to real-time operation) and the type and quality of available data (e.g., historical forecast errors vs probabilistic forecasts, or many vs few data points). In the following, we will show the exact expressions for $f_{\mathcal{P}}$ and $f_{\mathcal{P}}^{-1}$ for five different distributional assumptions that we believe are appropriate in the context of SCOPF. Each distributional assumption corresponds to a different set $\mathcal{P}$. Depending on which assumption 1) - 5) is deemed appropriate, we define $f_{\mathcal{P}}^{-1}(1 - \epsilon)$ according to either an inverse cumulative distribution function for known distributions 1), 2) or a probability inequality when only partial information is available 3) - 5).

2.5.1 Reformulations for Known Distributions

The two first reformulations rely on the assumption of one specific distribution, and are tight, i.e., the security level $1 - \epsilon$ is enforced with equality. In this case, the set $\mathcal{P}$ only contains a single distribution, since the distribution of δ_n is assumed to be known.

1) Normal Distribution (Φ)

The normal distribution is a good approximation for the distribution of δ_n in two different cases. First, when δu follows a multivariate normal distribution, which has typically been used as a model for, e.g., load uncertainty, δ_n will be normally distributed as well. Second, when the number of uncertainty sources is large and not highly correlated, arguments similar to the central limit theorem [57] imply that the distribution of δ_n (which is a weighted sum of δu) is likely to be close to a normal distribution. This holds even if δu is not normally distributed.

Assume δ_n is distributed according to the normal distribution $\mathcal{N}(0, 1)$. Then, we have that

$$f_\Phi(k) = \Phi(k) , \quad f_\Phi^{-1}(1 - \epsilon) = \Phi^{-1}(1 - \epsilon) , \quad (2.35)$$

where Φ, Φ^{-1} are cumulative distribution function of the standard normal and its inverse.

2) Student's t Distribution (t)

When the distribution is heavy tailed, the Student's t -distribution can be an appropriate representation. Particularly when considering small violation probabilities, i.e. $\epsilon < 0.03$, Student's t distribution provides additional robustness compared to the normal distribution.

Assume δ_n is distributed according to a Student's t distribution $t_{\nu, \sigma_T}(0, \sigma_T)$ with ν degrees of freedom and scale parameter $\sigma_T = \sqrt{(\nu - 2)/\nu}$. Then, we have that

$$f_t(k) = t_{\nu, \sigma_T}(k) , \quad f_t^{-1}(1 - \epsilon) = t_{\nu, \sigma_T}^{-1}(1 - \epsilon) , \quad (2.36)$$

where $t_{\nu, \sigma_T}, t_{\nu, \sigma_T}^{-1}$ are the cumulative distribution function of the Student's t distribution and its inverse.

2.5.2 Distributionally Robust Reformulations

In many cases, only limited knowledge about the distribution of δ_n is available. The three next reformulations assume only partial information about the distribution, and ensure that the chance constraint holds (i.e., the security level remains at or above $1 - \epsilon$) for the set of distributions $\mathcal{P}$ that share the assumed properties. Since these formulations hold not only for one, but for a family of distributions, they are called *distributionally robust*. We note that the distributionally robust reformulations 3)-5) are generally not tight, and will usually lead to empirical violation probabilities below ϵ .

3) Symmetric, Unimodal Distributions (S)

If, for similar reasons as above, the distribution is likely to be close to normal, but we do not know how close, we can resort to the general assumption of unimodal, symmetric distribution with known mean and covariance. A unimodal distribution is, roughly speaking, a distribution where the probability density function has only one peak, referred to as the mode. The formal definition of unimodality can be found in [60].

Let the set $\mathcal{P} = S$ correspond to the set of all symmetric, unimodal distributions S with zero mean and unit variance. Due to symmetry, the mode of the distributions coincides with the mean, and we know that for any $k > 0$, the equality $\mathbb{P}[\delta_n \geq k] = \frac{1}{2}\mathbb{P}[|\delta_n| \geq k]$ holds. With this, the classical Gauss bound [66, 60] can be used to establish

$$\begin{aligned}
 f_S(k) &= 1 - \sup_{\mathbb{P} \in S} \mathbb{P}[\delta_n \geq k] \\
 &= \begin{cases} 1 - \frac{1}{2} \sup_{\mathbb{P} \in U} \mathbb{P}[|\delta_n| \geq k] & \text{if } k > 0 \\ 0 & \text{otherwise} \end{cases} \\
 &= \begin{cases} \frac{9k^2 - 2}{9k^2} & \text{if } k \geq \sqrt{\frac{4}{3}} \\ \frac{1}{2} + \frac{k}{2\sqrt{3}} & k > 0 \\ 0 & \text{otherwise} \end{cases} . \quad (2.37)
 \end{aligned}$$

The inverse function f_S^{-1} can be found by inverting (2.37),

$$f_S^{-1}(1 - \epsilon) = \begin{cases} \sqrt{\frac{2}{9\epsilon}} & \text{for } 0 \leq \epsilon \leq \frac{1}{6} \\ \sqrt{3}(1 - 2\epsilon) & \text{for } \frac{1}{6} < \epsilon < \frac{1}{2} \\ 0 & \text{for } \frac{1}{2} \leq \epsilon \leq 1 \end{cases} . \quad (2.38)$$

4) Unimodal Distributions (U)

In systems where the forecast uncertainty is related mainly to load, wind and PV production, the distribution of δu is likely to be unimodal, with fluctuations centered around the forecasted value. Under such conditions, it is reasonable to assume that the distribution of δ_n is also unimodal.

The unimodal case, in which $\mathcal{P} = U$ consists of unimodal distributions with zero mean and unit variance, can be dealt with using the one-sided Vysochanskij–Petunin inequality [67], i.e.,

$$\begin{aligned}
 f_U(k) &= 1 - \sup_{\mathbb{P} \in U} \mathbb{P}[\delta_n \geq k] \\
 &= 1 - \begin{cases} \frac{4}{9(1+k^2)} & \text{if } k \geq \sqrt{\frac{5}{3}} \\ 1 - \frac{4}{3} \frac{k^2}{1+k^2} & \text{if } k \geq 0 \\ 1 & \text{otherwise} \end{cases} \\
 &= \begin{cases} 1 - \frac{4}{9(1+k^2)} & \text{if } k \geq \sqrt{\frac{5}{3}} \\ \frac{4}{3} \frac{k^2}{1+k^2} & k \geq 0 \\ 0 & \text{otherwise} \end{cases}. \quad (2.39)
 \end{aligned}$$

Inverting (2.39), we obtain the following expressions for the inverse function f_U^{-1} ,

$$f_U^{-1}(1 - \epsilon) = \begin{cases} \sqrt{\frac{4}{9\epsilon} - 1} & \text{for } 0 \leq \epsilon \leq \frac{1}{6} \\ \sqrt{\frac{3(1-\epsilon)}{1+3\epsilon}} & \text{for } \frac{1}{6} < \epsilon \leq 1 \end{cases}. \quad (2.40)$$

5) Known Mean and Covariance (C)

The most general reformulation assumes only known (and finite) mean and covariance. This might be the most applicable reformulation in systems where little is known about the distribution. For example in Europe, the market for intra-day electricity trading introduce uncertainty in the power injections from conventional power plants. These transactions might follow almost any probability distribution, and can even be discrete.

When the set $\mathcal{P}$ corresponds to the set C containing all distributions with zero mean and unit variance, the classical Cantelli inequality [68]

establishes that

$$\begin{aligned}
 f_C(k) &= 1 - \sup_{\mathbb{P} \in \mathcal{C}} \mathbb{P}[\delta_n \geq k] \\
 &= 1 - \begin{cases} \frac{1}{1+k^2} & \text{if } k \geq 0 \\ 1 & \text{otherwise} \end{cases} \\
 &= \begin{cases} \frac{k^2}{1+k^2} & \text{if } k \geq 0 \\ 0 & \text{otherwise} \end{cases} .
 \end{aligned} \tag{2.41}$$

Taking the inverse gives us the following result for f_C^{-1} ,

$$f_C^{-1}(1 - \epsilon) = \sqrt{\frac{1 - \epsilon}{\epsilon}} \quad \text{for } 0 \leq \epsilon \leq 1. \tag{2.42}$$

The expressions for $f_{\mathcal{P}}^{-1}(1 - \epsilon)$ for all five assumptions are summarized in Table 2.1 and plotted against the security level $1 - \epsilon$ in Fig. 2.1.

Table 2.1: Expressions for $f_{\mathcal{P}}^{-1}(1 - \epsilon)$. Φ : Cumulative distribution function of the standard normal distribution. t_{ν, σ_T} : Cumulative distribution function of the Student t distribution with zero mean, ν degrees of freedom and scale parameter $\sigma_T = (\nu - 2)/\nu$.

1) Normal	$f_{\Phi}^{-1}(1 - \epsilon) = \Phi^{-1}(1 - \epsilon)$
2) Student's t	$f_t^{-1}(1 - \epsilon) = t_{\nu, \sigma_T}^{-1}(1 - \epsilon)$
3) Symmetric, unimodal	$f_S^{-1}(1 - \epsilon) = \begin{cases} \sqrt{\frac{2}{9\epsilon}} & \text{for } 0 \leq \epsilon \leq \frac{1}{6} \\ \sqrt{3}(1 - 2\epsilon) & \text{for } \frac{1}{6} < \epsilon < \frac{1}{2} \\ 0 & \text{for } \frac{1}{2} \leq \epsilon \leq 1 \end{cases}$
4) Unimodal	$f_U^{-1}(1 - \epsilon) = \begin{cases} \sqrt{\frac{4}{9\epsilon} - 1} & \text{for } 0 \leq \epsilon \leq \frac{1}{6} \\ \sqrt{\frac{3(1-\epsilon)}{1+3\epsilon}} & \text{for } \frac{1}{6} < \epsilon \leq 1 \end{cases}$
5) Mean, covariance	$f_C^{-1}(1 - \epsilon) = \sqrt{\frac{1-\epsilon}{\epsilon}} \quad \text{for } 0 \leq \epsilon \leq 1$

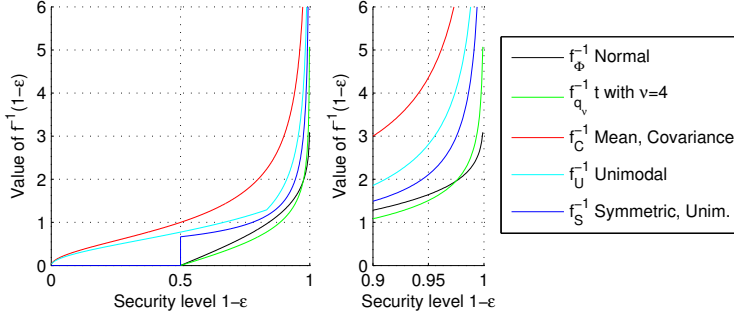


Figure 2.1: Values of $f_{\mathcal{P}}^{-1}(1 - \epsilon)$ for 1) the normal distribution (Normal), 2) the Student t distribution with 4 degrees of freedom (t with $\nu = 4$), 3) symmetric, unimodal distributions (Symmetric, Unim.), 4) unimodal distributions (Unimodal), and 5) the reformulation based on mean and covariance (Mean, Covariance). The left part shows security levels $0 \leq \epsilon \leq 1$, while the right part is a zoom in on high security levels $\epsilon \leq 0.05$.

2.5.3 Comparison

As shown through the derivation of (2.34), the chance constraint (2.25) can be reformulated to the following analytic expression for all distributional assumptions 1) - 5),

$$a(p_G) \leq c - b(p_G)\mu - f_{\mathcal{P}}^{-1}(1 - \epsilon) \| b(p_G) \Sigma_W^{1/2} \|_2 . \quad (2.43)$$

Analyzing (2.43), we see that the left part, $a(p_G) \leq c$, represents the *nominal* constraint, i.e., the constraint we would obtain if we neglect the forecast uncertainty. The second and third term represent an adjustment of the nominally available capacity, which is necessary to secure the system against forecast deviations. The second term is a correction related to the forecast error bias μ , which can be either positive or negative. The third term is always negative, i.e., it always leads to a reduction in the available capacity. This reduction can be interpreted as a security margin against uncertainty, i.e., an *uncertainty margin* s , defined by

$$\lambda = f_{\mathcal{P}}^{-1}(1 - \epsilon) \| b(p_G) \Sigma_W^{1/2} \|_2 . \quad (2.44)$$

Notice that the larger $f_{\mathcal{P}}^{-1}(1 - \epsilon)$, the larger the uncertainty margin. Since the reformulations 1) - 5) differ only in the definition of $f_{\mathcal{P}}^{-1}(1 - \epsilon)$,

we can compare them by comparing the value of $f_{\mathcal{P}}^{-1}(1 - \epsilon)$ for a given ϵ . In Fig. 2.1, $f_{\mathcal{P}}^{-1}(1 - \epsilon)$ is plotted against the security level $1 - \epsilon$.

First, we observe that all $f_{\mathcal{P}}^{-1}(1 - \epsilon)$ increase as ϵ decreases, indicating that a larger uncertainty margin is necessary to achieve a lower violation probability.

Second, as we move from reformulation 1) to 5), we are gradually decreasing the available information about the distribution. With decreasing information, the probabilistic bounds become more conservative and a higher value of $f_{\mathcal{P}}^{-1}(1 - \epsilon)$ is necessary to ensure the desired security level (i.e., $f_C^{-1} > f_U^{-1} > f_S^{-1}$). The lowest values are obtained when we assume knowledge of the actual distribution, i.e., for the normal and the Student's t distribution. Note that all reformulations assuming symmetry have f_S^{-1} , f_{Φ}^{-1} , $f_t^{-1} \leq 0$ for $\epsilon \geq 0.5$.

Finally, Student's t distribution has a more pronounced peak and heavier tails than the normal distribution. This is reflected in that for lower security levels, $f_t^{-1}(1 - \epsilon) < f_{\Phi}^{-1}(1 - \epsilon)$, while at high security levels, $f_t^{-1}(1 - \epsilon) > f_{\Phi}^{-1}(1 - \epsilon)$.

2.6 Reformulated Chance Constraints

With the reformulation presented above, we can represent the chance constraints (2.15) - (2.24) as

$$p_{G,i} \leq p_{G,i}^{max} - \alpha_i \mathbf{1}_{1,m} \mu - f_{\mathcal{P}}^{-1}(1 - \epsilon) \|\alpha_i \mathbf{1}_{1,m} \Sigma_W^{1/2}\|_2, \quad \forall i \in \mathcal{G} \quad (2.45)$$

$$p_{G,i} \geq p_{G,i}^{min} - \alpha_i \mathbf{1}_{1,m} \mu + f_{\mathcal{P}}^{-1}(1 - \epsilon) \|\alpha_i \mathbf{1}_{1,m} \Sigma_W^{1/2}\|_2, \quad \forall i \in \mathcal{G} \quad (2.46)$$

$$p_{G,i} + \alpha_i^k p_{G,k} \leq p_{G,i}^{max} - \alpha_i^k \mathbf{1}_{1,m} \mu - f_{\mathcal{P}}^{-1}(1 - \epsilon) \|\alpha_i^k \mathbf{1}_{1,m} \Sigma_W^{1/2}\|_2, \quad \forall i \in \mathcal{G}, k \in \mathcal{G} \quad (2.47)$$

$$p_{G,i} + \alpha_i^k p_{G,k} \geq p_{G,i}^{min} - \alpha_i^k \mathbf{1}_{1,m} \mu + f_{\mathcal{P}}^{-1}(1 - \epsilon) \|\alpha_i^k \mathbf{1}_{1,m} \Sigma_W^{1/2}\|_2, \quad \forall i \in \mathcal{G}, k \in \mathcal{G} \quad (2.48)$$

$$\mathbf{M}_{(ij, \cdot)}(p_G + u_f - d) \leq p_{ij}^{max} - \mathbf{M}_{(ij, \cdot)}(\mathbf{I} - \alpha \mathbf{1}_{1,m}) \mu - f_{\mathcal{P}}^{-1}(1 - \epsilon) \|\mathbf{M}_{(ij, \cdot)}(\mathbf{I} - \alpha \mathbf{1}_{1,m}) \Sigma_W^{1/2}\|_2, \quad \forall ij \in \mathcal{L} \quad (2.49)$$

$$\mathbf{M}_{(ij, \cdot)}(p_G + u_f - d) \geq -p_{ij}^{max} - \mathbf{M}_{(ij, \cdot)}(\mathbf{I} - \alpha \mathbf{1}_{1,m}) \mu + f_{\mathcal{P}}^{-1}(1 - \epsilon) \|\mathbf{M}_{(ij, \cdot)}(\mathbf{I} - \alpha \mathbf{1}_{1,m}) \Sigma_W^{1/2}\|_2, \quad \forall ij \in \mathcal{L} \quad (2.50)$$

$$\mathbf{M}_{(ij, \cdot)}(p_G^k + \alpha^k p_{G,k} + u_f - d) \leq p_{ij}^{max} - \mathbf{M}_{(ij, \cdot)}(\mathbf{I} - \alpha \mathbf{1}_{1,m}) \mu$$

$$-f_{\mathcal{P}}^{-1}(1-\epsilon) \|\mathbf{M}_{(ij, \cdot)}(\mathbf{I} - \alpha^k \mathbf{1}_{1,m}) \Sigma_W^{1/2}\|_2, \quad \forall_{ij \in \mathcal{L}, k \in \mathcal{G}} \quad (2.51)$$

$$\begin{aligned} \mathbf{M}_{(ij, \cdot)}(p_G^k + \alpha^k p_{G,k} + u_f - d) &\geq -p_{ij}^{max} - \mathbf{M}_{(ij, \cdot)}(\mathbf{I} - \alpha^k \mathbf{1}_{1,m})\mu \\ &+ f_{\mathcal{P}}^{-1}(1-\epsilon) \|\mathbf{M}_{(ij, \cdot)}(\mathbf{I} - \alpha^k \mathbf{1}_{1,m}) \Sigma_W^{1/2}\|_2, \quad \forall_{ij \in \mathcal{L}, k \in \mathcal{G}} \end{aligned} \quad (2.52)$$

$$\begin{aligned} \mathbf{M}_{(ij, \cdot)}^k(p_G + u_f - d) &\leq p_{ij}^{max} - \mathbf{M}_{(ij, \cdot)}^k(\mathbf{I} - \alpha \mathbf{1}_{1,m})\mu \\ &- f_{\mathcal{P}}^{-1}(1-\epsilon) \|\mathbf{M}_{(ij, \cdot)}^k(\mathbf{I} - \alpha \mathbf{1}_{1,m}) \Sigma_W^{1/2}\|_2, \quad \forall_{ij \in \mathcal{L}, k \in \mathcal{L}} \end{aligned} \quad (2.53)$$

$$\begin{aligned} \mathbf{M}_{(ij, \cdot)}^k(p_G + u_f - d) &\geq -p_{ij}^{max} - \mathbf{M}_{(ij, \cdot)}^k(\mathbf{I} - \alpha \mathbf{1}_{1,m})\mu \\ &+ f_{\mathcal{P}}^{-1}(1-\epsilon) \|\mathbf{M}_{(ij, \cdot)}^k(\mathbf{I} - \alpha \mathbf{1}_{1,m}) \Sigma_W^{1/2}\|_2, \quad \forall_{ij \in \mathcal{L}, k \in \mathcal{L}} \end{aligned} \quad (2.54)$$

$$(2.55)$$

where $\mathbf{I} \in \mathbb{R}^{m \times m}$ is the identity matrix. When comparing (2.45) - (2.54) to (2.34), we recognize the same structure. The first part represent the constraint we would obtain by neglecting the forecast uncertainty, whereas the second and third term on the right hand side represent the forecast bias correction and the uncertainty margin. We observe how the uncertainty margin leads to a reduction in the available transmission and generation capacity. Hence, a higher uncertainty margin will not only reduce the probability of violation, but also increase the nominal cost of operation (i.e., the cost of the CC-SCOPF). The acceptable violation probability ϵ and the distributional assumption which defines the function $f_{\mathcal{P}}^{-1}(1 - \epsilon)$ should therefore be chosen carefully to obtain a good trade-off between security against forecast errors and cost of operation.

A main advantage of the proposed formulation is that the stochastic problem does not increase the computational burden compared with the corresponding deterministic problem. The reformulated chance constraints (2.45) - (2.54) are linear, since the uncertainty margin is not dependent on any decision variables and can be pre-computed. The CC-SCOPF problem (2.13), (2.14), (2.45) - (2.53) is thus a linear program with the same computational complexity as a traditional DC SCOPF.

2.7 Case Study - Cost of Uncertainty

In this section, we investigate how the introduction of the chance constraints affects the solution of the OPF problem, and the resulting dispatch cost. We further compare the cost increase induced by the uncer-

tainty margins to the cost of enforcing the N-1 constraints. We assume a normal distribution for the forecast errors, and solve the CC-SCOPF with $f_{\mathcal{P}}^{-1}(1 - \epsilon) = \Phi^{-1}(1 - \epsilon)$.

We compare four different OPF formulations:

1. The OPF without consideration of security constraints (2.47), (2.48), (2.51) - (2.54), and without uncertainty ($\mu = 0$, $\Sigma_W = 0$).
2. The CC-OPF with chance constraints to account for uncertainty, but no security constraints.
3. The SCOPF with security constraints, but without consideration uncertainty ($\mu = 0$, $\Sigma_W = 0$).
4. The CC-SCOPF with both security and chance constraints.

2.7.1 Valuation Framework

The solution obtained by solving the OPF problem represents the most desirable generation dispatch, as it ensures efficient operation of both generation units and the transmission grid [29]. Incorporating additional constraints to account for security and uncertainty considerations not only leads to increased security but also to higher generation cost. Solving all four optimization problems allows for explicit analysis of the trade-off between operational security and cost. We propose to use two measures:

- Cost of Security (*CoS*): Cost difference between the OPF and the SCOPF (as defined in [29])

$$CoS = Cost_{SCOPF} - Cost_{OPF} .$$

- Cost of Uncertainty (*CoU*): Cost difference between a deterministic and a chance-constrained OPF

$$CoU_{OPF} = Cost_{CC-OPF} - Cost_{OPF} ,$$

$$CoU_{SCOPF} = Cost_{CC-SCOPF} - Cost_{SCOPF} .$$

When referring to both of these measures, the term *CoU* without subscript is used.

While *CoS* estimates the cost of enforcing the N-1 criterion, *CoU* measures the cost of integrating fluctuating generation resources and other sources of uncertainty.

Table 2.2: Generator Data

Bus	Generator #	Cost [\$/p.u.h.]	p_G^{max}
1, 2	1, 2, 5, 6	24.842	20
1, 2	3, 4, 7, 8	10.239	76
7	9-11	17.974	100
13	12-14	18.470	197
15	15-19	21.227	12
15, 16, 23	20, 21, 30, 31	9.537	155
18, 21	22, 23	5.230	400
22	24-29	1	50
23	32	9.587	350

2.7.2 Test System

Our case study is based on the 24-bus system shown in Fig. 2.2. The system corresponds to the IEEE One Area RTS-96 system, as described in [69], with some modifications as follows. The line capacities are reduced to 80 % of the steady state capacity listed in [69], since only active power flows are considered. The generation costs used in the OPF are the linear cost components from [70], and are listed with other generator data in Table 2.2. We consider a peak load situation, where the total system load d_{peak} equals 2850 MW and the total dispatchable generation capacity (not including wind energy) is 3405 MW.

Aggregations of wind power in-feed are located at bus 8 and bus 15, with an installed capacity of 500 and 700 MW, respectively. In the base case scenario, the forecasted wind power in-feed equals 25 % of installed wind power capacity, with 125 MW forecasted in-feed at bus 8 and 175 MW forecasted in-feed at bus 15. The wind power injections are assumed to be "must-run" generation, and are modelled as negative loads (i.e., the generation cost of wind is zero). Based on results from [71], the standard deviation σ_W of each wind power plant is assumed to be 7.5 % of the installed capacity. We assume that the forecast errors of the two wind power plants are independent, such that the covariance matrix Σ_W is diagonal, with σ_W^2 on the diagonal. Further, we assume that the forecast is unbiased, such that $\mu = 0$. The acceptable violation probability ϵ is set to 5 %. Since transmission lines can typically endure

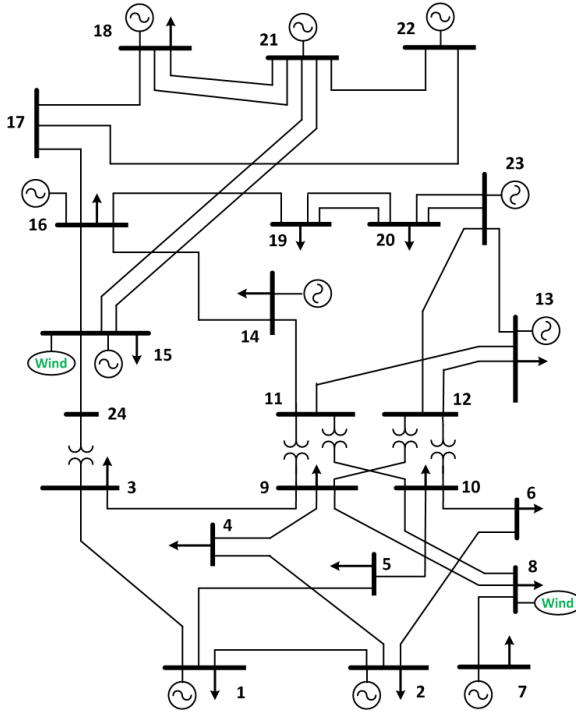


Figure 2.2: 24-bus test system with wind power in-feed at bus 8 and bus 15.

a higher load for a short amount of time, the post-contingency line limit is higher than the steady state limit. The short-term rating is assumed to be 10% higher than in steady state, such that $p_{ij}^{max,k} = 1.1p_{ij}^{max}$ for the security constraints. Each generator is assumed to have 25 % of their nominal capacity available as regulating power in case of a contingency k , such that $p_G^{max,k} = 1.25p_G^{max}$.

2.7.3 Investigations

The following aspects are investigated:

1) *Uncertainty margins:* An example of how the uncertainty margins reduce violation probability is shown to illustrate the purpose of the chance constraints. Further, as the influence of the wind power fluctuations is not the same throughout the system, the influence on different

lines is investigated. This is done by calculating the uncertainty margin for each of the lines and compare them to each other.

2) *Effect of uncertainty on cost:* The generation cost of the optimal dispatch from each of the four different optimization problems is calculated for the base case described above. The results allow us to assess CoS and CoU for the base case.

3) *Sensitivity with regard to load level:* The base case considers a load level corresponding to the peak system load. To find out how CoS and CoU depend on the load level, generation cost is calculated for a range of different load levels, where $d = \frac{X\%}{100\%} \cdot d_{peak}$ with $X = \{20, 40, 60, 80, 100\}$.

4) *Sensitivity with regard to forecast uncertainty:* To assess how the level of forecast uncertainty influences CoU , generation cost is computed for different standard deviations σ_W .

2.7.4 Results

1) Uncertainty Margins

One of the active transmission constraints in the base case OPF is the post-contingency constraint on line 23 (bus 14-16) after the outage of line 7 (bus 3-24). We illustrate the effect of the chance constraint by analyzing the post-contingency flows on line 23. Fig. 2.3 shows the expected values for the post-contingency line flow as obtained with SCOPF and CC-SCOPF, with the cumulative distribution function of the line flow superimposed. Due to the uncertainty margin in the CC-SCOPF, the expected line flow is shifted from 440 MW to 420 MW. This leads to a reduction of the probability of constraint violation from 50 % with the SCOPF to 5 % with the CC-SCOPF, and thus decreases the probability of post-contingency overloads.

The uncertainty margin for each line in the pre-contingency case is shown in Fig. 2.4. The largest uncertainty margins are necessary for lines that are related to power transfer from the buses where wind energy is produced (i.e. lines 12, 13 and 24), for some of the lines that transport power from the generators in the north to the loads in the south (i.e. lines 19 and 23 through bus 14), as well as for lines connected to large generators taking a major share in balancing the wind power deviations (i.e. line 29 between bus 18 and 17). These are the lines that are most influenced by deviations in the wind power in-feed.

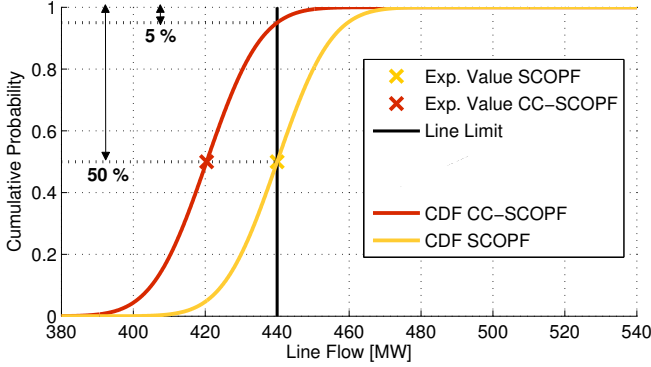


Figure 2.3: Expected value of the power flowing on line 23 (bus 14-16) after outage of line 7 (bus 2-3), as computed with the SCOPF and the CC-SCOPF. Superimposed on the expected value is the cumulative distribution function (CDF) of the line flow. Because of the uncertainty margin in the CC-SCOPF, the probability of violating the line limit constraint is reduced from 50 % to 5 %.

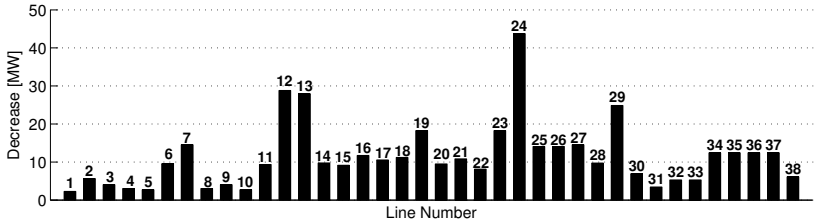


Figure 2.4: Uncertainty margins for each line in the pre-contingency case. The uncertainty margins represent the decrease of available capacity on the lines which is necessary to secure the system against uncertainty.

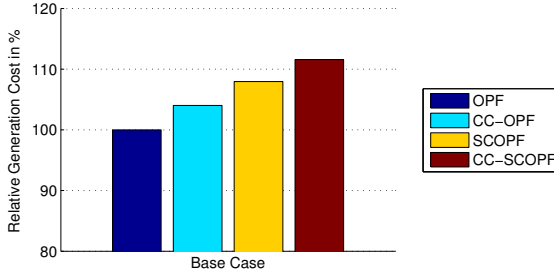


Figure 2.5: Total generation cost of the optimal dispatch computed with the OPF, CC-OPF, SCOPF and CC-SCOPF.

2) Effect of Uncertainty on Cost

In Fig. 2.5, the generation cost of the optimal dispatch from each of the four different OPF problems is shown for the base case described above. The difference in cost between the OPF and the SCOPF is approximately twice as high as the difference between the deterministic and probabilistic solutions, $CoS \approx 2 \cdot CoU$. The cost of imposing security constraints on the system, i.e. adding new constraints, is thus higher than the cost induced by constraint tightening due to uncertainty in this case.

3) Sensitivity with Regard to Load Level

The results obtained for different load levels are shown in Fig. 2.6. At very low load levels (20 %), the consideration of uncertainty and security constraints does not add any cost such that $CoS = CoU = 0$. In such low load situations, there is enough transfer capacity available to produce power at the cheapest power plants, even when security constraints and constraint tightening due to uncertainty is introduced. The overall cost is also very low, as much of the load is covered by wind generation. At intermediate load levels (40-60 %), the difference between the CoS and CoU is larger than in the peak load case, with $CoS > 4 \cdot CoU$. In high load situations (80-100 %), the $CoS \approx 2 \cdot CoU$. This suggests that the relative magnitude of CoU compared with CoS is larger in high load situation than in situations where the system is under less pressure. In a situation where the flows through the system are already high, decreas-

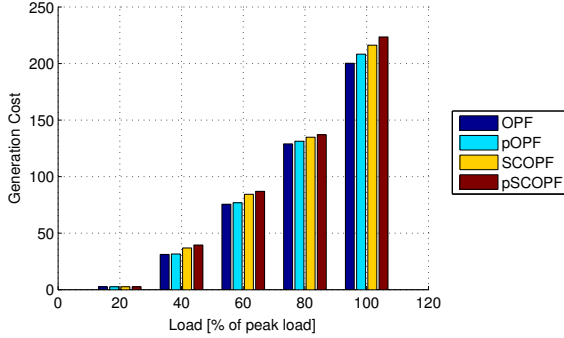


Figure 2.6: Total (absolute) generation cost for different load levels.

ing the transfer capacity has a higher impact on the generation cost.

4) Sensitivity with Regard to Forecast Uncertainty

In Fig. 2.7, the generation cost is shown for different standard deviations σ_W , given as % of the installed wind capacity. The cost increase is linearly related to the increase of the standard deviation up to a standard deviation of 12.5 %. With higher standard deviations, the cost of the CC-SCOPF dispatch increase more rapidly. This is because additional and more expensive generators (i.e. generators at bus 1 and 2) are dispatched to ensure feasibility. At the highest considered uncertainty level with $\sigma_W = 15\%$, the influence from the random variations is so high that $CoS < CoU$. The uncertainty has reached a level where the constraint tightening increases the cost more than the additional security constraints.

2.8 Case Study - Performance of the Analytical Reformulations

To investigate which of the analytical reformulations is the most appropriate, we assess their empirical performance in a case study for the

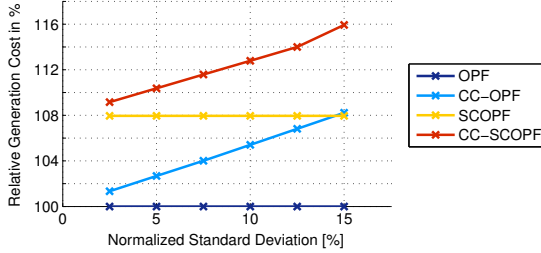


Figure 2.7: Total generation cost for different standard deviations σ_W .

IEEE 118 bus system. The uncertainties are modelled based on historical forecast errors from the Austrian Power Grid, reflecting the complicated distributions and interdependence of forecast errors observed in real data. We investigate the performance in terms of operational cost, and compare the empirical violation probabilities with the predefined acceptable violation probability. We further assess and discuss the accuracy of the distributional assumptions.

2.8.1 Test System

We base our study on the IEEE 118-bus system [72], with a few modifications as follows. The generation cost is assumed to be linear, and is based on the linear cost coefficients of the data provided with Matpower 4.1 [73]. Although the formulation could be extended to include unit commitment [39], it is not considered here. Therefore, the minimum generation output of the conventional generators is set to zero. The forecast uncertainty δu is modeled based on historical data for 1 year from the Austrian Power Grid (APG). We define the forecast error as the difference between the so-called Day-Ahead Congestion Forecast (DACF) and the snapshot (the real-time power injections) for all hours and buses with available data (8492 data points for 28 buses). Since the system is constantly evolving and might exhibit seasonal patterns, we assume that the power system operator only uses data from the past three months. We use two three-month periods to define the forecast uncertainty for this case study, such that we obtain 2207 data samples for a total of 54 buses.

The historical data was assigned to different load buses throughout the system, and modified such that the standard deviation corresponds to

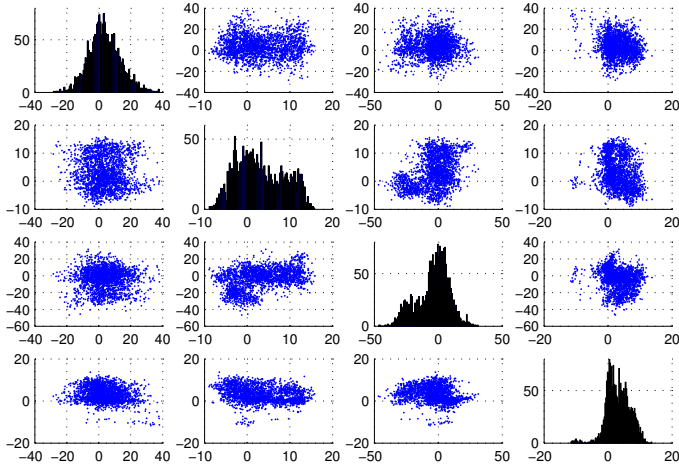


Figure 2.8: Forecast errors for 4 selected nodes of case study. The diagonal plots show the histograms of the forecast errors (x-axis: deviation in MW, y-axis: number of occurrences). The off-diagonal plots show the scatter plots between two corresponding forecast errors (x- and y-axis: deviation in MW).

20 % of the forecasted load. The mean μ and covariance Σ_W used in the CC-SCOPF were calculated based on this modified data (i.e, assuming perfect knowledge of μ , Σ_W). Fig. 2.8 shows the forecast errors from some representative nodes, including the histograms and pair-wise scatter plots of the forecast errors. By inspection, it is clear that the forecast errors are not normally distributed.

2.8.2 Operational Cost and System Security

To assess how the analytical reformulations perform in terms of operational cost and achieved empirical violation probability, we solve the CC-SCOPF for all five reformulations in Table 2.1. We consider a range of different acceptable violation probabilities $\epsilon = \{0.125, 0.1, 0.075, 0.05, 0.01, 0.005\}$. The results are compared with each other and with the solution of the corresponding deterministic SCOPF.

Operational Cost

The operational cost, normalized by the cost of the deterministic problem, is shown in the upper part of Fig 2.9. The results include the solution based on 1) a normal and 2) a Student t distribution, and the distributionally robust solutions 3)-5). Moving from brighter to darker bars, we thus assume lessened knowledge about the distribution. All chance-constrained solutions have higher cost than the deterministic solution, showing that the consideration of uncertainties increase the nominal cost of operation. The reformulations which assume more knowledge about the distribution 1), 2) lead to lower cost than the more general reformulations 3)-5). The most expensive solutions are obtained for reformulation 5), which only assumes knowledge of mean and covariance. In cases where ϵ is small, the distributionally robust are not able to find any feasible solution. For the reformulation based only on the mean and covariance 5), this happens already when $\epsilon \leq 0.05$. The reformulations assuming 4) unimodality or 3) unimodality and symmetry are infeasible when $\epsilon \leq 0.025$ and $\epsilon \leq 0.01$, respectively.

Empirical Violation Probability

To assess the performance of the chance constraint reformulation, we investigate how well the reformulations perform in terms of limiting the empirical violation probability ϵ_{emp} . We run a Monte Carlo simulation with the 2208 historical samples, and determine the empirical violation probability ϵ_{emp} by the number of violations per constraint. We also contrast the CC-SCOPF with the solution obtained with a deterministic SCOPF. The empirical violation probability per constraint is shown in the middle of Fig. 2.9. Note that we only show the *maximum* empirical violation probability ϵ_{emp} .

The maximum empirical violation probability of the deterministic SCOPF solution is very high, with $\max(\epsilon_{emp}) = 0.87$. This shows the need for probabilistic methods to avoid frequent violations of operational limits. All CC-SCOPF solutions have much lower violation probabilities than the deterministic. With the reformulation based on a normal distribution 1), there are always some constraints with an empirical violation probability $\epsilon_{emp} > \epsilon$, but the difference is never larger than $\epsilon_{emp} - \epsilon \leq 0.015$. Reformulation 2), which assumes the more peaked and heavy tailed Student's t distribution, has larger empirical violation

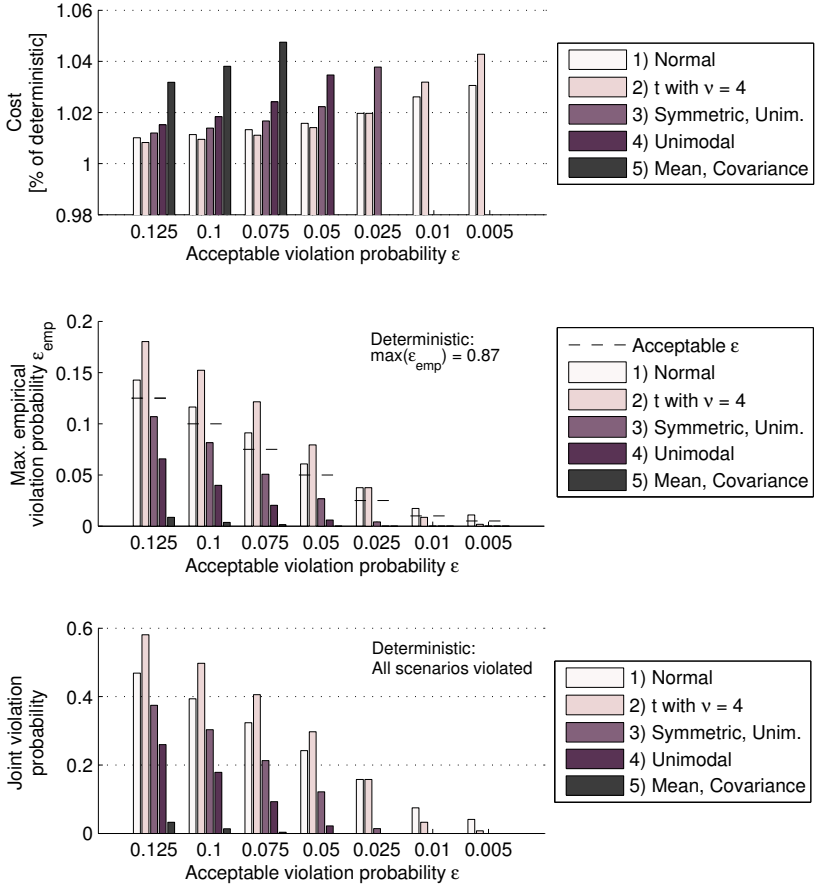


Figure 2.9: Results derived from the different SCOPF solutions, with different acceptable violation probabilities $\epsilon = \{0.125, 0.1, 0.075, 0.05, 0.01, 0.005\}$ from left to right. The bars denote (from bright to dark) the CC-SCOPF based on the assumption of 1) a normal distribution, 2) a Student t distribution, 3) a symmetric, unimodal distribution, 4) a unimodal distribution and 5) known mean and covariance. When a bar is missing, it indicates that the solution is infeasible. From top to bottom, the figure shows nominal dispatch cost (top), the maximum empirical violation probability ϵ_{emp} (middle) and the empirical joint violation probability (bottom).

probabilities than the normal distribution for $\epsilon > 0.025$. However, it offers a more conservative solution with fewer violations when $\epsilon < 0.025$, with $\epsilon_{emp} < \epsilon$. The distributionally robust solutions are well below the violation probability ϵ , with the symmetric, unimodal reformulation 3) being least conservative and the reformulation based only on mean and covariance 5) being the most conservative. We note that although the distributionally robust reformulations lead to $\epsilon_{emp} < \epsilon$, it does not necessarily imply that the underlying assumption (e.g., symmetry and unimodality) is accurate. We might get a low empirical violation probability, even if we assumed the wrong family of distributions.

Interestingly, both the reformulations based on the normal and Student's t distribution 1), 2) are able to find solutions with very small violation probabilities without causing infeasibility. In particular, the heavy tailed Student's t distribution is able to achieve empirical violation probabilities $\epsilon_{emp} < 0.005$. This indicates that the *original* CC-SCOPF is feasible for $\epsilon = 0.005$, although the *reformulated* CC-SCOPF based on 3) - 5) cannot guarantee a feasible solution.

The above results highlight two important aspects of the CC-SCOPF. First, we need to define ϵ such that it reflects a reasonable trade-off between cost and security, as enforcing a smaller ϵ increases cost. Second, we want to achieve an empirical violation probability ϵ_{emp} as close as possible to the accepted violation level ϵ . A reformulation with unjustified assumptions might lead to too many violations ($\epsilon_{emp} \gg \epsilon$) and insecure operations. At the same time, a too cautious reformulation might lead to very conservative solution ($\epsilon_{emp} \ll \epsilon$) and unnecessary high cost. A conservative reformulation might even cause infeasibility of the reformulated CC-SCOPF although the original CC-SCOPF problem is feasible.

Empirical Joint Violation Probability

Solving a problem with separate chance constraints does not provide any guarantees for the joint violation probability ϵ_J , i.e. the probability of observing at least one constraint violation. However, the joint violation probability represents the overall system security, and is therefore also of interest. We determine the empirical joint violation probability from the number of violated samples in the Monte Carlo simulation. The results are shown at the bottom of Fig. 2.9.

As expected, the joint violation probability is substantially larger than the violation probability per constraint. However, the ratio between the

joint violation probability and $\max(\epsilon_{emp})$ is approximately 3 – 4.5 in all cases, indicating that it is possible to at least partially control the joint violation probability by choosing an appropriate ϵ .

2.8.3 Accuracy of Chance Constraint Assumptions

Since the transmission and generation constraints are enforced as separate chance constraints, each constraint has a univariate probability distribution related to it. To assess whether these distributions satisfy the assumptions behind the analytical reformulations, we run statistical tests for each constraint. In particular, we use the Shapiro-Wilk test [74] to test if the distribution is normal, and Hartigans dip test [75] to test unimodality, using the implementation in R [76]. The test output is a p-value between 0 and 1, which indicates how probable it is that the data comes from a normal or a unimodal distribution, respectively. Typically, the hypothesis (normality or unimodality) is accepted for p-values above $p > 0.95$, and rejected for p-values $p < 0.05$. In between, we can neither reject nor confirm the hypothesis.

In Fig. 2.10, the p-values from both tests are plotted as a histogram. The bars show the percentage of constraints with p-values in the indicated p-value interval. We observe that the p-values from the Shapiro-Wilk test are below the threshold $p < 0.05$ for most constraints, while the p-values from Hartigans Dip Test are above $p > 0.95$ for the majority of the constraints. We thus conclude that unimodality is a reasonable assumption for most constraints, while the assumption of a normal distribution can be rejected in most cases.

Although the statistical test rejects normality, the normal distribution might still be a good assumptions for the tails of the distribution, which is what we are interested in. In Fig. 2.11, the empirical distribution of the line flow deviations for one active transmission constraint is shown for the case with $\epsilon = 0.1$. This constraint had the lowest p-value among the active constraints in the Shapiro-Wilk test. Fig. 2.11 further shows the empirical uncertainty margin corresponding to an empirical violation probability $\epsilon_{emp} = 0.1$, as well as the uncertainty margins obtained with the five distributional assumptions. We observe that the uncertainty margins based on the normal distribution (plotted in red) match very closely to the empirical margins (plotted in green). The Student t distribution underestimates the margin, while the distributionally robust reformulations overestimate it.

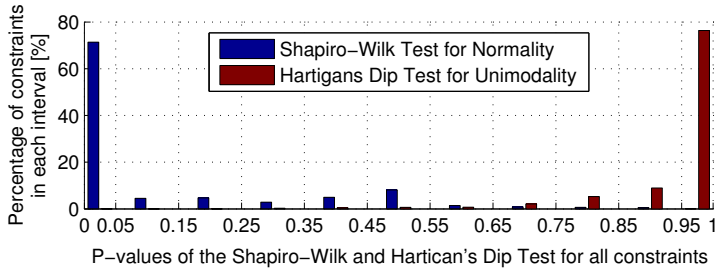


Figure 2.10: P-values obtained from Hartigan's dip test for unimodality and the Shapiro-Wilk test of normality. The histogram shows the percentage of all constraints in each interval. A high p-value indicates a high probability that the distribution is unimodal or normal, respectively. Based on the results, it seems highly probable that most constraints have distributions that are unimodal (p-values > 0.95), while it is unlikely that the distributions are normal (p-values < 0.05).

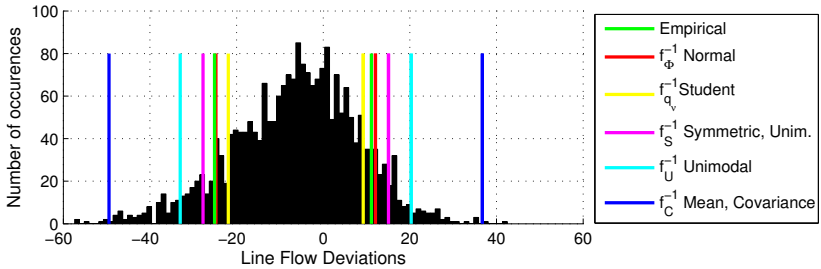


Figure 2.11: Histogram of post-contingency line flow deviations. The uncertainty margins are computed empirically (green), and for 1) a normal distribution (red), 2) a Student t distribution (yellow), 3) a symmetric, unimodal distribution (magenta), 4) a unimodal distribution (light blue) and 5) a distribution where only the mean and covariance are known (dark blue).

2.8.4 Discussion

In the above simulations, we observe that the distribution of power flows can be close to normally distributed, although the power injections are not. This is as expected, as the weighted sum of a large number of random variables will often be close to normally distributed [57]. We conclude that the reformulation based on a normal distribution can provide a good trade-off between cost and security when there is a significant number of uncertainty sources. Although assuming a normal distribution does not guarantee empirical violation probabilities $\epsilon_{emp} < \epsilon$, the assumption might be useful when ϵ can be interpreted as a guideline rather than a hard limit. Statistical tests and assessments as in Fig. 2.11 can be used to assess whether the normal distribution is a reasonable approximation. If the system operator wants a higher confidence in enforcing the acceptable violation probability ϵ and is willing to tolerate a larger increase in operational cost, assuming a unimodal distribution would be reasonable.

Further, note that the distributional assumption, as well as the violation probability, can be chosen differently for each constraint. If a specific line is known to have a special distribution (e.g., line connecting a wind power plant to the rest of the grid), the function $f_{\mathcal{P}}^{-1}$ can be specifically tailored to this line. In particular, this is true for the generator constraints (2.15)-(2.18). All generator constraints depend only on the total power mismatch Ω , which is a scalar random variable. The total power mismatch is typically well known to the operator, as it is recorded on a minute-to-minute basis in order to send appropriate AGC signals to the generators. It is therefore possible to obtain relatively accurate estimates for the empirical quantiles for the distribution of Ω . Swiss-grid currently uses an empirical distribution based on historical data for the Area Control Error (ACE) to enforce a chance-constrained reserve dimensioning criterion when clearing their reserve market [56]. The knowledge about this distribution could be applied to get an accurate reformulation of (2.15)-(2.18).

2.9 Summary and Conclusions

In this chapter, we formulated a CC-SCOPF with N-1 constraints to secure the system against outages and with chance constraints to secure the system against uncertainties. We further discussed different

analytic reformulations for chance constraints and their applicability in the CC-SCOPF context. The chance constraints are reformulated either by assuming a known probability distribution (such as normal or Student t distribution) or by using distributionally robust reformulations assuming general properties of the distribution (i.e., known mean and variance, symmetry, and unimodality).

The reformulated chance constraints all have a similar form, and are easily comparable. They are similar to the nominal constraints of the deterministic problem, except for the uncertainty margin (a security margin against forecast deviations), which represents a reduction of the transmission or generation capacity. With a larger uncertainty margin, the probability of violations decreases, but the nominal operational cost increases. Therefore, it is desirable to find a reformulation which leads to an uncertainty margin which is sufficiently large, yet as small as possible.

The analytically reformulated chance constraints offer a transparent and scalable way of ensuring security with a large number of uncertainty sources. Under the conditions presented here, i.e. pre-defined participation factors for reserve activation, the CC-SCOPF has the same computational complexity as the corresponding deterministic problem, as it remains linear and has the same number of constraints.

In the case study for the IEEE RTS 96 system, a valuation framework to estimate cost related to secure grid integration of renewables, so called *Cost of Uncertainty (CoU)*, was proposed. It was used together with the measure *Cost of Security (CoS)*, which estimates the cost of ensuring N-1 security, to evaluate the relative costs of securing the system against uncertainties from RES and contingencies. In the base case, *CoS* was found to be approximately twice as high as *CoU*. Sensitivities with regards to the load level and the level of uncertainty in the system show that this is not a general result, but that the *CoS* and *CoU* depend strongly on the network configuration.

In the case study based on the IEEE 118 bus system and forecast errors from APG in Austria, the trade-off between security and cost is highlighted and the different reformulations are compared with each other. We show that the normal distribution is a plausible approximation in systems with a large number of uncertainty sources, particularly when the acceptable violation probability can be interpreted as a guideline, rather than a hard constraint. If the transmission system operator wants to enforce the violation probability as a strict limit, choosing a

more conservative, distributionally robust reformulation based on unimodality will provide more confidence.

In general, the CC-SCOPF with analytically reformulated chance constraints provides a simple, yet efficient approach to assess and mitigate the effect of uncertainty in power system operational planning. The methods presented in the subsequent chapters will extend the above formulation to include more general balancing policies and corrective control, and apply similar ideas to the OPF problem with AC power flow constraints.

Chapter 3

Integrated Balancing and Congestion Management under Forecast Uncertainty

In this chapter, we investigate how different strategies for reserve activation influence the cost of integrating uncertain in-feeds. In particular, we assess under which conditions it is beneficial for the TSOs to balance across a larger region and when deviations should be handled close to where they originate. To achieve this, we propose different policies for reserve activation to balance the system, which are included in the chance-constrained framework proposed above and reformulated in a similar way. We demonstrate the feasibility of our approach on the IEEE 118 bus system. The case study illustrates the influence of the different balancing policies on the overall system cost as well as on the utilization of the transmission system. Comparisons are made between the deterministic and the probabilistic approaches in conjunction with the proposed balancing and congestion management schemes. The results show how accounting for congestion when activating reserves can help reducing the overall cost of handling uncertainty.

3.1 Motivation and Related Work

For the system operator, increased uncertainty has several implications. First, larger and more frequent fluctuations in the power production lead to an increased need for reserve capacity and reserve activation to keep the power balance. Second, uncertainty in the power injections impacts the power flows in the system, and hence new approaches to handle congestions are required.

Since fluctuations in the power production from renewable sources are not perfectly correlated, deviations across a large geographical area partially cancel each other out. The total power mismatch across all areas thus tends to be smaller than the sum of the absolute power mismatches for each area. To achieve power system balance with minimum reserve activation, the balancing area should therefore be as large as possible. This effect has led to the implementation of inter-TSO balancing schemes such as imbalance netting [77], [52], where active power imbalances between TSOs are netted (i.e., cancelled out against each other) before the control reserves are activated. As an example, the European initiative International Grid Control Cooperation (IGCC) started in 2011 with imbalance netting between the four German and the Danish TSOs, and has later gradually expanded to include 10 TSOs in 7 countries [78]. By January, IGCC had avoided activation of more than 3.8 TWh worth of control reserves [78].

However, while larger balancing areas reduce the amount of reserve activation, a power mismatch might be balanced far away from where it occurred. This leads to changes in the power flows, and complicates congestion management. For example, an unscheduled increase in the wind power close to the north of Europe and subsequent down-regulation by a power plant in the south might induce additional loop flows and overloads on transmission lines transporting power between the two different parts of the grid. If not planned appropriately, imbalance netting might thus necessitate additional (costly) redispatch measures, which undermine part of the the cost saving reasoning.

The TSO thus faces a trade-off: While balancing across a large region, i.e. *global balancing*, reduces the requirement on reserves, it might increase the cost of congestion management. An approach where deviations are balanced close to where they originate, i.e. *local balancing*, simplifies congestion management, but requires more reserves. The purpose of the study presented here is to investigate under which conditions each of the two approaches (local or global balancing) are cost-effective.

The importance of accounting for grid constraints in inter-zonal balancing operations was also investigated in, e.g., [79], but with a focus on market integration and only a limited set of transmission constraints. Here, we focus on security aspects in transmission system operation, and incorporate a more detailed model of both the transmission grid, the forecast uncertainty and the available reserves. We achieve this by extending the CC-OPF from Section 2.4 to model balancing reserves. We compare three different policies for reserve activation, where i) the generators provide balancing based on predefined participation factors α_G as in Section 2.3.1, ii) the participation vector α is optimized to provide best possible balancing of the total deviation Ω as in [18], and iii) we define and optimize a separate participation factor for each fluctuation ω .

In the case study, we assess and explain the differences in cost and congestion between the different balancing approaches, and show how the local balancing policy combines the benefits of imbalance netting and congestion management.

The presented results are based on work published in [21].

3.2 Modelling Reserve Policies

For a quantitative comparison of different balancing and congestion management approaches, we extend the CC-OPF by explicit modelling of reserve capacities and three different policies for reserve activation. To focus on the handling of fluctuations, the formulation presented here omits consideration of generator and transmission line outages. We base our modelling on the uncertainty representation (2.3), which expresses the actual power production $\tilde{u}$ in terms of the expected power production u and a zero mean fluctuation ω .

As above, we assume that the generators adjust their in-feeds to balance the system while accommodating the fluctuations ω . However, we now explicitly model the available up- and down-reserve capacities used for balancing, denoted by $r^+, r^- \in \mathbb{R}^m$. The reserve capacities can either be pre-defined, or be co-optimized along with the schedules for energy provision p_G .

In the following, we present three different *balancing policies*, which define the specific ways in which generators react to imbalances. The two first policies are based on *global* balancing, where the generators

only react to the overall power deviation Ω , while the last policy is based on *local* balancing, where the generators react directly to the local deviations ω .

Policy I - Global Balancing with Predefined Policy

As in 2.3.1, the contribution of balancing energy from each generator is a predetermined input to the optimization problem. Previously, we assumed that each generator contributes relative to their maximum output. Here, we base the definition on the relative share in reserve provision. We assume a European market setting where reserve markets are typically cleared at a weekly or daily basis, such that r^+ , r^- are known a-priori. The reserve procurement is assumed to be conservative such that there is sufficient reserves r^+ , r^- to cover relevant fluctuations, and symmetric such that $r^+ = r^-$. Further, we assume that generators have passed prequalification tests which ensure that their ramping capabilities enable sufficiently fast activation of reserves. The actual generator output $\tilde{p}_G$ is then given by

$$\tilde{p}_G(\Omega) = p_G - \alpha_1 \Omega \quad \text{where} \quad \alpha_1 = \frac{r^+}{\sum_{i \in \mathcal{G}} r_i^+} \in \mathbb{R}^m. \quad (3.1)$$

By definition, we have $\sum_{i \in \mathcal{G}} \alpha_{1,i} = 1$, which ensures balanced operation. This balancing policy corresponds to the situation where the system operator procures reserves without consideration of where the reserves are located.

Policy II - Optimal Global Balancing

As in [18], we do not predetermine the balancing policy, but allow the CC-OPF determine the optimal balancing location. The contribution of each generator to balance the total power mismatch Ω is given by

$$\tilde{p}_G(\Omega) = p_G - \alpha_2 \Omega, \quad (3.2)$$

where the balancing vector $\alpha_2 \in \mathbb{R}^m$ is subject to optimization. To ensure power balance during fluctuations and activation of only up or down reserves, we enforce the following constraints on α_2 :

$$\sum_{i \in \mathcal{G}} \alpha_{2,i} = 1, \quad \alpha_2 \geq 0. \quad (3.3)$$

This balancing policy corresponds to a situation where the TSO is able to decide where the balancing should happen in the system, and can use this ability to influence how reserve activation changes the power flows in the system.

Policy III - Optimal Local Balancing

For the local balancing case, the generators are reacting to each wind power plant separately. The contribution of each generator is given by

$$\tilde{p}_G(\omega) = p_G - \alpha_3 \omega, \quad (3.4)$$

where $\alpha_3 \in \mathbb{R}^{m \times m}$ is a matrix describing the individual balancing of each uncertainty source ω . As for Policy II, the balancing matrix α_3 is subject to optimization. To enforce power balance during deviations and activation of only up- or down-reserves, we add the following constraints:

$$\sum_{i \in \mathcal{G}} \alpha_{3(i,j)} = 1 \quad \forall_{j \in \mathcal{U}}, \quad \alpha_3 \geq 0. \quad (3.5)$$

Here, the first equation ensures that the activation of balancing energy for any individual fluctuation ω_j is equal to the fluctuation itself, i.e., $\sum_{i \in \mathcal{G}} \alpha_{3(i,j)} \omega_j = \omega_j$. By optimizing the full matrix α_3 , the TSO has more influence on how balancing changes power flows. In particular, the deviations can be balanced locally in parts of the system with significant congestion. On the other hand, the optimization can let the deviations cancel each other in parts of the grid, e.g. in a geographically close region without congestion, by choosing balancing vectors $\alpha_3(i, \cdot) = \alpha_3(j, \cdot)$. Policy III thus allows us to evaluate the trade-off between local and global balancing, and to quantify the benefits of moving from current balancing practice towards a more complex policy based on the location of both reserves and fluctuations.

For the remainder of this section, we will use the following short-hand notation to express Policies I-III,

$$\tilde{p}_G = p_G - \alpha \omega, \quad (3.6)$$

where $\alpha \in \mathbb{R}^{m \times m}$ is given by

$$\begin{aligned} \alpha &= [\alpha_1 \quad \alpha_1 \quad \dots \quad \alpha_1] && \text{for Policy I,} \\ \alpha &= [\alpha_2 \quad \alpha_2 \quad \dots \quad \alpha_2] && \text{for Policy II,} \\ \alpha &= \alpha_3 && \text{for Policy III,} \end{aligned}$$

in the three different cases. Note that α_1 and α_2 are special cases of α_3 .

3.3 Chance-Constrained Optimal Power Flow with Reserve Activation Policies

With the above modelling considerations, the CC-OPF is stated as follows:

$$\min_{p_G, \alpha, r^+, r^-} \sum_{i \in \mathcal{G}} (c_i p_{G,i} + c_i^+ r_i^+ + c_i^- r_i^-) \quad (3.7)$$

$$\text{s.t.} \quad (3.8)$$

$$\sum_{i \in \mathcal{N}} p_{G,i} - d_i + u_i = 0 \quad (3.9)$$

$$\sum_{i \in \mathcal{G}} \alpha_{(i,j)} = 1 \quad \forall j \in \mathcal{U}, \quad \alpha \geq 0 \quad (3.10)$$

$$p_G + r^+ \leq p_G^{max} \quad (3.11)$$

$$p_G - r^- \geq p_G^{min} \quad (3.12)$$

$$\mathbb{P} [-\alpha_{(i,\cdot)} \omega \leq r_i^+] \geq 1 - \epsilon_G \quad \forall i \in \mathcal{G} \quad (3.13)$$

$$\mathbb{P} [-\alpha_{(i,\cdot)} \omega \geq r_i^-] \geq 1 - \epsilon_G \quad \forall i \in \mathcal{G} \quad (3.14)$$

$$0 \leq r^+ \leq \bar{r}^+, \quad 0 \leq r^- \leq \bar{r}^- \quad (3.15)$$

$$\mathbb{P} [\mathbf{M}_{(ij,\cdot)} (p_G - \alpha \omega - d + u + \omega) \leq p_{ij}^{max}] \geq 1 - \epsilon_L \quad \forall ij \in \mathcal{L} \quad (3.16)$$

$$\mathbb{P} [\mathbf{M}_{(ij,\cdot)} (p_G - \alpha \omega - d + u + \omega) \geq -p_{ij}^{max}] \leq 1 - \epsilon_L \quad \forall ij \in \mathcal{L} \quad (3.17)$$

The objective (3.7) is to minimize the cost of generation p_G and reserve capacity r^+ , r^- . The vectors c , c^+ , c^- contain costs, i.e. bids, from the generators for the provision of energy and reserves. Eq. (3.9) reflects the power balance during nominal operation, while (3.10) enforces the power balance during fluctuations and a positive α . Constraints (3.11) - (3.12) ensure that the scheduled generation p_G , in combination with the reserve capacities r^+ , r^- , remain within the upper and lower generation limits p_G^{max} , p_G^{min} . Eq. (3.13), (3.14) ensure that the generators assigned to provide reserves have sufficient reserve capacity to cover their share of the fluctuations. Eq. (3.15) place upper bounds $\bar{r}^+$, $\bar{r}^-$ on the reserves that can be scheduled from each generator to ensure that the ramping capabilities of the generators are respected. Eq. (3.16), (3.17) enforce the transmission constraints.

Since the reserve activation and power flows depend on the fluctuations ω , the reserve constraints (3.13), (3.14) and transmission constraints

(3.16), (3.17) are formulated as separate chance constraints with violation probability ϵ_G for the reserve constraints and ϵ_L for the line constraints.

3.3.1 Reformulated Chance Constraints

The reformulation of the chance constraints can be handled in a similar way as in Section 2.5. Assuming normally distributed uncertainty, we can reformulate the reserve constraints (3.13), (3.14) and the line constraints (3.16), (3.17) as

$$\Phi^{-1}(1 - \epsilon_G) \|\alpha_{(\cdot, i)}^T \Sigma_W^{1/2}\|_2 \leq r^+, \quad \forall_{i \in \mathcal{G}} \quad (3.18)$$

$$-\Phi^{-1}(1 - \epsilon_G) \|\alpha_{(\cdot, i)}^T \Sigma_W^{1/2}\|_2 \geq -r^-, \quad \forall_{i \in \mathcal{G}} \quad (3.19)$$

$$\begin{aligned} \mathbf{M}_{(ij, \cdot)}(p_G - d + u) &\leq p_{ij}^{max} \\ &\quad - \Phi^{-1}(1 - \epsilon_L) \|\mathbf{M}_{(ij, \cdot)}(\mathbf{I} - \alpha) \Sigma_W^{1/2}\|_2, \quad \forall_{ij \in \mathcal{L}} \end{aligned} \quad (3.20)$$

$$\begin{aligned} \mathbf{M}_{(ij, \cdot)}(p_G - d + u) &\geq -p_{ij}^{max} \\ &\quad + \Phi^{-1}(1 - \epsilon_L) \|\mathbf{M}_{(ij, \cdot)}(\mathbf{I} - \alpha) \Sigma_W^{1/2}\|_2. \quad \forall_{ij \in \mathcal{L}} \end{aligned} \quad (3.21)$$

Since α is now an optimization variable and appear inside the 2-Norm, (3.18) - (3.21) are SOC constraints.

For the line constraints, we obtain the uncertainty margin

$$\lambda_{ij} = \Phi^{-1}(1 - \epsilon_L) \|\mathbf{M}_{(ij, \cdot)}(\mathbf{I} - \alpha \omega) \Sigma_W^{1/2}\|_2, \quad (3.22)$$

which has a direct interpretation as a reduction of the available transmission capacity. Based on the uncertainty margin (3.22) it is clear how Policies I-III have different effects on congestion. For Policy I, the variables α are predetermined, which implies that (3.22) is a fixed quantity. For Policy II and III, the values of α can be optimized to reduce the uncertainty margin on congested lines (thus increasing nominal transmission capacity). Policy III has a higher potential to reduce the uncertainty margins than Policy II, since it reacts to the individual fluctuations.

3.3.2 Problem Complexity

For Policy I where α is fixed, the uncertainty margins (3.22) are known quantities that can be pre-calculated. Moreover, with α_1 defined according to (2.7), the reserve constraints (3.18), (3.19) will be feasible if an

only if enough reserves $\sum_{i \in \mathcal{G}} r^+ = \sum_{i \in \mathcal{G}} r^- \leq \Omega$ have been procured. Assuming that the reserve procurement is appropriately dimensioned, these constraints can thus be omitted from the optimization problem. The CC-OPF with Policy I thus remains a linear program with the same solution complexity as the corresponding deterministic OPF.

For Policy II, the generation constraints (3.18), (3.19) can be simplified [80]. We express $\alpha = \alpha \cdot \mathbf{1}_{1,m}$, such that (3.18) reduces to the following linear constraint:

$$\alpha_i \Phi^{-1}(1 - \epsilon_G) \sigma_\Omega \leq r^+ \quad \forall_{\{i\} \in \mathcal{G}} , \quad (3.23)$$

$$\alpha_i \Phi^{-1}(1 - \epsilon_G) \sigma_\Omega \leq r^- \quad \forall_{\{i\} \in \mathcal{G}} . \quad (3.24)$$

The transmission constraints (3.16), (3.17) remain SOC constraints, increasing the computational complexity of the problem. The SOC constraints are however convex, which allows for the design of efficient solution algorithms along the lines of [18].

For Policy III, both generation and transmission constraints (3.18) - (3.21) are true SOC constraints. In addition, the solution complexity increases due to the larger number of optimization variables in α_3 . Hence, while Policy III is the most general policy and can be expected to deliver the lowest cost solution, this advantage comes at a higher computational cost. For the case study implemented here, it was possible to solve the problem with Policy III as a one-shot optimization problem. However, if the computational burden of solving the problem as a one-shot optimization is too high, the sequential SOCP algorithm discussed in the next chapter has been shown to solve similar problems even for large scale systems.

3.4 Case Study - Comparison of Balancing Policies

In this case study, we investigate how the different balancing policies impacts system operation. To isolate the impacts of the balancing policy on congestion, we assume that the reserve capacities r^+ , r^- have been pre-procured and are given a-priori as inputs to the optimization.

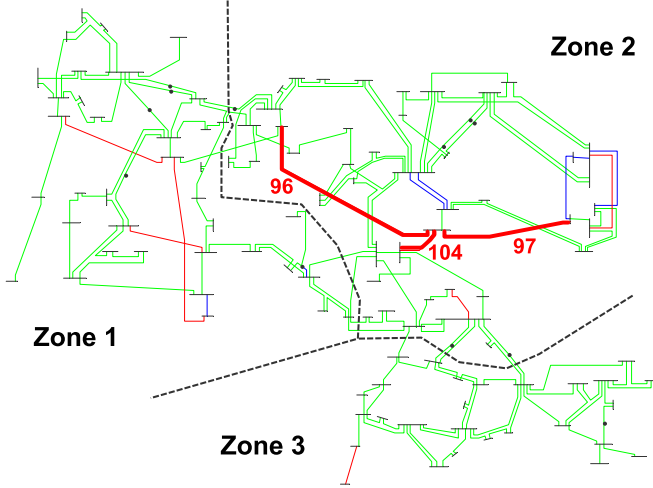


Figure 3.1: IEEE 118 bus system with 3 Zones and marking of the lines 96, 97 and 104. The color of the lines reflect the loading in the deterministic OPF solution, with green $< 0.9 p_{ij}^{max}$, blue $0.9 - 0.99 p_{ij}^{max}$ and red $> 0.99 p_{ij}^{max}$.

3.4.1 Test System

We use the IEEE 118 bus test system as defined in [72], with some modifications to make the system more congested. The system is divided into three zones, as seen in Fig. 3.1. The load is increased by a factor of 1.5 in zone 1 and 2, and by a factor of 2 in zone 3. The transfer capacities of the transmission lines are decreased to 75% of the original capacity. The generation capacity of all generators is increased by a factor of 3, and the available reserves capacities r^+ , r^- are set to 15% of the total generation capacity, i.e. $r^+ = r^- = 0.15 p^{max}$.

The system loads are interpreted as a mix between load and renewable energy sources connected at a lower voltage level. Instead of considering that particular wind in-feeds are uncertain, we assume that all 91 loads fluctuate around their forecasted consumption, and that the standard deviation σ_W of each load is equal to 20% of the forecasted consumption. For simplicity, we further assume that the fluctuations are uncorrelated, such that $\Sigma_W = \text{diag}(\sigma_W^2)$, although the method can han-

dle correlation by introducing off-diagonal elements in Σ_W . We choose acceptable violation probabilities $\epsilon_G = \epsilon_L = 0.05$.

We compare the CC-OPF with a deterministic OPF, which corresponds to the CC-OPF with $\omega = 0$. The CC-OPF is implemented in MatLab using Yalmip [81] to formulate the SOC constraints, and is solved using CPLEX. For the 118 bus test system, we were able to solve the problem with Policies I-III in one shot, without requiring the implementation of specialized solution algorithms.

3.4.2 Results

We first compare the nominal cost between the deterministic solution and the three chance-constrained solutions corresponding to Policy I, II and III. Second, we explain the differences in cost by looking at the uncertainty margins of different congested lines. Finally, we discuss how balancing is distributed between generators for Policies I, II and III.

Cost of uncertainty

In Fig. 3.2, the nominal cost of the generation dispatch (i.e., the value of the objective function) is shown for the deterministic solution, as well as for the chance-constrained problem with the three different policies. All costs are normalized by the cost of the deterministic solution. We observe that all chance-constrained solutions lead to a cost increase compared with the deterministic solution. With Policy I, the increase amounts to +2.5%, while Policy II is slightly less expensive with +2.3%. Policy III only results in higher costs of +0.2%.

While the nominal cost of the generation dispatch increases when accounting for uncertainty, the solution obtained through either of the chance-constrained approaches is more secure and will lead to significantly fewer violations of generation and transmission constraints in real time. The difference in the cost increase between Policies I-III show that the cost of integrating uncertainty in power system operations significantly varies dependent on different balancing strategies. In the following, we explain the reasons for the difference in cost.

Uncertainty margins

Accounting for uncertainty leads to a reduction in the available transmission capacity. Being able to influence the uncertainty margins

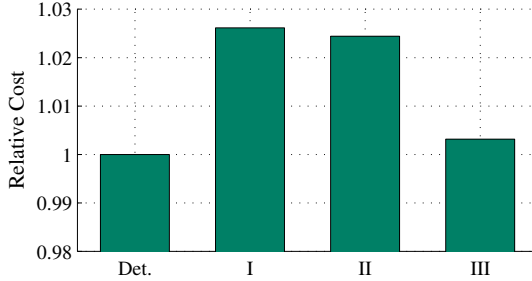


Figure 3.2: Cost of optimal power flow solution, normalized by the cost of the deterministic solution. Det.: Deterministic OPF. Policy I: CC-OPF with predetermined, global balancing. Policy II: CC-OPF with optimal, global balancing. Policy III: CC-OPF with optimal, local balancing.

through the choice of α_2 (with Policy II) or α_3 (with Policy III) is however useful to reduce the uncertainty margins on congested lines. Examples of such lines are line 96, 97 and 104 in Fig. 3.1. The uncertainty margins λ_{ij} for these lines are shown in Fig. 3.3. We observe that compared with the predefined Policy I, both Policy II and III reduce the necessary uncertainty margins. For Policy III, the uncertainty margins on the critical lines are almost zero.

With smaller uncertainty margins, more transmission capacity is available for transportation of energy from low-cost zones to high-cost zones, which decreases overall cost. The differences in the uncertainty margins therefore explain the cost differences seen in Fig. 3.2.

Control policy

Finally, we assess how the control policy α behaves for the different policies. Fig. 3.4 shows the value of the different entries of α_1 , α_2 (vectors) and α_3 (matrix). For the global Policies I and II, the balancing is spread relatively evenly among a large number of generators. The largest entries are $\max\{\alpha_1\} = 0.05$ and $\max\{\alpha_2\} = 0.22$, respectively. For the local Policy III, the majority of any single fluctuation ω_j is typically covered by a single generator. Further, we observe that the uncertain loads located geographically close to each other are typically balanced

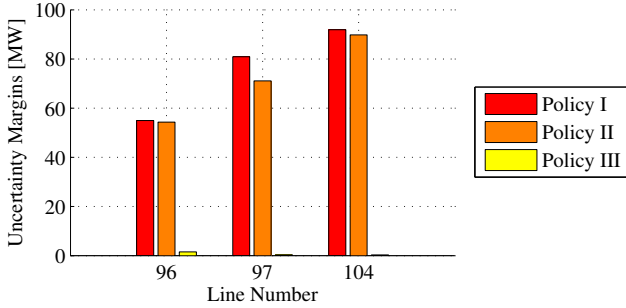


Figure 3.3: Uncertainty margins λ_{ij} for lines 96, 97 and 104 with the three different policies.

through the same generators, such that α_3 takes on a block-like structure. For example, loads 58-87 (located in zone 3) are covered mainly by generators 35, 36 (connected at bus 77 and 80 in zone 3).

As discussed in the introduction, global balancing reduces the need for activation of balancing energy, while local balancing is more effective in reducing congestion. Policy III brings together the benefits in an optimal way by combining global and local balancing into "regional" balancing. On a regional level, imbalance netting is applied by choosing similar balancing vectors $\alpha_{3(j, \cdot)}$ for all ω_j in a given region (e.g., zone 3). However, the balancing vectors $\alpha_{3(j, \cdot)}$ differ between regions that are separated by congestion, in order to reduce the uncertainty margins on the congested lines.

3.5 Summary and Conclusions

In this chapter, we investigated different approaches to integrated balancing and congestion management in systems with significant levels of forecast uncertainty. We compared global balancing approaches which only account for the total system deviation with a local balancing approach where fluctuations are balanced based on their location in the system. While the global balancing policy requires less reserve activation, the local balancing policy can handle congestion more efficiently.

The approaches were quantitatively compared within the chance-constrained OPF framework, which was extended to account for local

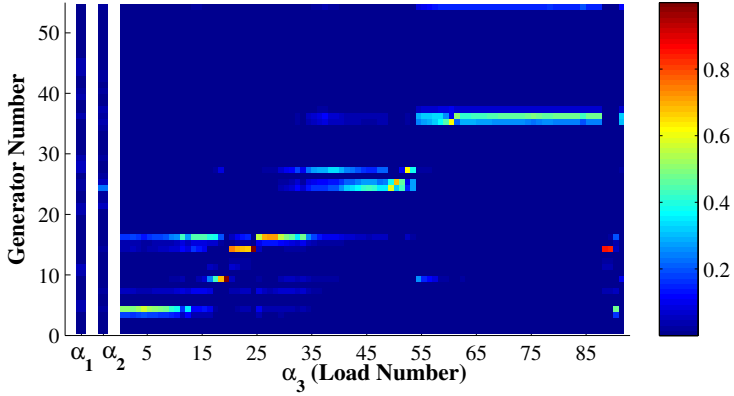


Figure 3.4: Value of α for the three different balancing policies. To the left: The balancing vectors α_1 , α_2 , where the generators react only to the overall power mismatch (one entry per generator). To the right: The balancing matrix α_3 , where the generators react to each uncertain in-feed separately. The sum for all entries in each vertical column is one to ensure that the system is always balanced.

balancing policies. In a case study based on the IEEE 118 bus system with 91 uncertain loads, we demonstrated that the local balancing policy reduces the cost of integrating wind power by reducing congestion due to forecast errors. Further, we observed that the optimization automatically chooses to perform imbalance netting in parts of the system without internal congestion.

We would like to remark that the cost of procuring reserve capacity has a significant influence on the benefit of local balancing which requires higher reserve capacities. If the reserve capacities are included as optimization variables in the problem instead of being fixed a-priori as in the case study, the benefit of the local policy would become less pronounced. In particular, the local policy would become more similar to the global policy if the cost of reserve capacity c^+ , c^- is high.

Finally, we note that the above results make *regional* balancing, where fluctuations within a region specified by congestion are balanced together, seem interesting. Regional balancing might be a useful trade-off between computational complexity of the CC-OPF and more flexible use of reserves.

Chapter 4

Corrective Control to Handle Forecast Uncertainty

While higher shares of electricity generation from renewable energy sources and market liberalization increase uncertainty in power systems operation, operation becomes more flexible with improved control systems and installation of equipment such as PST and HVDC connections. The use of corrective control in response to outages is known to reduce the cost of N-1 security. In this chapter, we propose a method to extend the use of corrective control of PSTs and HVDC to react to uncertainty. We characterize the uncertainty as continuous random variables, and define the corrective control actions through affine control policies. This allows us to efficiently model control reactions to a large number of uncertainty sources. The control policies are then included in the CC-SCOPF, for which we develop an efficient solution algorithm. In a case study for the IEEE 118 bus system, we show that corrective control for uncertainty decreases in operational cost, while maintaining the same level of system security. Further, we demonstrate the scalability of the method by solving the problem for the IEEE 300 bus system and the Polish system with 2383 buses.

4.1 Motivation and Related Work

In power systems terminology, a *preventively* secure system state refers to a situation in which the power system remains secure after any credible contingency, typically defined as an N-1 situation, without any post-contingency control action. A *correctively* secure state refers to a situation where post-disturbance controls might be required to bring the system back to a secure operating point.

The use of corrective control has increased over the past years [30]. On the one hand, the use of corrective actions is driven by increased uncertainty from renewable electricity generation, economic considerations due to market liberalization and a system operated closer to its limits. On the other hand, better control systems, more situational awareness and the installation of devices such as phase-shifting transformers (PSTs) and high-voltage direct current (HVDC) connections provide the system operator with new possibilities to control power flows and react to changes in the system in real-time.

While corrective control can reduce operational cost [29] and is applied routinely in real-time system operation, the real-time set-point changes of HVDC and PSTs are typically chosen in an ad-hoc fashion. This is partially due to the difficulty of planning corrective control actions in response to forecast uncertainty, as it requires the consideration of a large number possible uncertainty scenarios in addition to consideration of contingencies. However, the need to ensure that corrective control will be sufficient in real-time calls for efficient ways of modelling both the possible system states and the corresponding corrective control actions.

Only few methods for power system operational planning under uncertainty have considered the application of corrective control actions or the existence of power flow control devices such as PSTs or HVDC. Refs. [4, 5] propose a three-stage OPF framework where corrective control actions are used to ensure feasibility during worst-case combinations of contingencies and uncertainty, which is extended to the definition of optimal corrective control actions in [82]. The OPF formulation in [11] accounts for uncertainty and includes corrective control for a limited number of pre-selected, critical uncertainty scenarios. In [16, 83], *post-contingency* corrective control by devices such as HVDC and PSTs is applied within a CC-OPF. However, corrective control actions *in reaction to the uncertainty realizations* have not been considered in either of these works. Here, we propose a framework with combined corrective

control in reaction to *both uncertainty and contingencies*. Related ideas considering corrective control of HVDC has been discussed in [84, 85], although without discussion of PSTs or addressing issues related to scalability.

Corrective control for uncertainty differs from post-contingency corrective control in several ways. While contingencies are typically low probability, discrete events that induce possibly large and sudden impacts on the system, uncertainty realizations occur frequently and develop in a more continuous fashion (although ramping due to, e.g., clouds, sunset or fog can be fast). These differences impact the time available for implementation, as well as the type and modelling of control reactions. However, we refer to both as *corrective control*, since they act to mitigate the impact of already occurred events (as opposed to preventive measures).

We focus on corrective control of HVDC and PSTs, which are typically controlled by the transmission system operator and are cheap to control, as opposed to the use of generation redispatch, which interferes with market operation and could incur significant cost. Since uncertainty from, e.g., renewable energy production are naturally characterized as continuous random variables, we propose to model the corrective control through a continuous, affine control policy. The corrective control actions are included in an extended version of the CC-SCOPF in Section 2.4, and reformulate the chance constraints into SOC constraints. We then present a solution algorithm which allows us to solve the CC-SCOPF for large systems. The proposed algorithm is based on solving a sequence of SOCPs, which in our case outperforms the cutting-plane algorithm proposed in [18].

The benefits of the method are demonstrated in a case study based on the IEEE 118 bus system. The results show that corrective control in reaction to uncertainty can reduce operational cost, while maintaining system security. Further, we demonstrate and discuss the scalability of the method in a case study including both the IEEE 300 bus test system and the large-scale Polish test case with 2383 buses.

The material presented in this chapter is based on [22].

4.2 Modelling Framework for Corrective Control

We focus on corrective control as a means to handle transmission line congestion. We distinguish between two types of corrective control: Corrective control to handle contingencies and corrective control to handle forecast uncertainty. In the following, we present how we model corrective control for a generic power flow controller, which changes set-point in response to transmission line outages or fluctuations in the power injections. We do not consider corrective control for generation outages, but this can be incorporated within the proposed modelling framework. Modelling considerations related specifically to HVDC and PSTs, as well as details on how the corrective control influences the power flows on the transmission lines are given in Section 4.3.

4.2.1 Corrective Control for Contingencies

Post-contingency control is modeled as a change in the set-points of power flow control devices, such as the flow across HVDC connection or the phase angles of PSTs, following an outage. Since outages can be characterized as a set of discrete events, we model the set-point changes by introducing additional optimization variables δ^{ij} for every outage as in [29, 16]. The set-point of a power flow controller following the outage of line ij is thus given by

$$\tilde{\pi}(ij) = \pi + \delta^{ij}, \quad (4.1)$$

where π , $\tilde{\pi}(ij)$ represents the pre- and post-contingency set-points, respectively, and δ^{ij} represents the set-point change following the contingency.

4.2.2 Corrective Control for Forecast Uncertainty

While post-contingency corrective control describes reactions of the controllers to an outage *after the outage has happened*, corrective control for uncertainty describes reactions of the controllers *after the uncertainty has realized*. We model these corrective control actions through affine control policies, where the controllers adjust their set-point proportional

to the forecast error. The controller set-points, including corrective control for forecast uncertainty, are given by

$$\tilde{\pi}(\omega) = \pi + \alpha_{corr}\omega, \quad (4.2)$$

where π , $\tilde{\pi}(\omega)$ represents the scheduled and real-time controller set-points, and the vector ω denotes the random fluctuations. The matrix α_{corr} defines the response of the controllers, and control parameters α_{corr} are subject to optimization.

Restricting the corrective action to an affine response leads to solutions that are less optimal than if we use, e.g., an arbitrary policy or define optimal set-points based on specific uncertainty scenarios. However, representing arbitrary policies or a reasonable set of uncertainty scenarios within an optimization problem is challenging, particularly when scaling to large systems. Therefore, while an affine response policy might lead to less optimal solutions, it also has several advantages. First, it allows us to optimize over the corrective control actions without compromising computational tractability. Second, a response based on a control policy allows us to treat the fluctuations as continuous variables (as opposed to a representation through a finite number of scenarios). Third, the affine policy can be easily implemented by the TSO. Fourth, even if the affine policy is not directly implemented, it guarantees that a feasible solution will exist.

Since we apply corrective control to manage power flows and congestion, the specific location of both the fluctuations and the control reactions in the grid matters. We therefore define a control parameters $\alpha_{(i,j)}$ for each pair of controller i and uncertainty source j , similar to the most general generation control policy in section 3.2. In this way, the controllers can react to fluctuation ω_j based on its location in the grid.

4.2.3 Combined Corrective Control

Assuming that each controller is able to provide corrective control for both contingencies and forecast uncertainty, we obtain the combined corrective control policy,

$$\tilde{\pi}(\omega, ij) = \pi + \alpha_{corr}\omega + \delta^{ij}. \quad (4.3)$$

Note that the reaction to forecast uncertainty is decoupled from the reaction to contingencies, i.e., α_{corr} does not depend on whether the

system is in post- or pre-contingency state. This is because the number of additional control variables would otherwise be very significant and difficult to handle.

4.2.4 Corrective Control in Real-Time Operation

The corrective control actions δ^{ij} , α_{corr} are necessary to ensure secure system operation in real time. The post-contingency corrective control actions δ^{ij} can be made available to the operator in form of a look-up table, such that they can be implemented fast following a contingency. The corrective control actions due to uncertainty, defined by a combination of the real time realization of the fluctuations ω and the control parameters α_{corr} , can be implemented either as continuous, automatic adjustments (e.g., similar to AGC control) or as a manual change in set-points which is implemented only when a constraint violation occurs in the system. However, we do not necessarily need to restrict the control actions to an affine policy in real-time operation. In real-time, the realization of the uncertainty ω is known, and the operator can rerun the SCOPF as a deterministic problem. In this problem, the set-points of the HVDC and PSTs can be optimally chosen based on the current operating point. The real-time control response can therefore take a more general form than the affine policy, and we solve an SCOPF with less restrictive constraints. This extends the feasible space of the optimization problem. If the operational planning problem was feasible with an affine policy, we are hence guaranteed to find a solution in real-time operation.

4.3 Power System Modelling with Corrective Control

In this section, we extend the CC-SCOPF to incorporate corrective control. We use the DC power flow model based on nodal power balance constraints and bus voltage angles, instead of the DC power flow model based on total power balance and PTDFs as in Section 2.3. The nodal formulation preserves sparsity and improves scalability of the problem.

The power flow control devices used for corrective control are the set of HVDC connections $\mathcal{H}$ and the set of PSTs $\mathcal{S}$, with $|\mathcal{H}| = h$ and $|\mathcal{S}| = s$, respectively. We consider the outage of any single line for the

N-1 security constraints, but leave out lines that lead to splitting of the system. Generation outages are handled in a simplified way through a pre-determined reserve requirement. We base our modelling on the uncertainty representation (2.3), which expresses the actual power production $\tilde{u}$ in terms of the expected power production u and a zero mean fluctuation ω .

4.3.1 HVDC and PST Set-Points

We assume that all HVDC installations are point-to-point connections such that the power flow p_{DC} is controllable within the limits $p_{DC}^{max}, p_{DC}^{min}$.

The difference between subsequent PST tap positions are assumed to be small enough for the angles ϑ to be well approximated as continuous variables. The upper and lower limit on the PST angles are given by $\vartheta^{max}, \vartheta^{min}$. As outlined in Section 4.2, the post-contingency corrective control of HVDC and PSTs are modelled through additional variables $\delta_{DC}^{ij}, \delta_{\vartheta}^{ij}$, while the corrective control for uncertainties are modelled through the matrices $\alpha_{DC} \in \mathbb{R}^{h \times m}$, $\alpha_{\vartheta} \in \mathbb{R}^{s \times m}$, respectively. The set-points of the HVDC and the PSTs with corrective control are thus given by

$$\tilde{p}_{DC,i}(\omega, ij) = p_{DC,i} + \delta_{DC,i}^{ij} - \alpha_{DC(i,\cdot)}\omega, \quad \forall i \in \mathcal{H}, ij \in \{0, \mathcal{L}\}, \quad (4.4)$$

$$\tilde{\vartheta}_{DC,i}(\omega, ij) = \vartheta_i + \delta_{\vartheta,i}^{ij} - \alpha_{\vartheta(i,\cdot)}\omega, \quad \forall i \in \mathcal{S}, ij \in \{0, \mathcal{L}\}, \quad (4.5)$$

where $ij = 0$ refers to the set-points in the pre-contingency state where $\delta_{DC}^0 = \delta_{\vartheta}^0 = 0$.

4.3.2 Power Balance and Generation Control

We split the modelling of the power balance into the case with and without deviations ω from the schedule. For the base case (with $\omega = 0$ and no outages), we enforce the nodal power balance constraints,

$$p_G - d + u + \mathbf{C}_{DC} p_{DC} = \mathbf{B}_{bus} \cdot \theta + \mathbf{B}_{\vartheta} \cdot \vartheta. \quad (4.6)$$

The vector $\theta \in \mathbb{R}^m$ refers to the voltage angles at each bus. The matrix $\mathbf{C}_{DC} \in \mathbb{R}^{m \times h}$ is the incidence matrix of the HVDC connections, with -1 at the bus defined as the "from" end and +1 at the bus defined as

the "to" end of the connection. The matrix $\mathbf{B}_\vartheta$ describes the influence of the PSTs and has non-zero entries only for buses connected to lines with PSTs. For PST k located at line ij (which leaves from bus i and enters bus j), we have the following non-zero elements,

$$\mathbf{B}_{\vartheta(i,k)} = \frac{1}{x_{ij}}, \quad \mathbf{B}_{\vartheta(j,k)} = -\frac{1}{x_{ij}}. \quad (4.7)$$

To balance fluctuations ω , we assume that the generators adjust their in-feeds according to the affine policy,

$$\tilde{p}_G(\omega) = p_G - \alpha_G \omega. \quad (4.8)$$

The participation vector $\alpha_G \in \mathbb{R}^m$ is an optimization variable subject to the usual constraints:

$$\sum_{i \in \mathcal{G}} \alpha_{G,i} = 1, \quad \alpha_G \geq 0. \quad (4.9)$$

Note that we do not include corrective control by generators for neither contingencies nor uncertainties. While the formulation itself can easily be extended by variables denoting post-contingency redispatch d_G^{ij} or a more general matrix α_G , as in Chapter 3, we choose to focus on corrective control with inexpensive control actions from HVDC and PSTs.

4.3.3 Power Flows

We compute the pre-contingency power flows as the sum of the base case flow p_{ij} (with $\omega = 0$ and no contingencies) and changes due to fluctuations δp_{ij}^ω :

$$p_{ij}^\omega = p_{ij} + \delta p_{ij}^\omega, \quad (4.10)$$

The base case power flow p_{ij} is given by

$$p_{ij} = \frac{1}{x_{ij}}(\theta_i - \theta_j) + \mathbf{b}_{\vartheta(ij, \cdot)} \vartheta, \quad (4.11)$$

where x_{ij} is the reactance of lines ij . The matrix $\mathbf{b}_\vartheta \in \mathbb{R}^{l \times s}$ maps the PST angles to the lines where the PSTs are located, with entries $\mathbf{b}_{\vartheta(ij,s)} = \frac{1}{x_{ij}}$ if the s^{th} PST is located at line ij and $\mathbf{b}_{\vartheta(ij,s)} = 0$ otherwise.

The power flow change δp_{ij}^ω is a result of both the fluctuations ω , the corrective control from HVDC and PSTs and the balancing by the generators. It is given by

$$\begin{aligned}\delta p_{ij}^\omega &= \mathbf{M}_{(ij, \cdot)} (-\alpha_G \mathbf{1}_{1,m} + \mathbf{I} + \mathbf{C}_{DC} \alpha_{DC} - \mathbf{B}_\vartheta \alpha_\vartheta) \omega + \mathbf{b}_{\vartheta(ij, \cdot)} \alpha_\vartheta \omega \\ &= \mathbf{A}_{(ij, \cdot)} \omega,\end{aligned}\tag{4.12}$$

The term in brackets is the effective change in the power injections due to fluctuations ω , and the last term is the direct influence of the changes in the PST set-points on the lines where they are located. Note that the matrix $\mathbf{A} = \mathbf{A}(\alpha_G, \alpha_{DC}, \alpha_\vartheta)$ is a linear function of the policies for balancing and corrective control. By allowing corrective control of HVDC and PSTs in reaction to uncertainties, it is possible to influence the change in the power flow and reduce line congestion due to deviations.

With line outages, the power flow changes both due to corrective control actions from HVDC and PSTs, and due to the change in system topology. The change in the power flow due to the corrective control actions alone (not yet considering the outage and including the change that would have occurred on the outaged line itself) can be modelled as

$$p_{ij}^{\delta_{kl}} = p_{ij} + \mathbf{M}_{(ij, \cdot)} \mathbf{C}_{DC} \delta_{DC}^{kl} + (\mathbf{b}_{\vartheta(ij, \cdot)} - \mathbf{M}_{(ij, \cdot)} \mathbf{B}_\vartheta) \delta_\vartheta^{kl}.$$

The effect of change in system topology can be accounted for using line outage distribution factors (LODFs) [86], with LF_{ij}^{kl} denoting the fraction of the power flow on the line kl that is shifted to line ij when line kl is outaged. The flow on a line ij with outage kl and fluctuation ω is given by

$$\begin{aligned}p_{ij}^{kl, \omega} &= p_{ij}^{\delta_{kl}} + \delta p_{ij}^\omega + LF_{ij}^{kl} (p_{kl}^{\delta_{kl}} + \delta p_{kl}^\omega) \\ &= p_{ij}^{\delta_{kl}} + \mathbf{A}_{(ij, \cdot)} \omega + LF_{ij}^{kl} (p_{kl}^{\delta_{kl}} + \mathbf{A}_{(kl, \cdot)} \omega) \\ &= p_{ij}^{\delta_{kl}} + LF_{ij}^{kl} p_{kl}^{\delta_{kl}} + (\mathbf{A}_{(ij, \cdot)} + LF_{ij}^{kl} \mathbf{A}_{(kl, \cdot)}) \omega.\end{aligned}$$

4.4 Chance-Constrained Optimal Power Flow with Corrective Control

In this section, we formulate the CC-SCOPF with corrective control, before showing the reformulated constraints.

4.4.1 Objective and Constraints

Objective

The objective of the CC-OPF is to minimize the total cost of energy and reserves as in Chapter 3.

$$\min_{\substack{p_G, \alpha_G, r^+, r^-, \\ p_{DC}, \alpha_{DC}, \delta_{DC}, \\ \vartheta, \alpha_\vartheta, \delta_\vartheta}} \sum_{i \in \mathcal{G}} (c_i p_{G,i} + c_i^+ r_i^+ + c_i^- r_i^-) \quad (4.13)$$

Power Balance and Generation Constraints

The power balance and generator constraints are given by

$$p_G - d + u + \mathbf{C}_{DC} p_{DC} = \mathbf{B}_{bus} \cdot \theta + \mathbf{B}_\vartheta \cdot \vartheta, \quad (4.14)$$

$$\sum_{i \in \mathcal{G}} \alpha_{G,i} = 1, \quad \alpha_G \geq 0, \quad (4.15)$$

$$p_G + r^+ \leq p_G^{max}, \quad p_G - r^- \geq p_G^{min}, \quad (4.16)$$

$$0 \leq r^+ \leq \bar{r}^+, \quad 0 \leq r^- \leq \bar{r}^-, \quad (4.17)$$

$$\mathbb{P} [-\alpha_{G,i} \Omega \leq r_i^+] \geq 1 - \epsilon_G, \quad \forall i \in \mathcal{G} \quad (4.18)$$

$$\mathbb{P} [-\alpha_{G,i} \Omega \geq r_i^-] \geq 1 - \epsilon_G, \quad \forall i \in \mathcal{G} \quad (4.19)$$

$$\sum_{i \in \mathcal{G}} r_i^+ \geq R^+, \quad \sum_{i \in \mathcal{G}} r_i^- \geq R^-, \quad (4.20)$$

Here, (4.14) defines the power balance in normal operation, while (4.15) ensures active power balance during wind power fluctuations and non-negativity of α_G . The generator constraints (4.16)-(4.19) have a similar interpretation as Eqs. (3.11) - (3.15), and ensure that nominal generation and scheduled reserves remain within the technical limits of the generators. Eq. (4.20) ensures that the total amount of reserves fulfill a predetermined reserve requirement R^+, R^- , which ensures sufficient reserves in case of generator outages. Eqs. (4.18), (4.19) enforce that the reserve activation requested from each generator can be covered by the corresponding reserves r^+, r^- . Since the reserve activation depends on the fluctuations Ω , these constraints are formulated as chance constraints, which require the probability of constraint violation to remain below an acceptable level ϵ_G .

Note that all reserve chance constraints depend only on the total mismatch Ω , which is a scalar random variable. Assuming that we enforce all generator constraints with the same acceptable violation probability ϵ_G , the reserve constraints (4.18), (4.19) are jointly enforced, i.e., the probability that *none* of the generation constraints will be violated is $1 - \epsilon_G$. To see this, we observe that the reserves of all generators will be fully utilized when the $1 - \epsilon_G$ quantile $\Omega_{1-\epsilon_G}$ of Ω is reached. For $\Omega < \hat{\Omega}_{1-\epsilon_G}$, *none* of the constraints (4.19) are violated. For $\Omega > \hat{\Omega}_{1-\epsilon_G}$, *all* constraints (4.19) are violated. To see this, consider constraint (4.18),

$$\mathbb{P} [-\alpha_{G,i}\Omega \leq r_i^+] = \mathbb{P} \left[-\Omega \leq \frac{r_i^+}{\alpha_{G,i}} \right] \geq 1 - \epsilon_G.$$

Since there is a non-zero cost c_i^+ associated with the reserve capacity r_i^+ in (4.13), at optimality we will have

$$\frac{r_i^+}{\alpha_{G,i}} = \Omega_{1-\epsilon_G} \quad \forall i \in \mathcal{G}.$$

Hence the *joint* probability of reserve insufficiency can be simplified as

$$\mathbb{P} [-\alpha_{G,i}\Omega \leq r_i^+, \quad \forall i \in \mathcal{G}] = \mathbb{P} [-\Omega \leq \Omega_{1-\epsilon_G}] = 1 - \epsilon_G.$$

The violation probability ϵ_G thus has a direct interpretation as the probability of having insufficient reserves, which is a commonly used risk measure in real power systems. In the Swiss power system, for example, the acceptable probability of reserve deficiency is 0.2% [56], which corresponds to $\epsilon_G = 0.001$ for up and down reserves.

Constraints for HVDC and PSTs

The constraints enforcing the upper and lower bounds on HVDC and PSTs, after activation of corrective control, are given by (4.21) - (4.24), while (4.25) allows the operator to limit the amount of post-contingency

control:

$$\mathbb{P} [p_{DC,i} + \delta_{DC,i}^{kl} - \alpha_{DC(i,\cdot)}\omega \leq p_{DC,i}^{max}] \geq 1 - \epsilon_{DC}, \quad (4.21)$$

$$\mathbb{P} [p_{DC,i} + \delta_{DC,i}^{kl} - \alpha_{DC(i,\cdot)}\omega \geq p_{DC,i}^{min}] \geq 1 - \epsilon_{DC}, \quad (4.22)$$

$$\mathbb{P} [\vartheta_j + \delta_{\vartheta,j}^{kl} - \alpha_{\vartheta(j,\cdot)}\omega \leq \vartheta_j^{max}] \geq 1 - \epsilon_{\vartheta}, \quad (4.23)$$

$$\mathbb{P} [\vartheta_j + \delta_{\vartheta,j}^{kl} - \alpha_{\vartheta(j,\cdot)}\omega \geq \vartheta_j^{min}] \geq 1 - \epsilon_{\vartheta}, \quad (4.24)$$

$$-\bar{\delta}_{DC,i}^{kl} \leq \delta_{DC,i}^{kl} \leq -\bar{\delta}_{DC,i}^{kl}, \quad -\bar{\delta}_{\vartheta,i}^{kl} \leq \delta_{\vartheta,i}^{kl} \leq -\bar{\delta}_{\vartheta,i}^{kl} \quad (4.25)$$

$$\forall i \in \mathcal{H}, j \in \mathcal{S}, kl \in \{0, \mathcal{L}\}.$$

Note that we do not explicitly limit the total power flowing through the PSTs. We assume that the power limit of the PST is accounted for when defining the transmission capacity of the line where it is located. The constraints (4.21) - (4.24) depend on the uncertainty ω , and are thus formulated as chance constraints with acceptable violation probability ϵ_{DC} , ϵ_{ϑ} . Since ϵ_{DC} , $\epsilon_{\vartheta} > 0$, there will be cases in which HVDC and PSTs reach their limit and are not able to continue to provide corrective control according to (4.4), (4.5). This saturation can be expressed as

$$\tilde{p}_{DC,i} = \min (p_{DC,i} + \delta_{DC,i}^{kl} - \alpha_{DC(i,\cdot)}\omega, \quad p_{DC,i}^{max}), \quad (4.26)$$

$$\tilde{\vartheta}_j = \min (\vartheta_j + \delta_{\vartheta,j}^{kl} - \alpha_{\vartheta(j,\cdot)}\omega, \quad \vartheta_j^{min}). \quad (4.27)$$

When such saturation occurs, we get different power flows than expected and possibly additional overloading on the transmission lines.

Power Flow Constraints

The power flow constraints can be expressed as

$$\begin{aligned} \mathbb{P} [p_{ij}^{\delta_{kl}} + LF_{ij}^{kl} p_{kl}^{\delta_{kl}} + (\mathbf{A}_{(ij,\cdot)} + LF_{ij}^{kl} \mathbf{A}_{(kl,\cdot)}) \omega \leq p_{ij}^{max}] &\geq 1 - \epsilon_L, \\ \mathbb{P} [p_{ij}^{\delta_{kl}} + LF_{ij}^{kl} p_{kl}^{\delta_{kl}} + (\mathbf{A}_{(ij,\cdot)} + LF_{ij}^{kl} \mathbf{A}_{(kl,\cdot)}) \omega \geq -p_{ij}^{max}] &\geq 1 - \epsilon_L, \end{aligned} \quad \forall ij \in \mathcal{L}, kl \in \{0, \mathcal{L}\} \quad (4.28)$$

and are enforced as chance constraints with acceptable violation probability ϵ_L . Index $kl = 0$ refers to the pre-contingency constraint with $LF_{ij}^{kl} = 0$.

Note that the problem does not include any consideration of the intermediate post-contingency, pre-corrective control system state, as described in e.g. [87]. Constraints to ensure feasibility of this state can however be added without any conceptual change to the method.

4.4.2 Reformulation of Chance Constraints

The chance constraints in (4.18)-(4.28) are reformulated into tractable SOC constraints using the analytical approach from Section 2.5 and the assumption of a normal distribution:

$$\alpha_{G,i} \Phi^{-1}(1 - \epsilon_G) \sigma_\Omega \leq r_i^+, \quad (4.29)$$

$$\alpha_{G,i} \Phi^{-1}(1 - \epsilon_G) \sigma_\Omega \geq -r_i^-, \quad (4.30)$$

$$p_{DC,i} + \delta_{DC,i}^{kl} + \Phi^{-1}(1 - \epsilon_{DC}) \|\Sigma_W^{1/2} \alpha_{DC(i,\cdot)}^T\|_2 \leq p_{DC,i}^{max} \quad (4.31)$$

$$p_{DC,i} + \delta_{DC,i}^{kl} - \Phi^{-1}(1 - \epsilon_{DC}) \|\Sigma_W^{1/2} \alpha_{DC(i,\cdot)}^T\|_2 \geq p_{DC,i}^{min} \quad (4.32)$$

$$\vartheta_j + \delta_{\vartheta,j}^{kl} + \Phi^{-1}(1 - \epsilon_\vartheta) \|\Sigma_W^{1/2} \alpha_{\vartheta(j,\cdot)}^T\|_2 \leq \vartheta_j^{max} \quad (4.33)$$

$$\vartheta_j + \delta_{\vartheta,j}^{kl} - \Phi^{-1}(1 - \epsilon_\vartheta) \|\Sigma_W^{1/2} \alpha_{\vartheta(j,\cdot)}^T\|_2 \geq \vartheta_j^{min} \quad (4.34)$$

$$p_{ij}^{\delta_{kl}} + LF_{ij}^{kl} p_{kl}^{\delta_{kl}} + \quad (4.35)$$

$$\Phi^{-1}(1 - \epsilon_L) \|\Sigma_W^{1/2} (\mathbf{A}_{(ij,\cdot)} + LF_{ij}^{kl} \mathbf{A}_{(kl,\cdot)})^T\|_2 \leq p_{ij}^{max} \\ p_{ij}^{\delta_{kl}} + LF_{ij}^{kl} p_{kl}^{\delta_{kl}} - \quad (4.36)$$

$$\Phi^{-1}(1 - \epsilon_L) \|\Sigma_W^{1/2} (\mathbf{A}_{(ij,\cdot)} + LF_{ij}^{kl} \mathbf{A}_{(kl,\cdot)})^T\|_2 \geq -p_{ij}^{max}$$

As observed in Section 3.3.2, the generator constraints are linear since they only depend on the total power mismatch Ω with standard deviation σ_Ω [80]. The remaining constraints (4.31) - (4.36) are Second Order Cone (SOC) constraints, which are convex for $\epsilon \leq 0.5$ [88, 18].

4.4.3 Exploiting Symmetry of SOCs

Each pair of upper and lower SOC constraints in (4.31)-(4.36) can be reduced to a pair of linear constraints and a single SOC constraint by exploiting the symmetry of the normal distribution [18]:

$$p_{DC} + \delta_{DC}^{kl} + \lambda_{DC} \leq p_{DC}^{max} \quad (4.37)$$

$$p_{DC} + \delta_{DC}^{kl} - \lambda_{DC} \geq p_{DC}^{min} \quad (4.38)$$

$$\lambda_{DC,i} \geq \Phi^{-1}(1 - \epsilon_{DC}) \|\Sigma_W^{1/2} \alpha_{DC(i,\cdot)}^T\|_2, \quad \forall i \in \mathcal{H} \quad (4.39)$$

$$\vartheta + \delta_\vartheta^{kl} + \lambda_\vartheta \leq \vartheta^{max} \quad (4.40)$$

$$\vartheta + \delta_\vartheta^{kl} - \lambda_\vartheta \geq \vartheta^{min} \quad (4.41)$$

$$\lambda_{\vartheta,j} \geq \Phi^{-1}(1 - \epsilon_\vartheta) \|\Sigma_W^{1/2} \alpha_{\vartheta(j,\cdot)}^T\|_2, \quad \forall j \in \mathcal{S} \quad (4.42)$$

$$p_{ij}^{\delta_{kl}} + LF_{ij}^{kl} p_{kl}^{\delta_{kl}} + \lambda_{ij}^{kl} \leq p_{ij}^{max}, \quad \forall_{ij \in \mathcal{L}, kl \in \{0, \mathcal{L}\}} \quad (4.43)$$

$$p_{ij}^{\delta_{kl}} + LF_{ij}^{kl} p_{kl}^{\delta_{kl}} - \lambda_{ij}^{kl} \geq -p_{ij}^{max}, \quad \forall_{ij \in \mathcal{L}, kl \in \{0, \mathcal{L}\}} \quad (4.44)$$

$$\lambda_{ij}^{kl} \geq \Phi^{-1}(1 - \epsilon_L) \parallel \Sigma_W^{1/2} (\mathbf{A}_{(ij, \cdot)} + LF_{ij}^{kl} \mathbf{A}_{(kl, \cdot)})^T \parallel_2 \quad (4.45)$$

The above reformulation cuts the number of SOC constraints in half and thus improves efficiency. Note that the SOC terms (4.39), (4.42), (4.45) are always positive, and introduce a tightening of the original, deterministic constraints. They correspond to the extended version of the uncertainty margins in (2.44).

4.5 Solution Algorithm

The full OPF problem with security and chance constraints in Section 4.4 is a Second Order Cone Program (SOCP), with the SOC constraints given by (4.39), (4.42) and (4.45). The problem has a large number of linear and SOC constraints, due to consideration of both the pre- and post-contingency situations. Although the SOC constraints are convex, it has been observed in the literature [18] that attempting to solve the entire optimization problem at once by using a non-linear SOCP solver can result in unacceptably long convergence times and numerical difficulties. The state of the art method in the literature for solving such CC-OPF problems is the sequential outer approximation cutting planes algorithm [18]. In this algorithm, a relaxed version of the problem is solved by eliminating all the SOC constraints. Then the relaxation is successively tightened using a sequence of polyhedral outer approximations obtained by adding separating hyperplanes (cutting-planes) to the violated SOC constraints until the problem is solved to optimality with desired accuracy. The success of the algorithm relies on exploiting three salient properties prevalent in the CC-OPF [18]: (i) only a small fraction of the non-linear chance constraints (SOCs) are active at optimality, and these correspond to critical/congested lines, (ii) each SOC term depends only on a limited number of decision variables, and (iii) linear programming solvers have better speed and stability compared to non-linear solvers.

However, there are some critical differences in the features of the CC-SCOPF considered in this paper that makes the cutting-plane algorithm unsuitable:

Feature 1: Due to the SOC constraints for the HVDC and PSTs, that are often tight at optimality, as well as the security constraints for the lines, more constraints must be approximated through polyhedral constraints.

Feature 2: Each SOC term, in particular the SOC terms for the lines, depend on a large number of decision variables since α_G , α_{DC} , α_θ are potentially large matrices. For those constraints, we found that the cutting-plane algorithm requires a large number of iterations to obtain a good polyhedral approximation and a feasible solution within acceptable tolerance. This both implies that the optimization problem needs to be solved a many times, and that a significant amount of memory is required to store the large and increasing number of linear constraints.

Feature 3: Due to security constraints, there is a very large number of SOC terms present in the problem. Evaluating the SOC constraints at any given candidate solution is therefore time consuming. Since the cutting-planes algorithm requires a large number of iterations with an SOC evaluation in each iteration, the problem solves very slowly.

To overcome these difficulties and obtain an efficient implementation, we develop a sequential SOCP algorithm. As in [18], we first solve a relaxed version of the problem involving only linear constraints. Instead of adding cutting-planes, we then add the full SOC terms for the most violated constraints. We continue solving a sequence of SOCPs until all constraints are satisfied, reaching the globally optimal solution of the original problem. In the following we describe the details of the algorithm and the reasoning behind.

Preprocessing: Solving the SCOPF Without Uncertainty

As a pre-processing step, we solve the SCOPF without consideration of uncertainty. This allows us to obtain a fast, first estimate of the active constraints.

Step 1: Solving the CC-OPF Without Security Constraints

- (a) We first solve a base case problem consisting of the power balance and generator constraints (4.14)-(4.17), (4.29), (4.30), the full PST and HVDC constraints (4.37) - (4.42) as well as the linear pre-contingency line constraints (4.43), (4.44). Since most of the SOC constraints for HVDC and PST are tight and there are few of them (i.e., no additional SOC terms for the security constraints), adding the full SOC upfront eliminates unnecessary iterations. In addition, the SOC terms belonging to the pre-contingency constraints violated in the SCOPF are added.

- (b) The line SOC constraints for the base case, i.e., (4.45) with $kl = 0$ are then checked for violations, and the most violated ones are added to the problem with a warm start from the previous iteration. This process is repeated until all the base case constraints are satisfied.

Step 2: Solving the Full CC-SCOPF With Security Constraints

We check for violation of the security constraints and add them sequentially using warm start. This part of the algorithm runs in three phases:

- (c) In the first iteration, we add all post-contingency constraints that were active in the SCOPF without uncertainty, as these most likely will be active in the CC-SCOPF as well. We include both the linear constraints and the corresponding SOC constraint to the problem.
- (d) After the first iteration, we check for violation of only the linear security constraints (4.43), (4.44), which is much faster than evaluating the full SOC constraints. However, since the SOC constraint implies a tightening of the linear constraint, a violation of the linear constraint always means that the corresponding SOC is violated as well. For the most violated constraints, we therefore add both the linear constraints and the corresponding SOC constraint to the problem. This process is repeated until all the linear security constraints are satisfied.
- (e) We then check for the violated SOC terms for the line security constraints (4.45), and add the most violated ones to the problem. As mentioned in *Feature 3*, the number of post-contingency SOC terms are rather large, and hence inefficient to evaluate. However, we observe that this stage of the algorithm only requires few iterations until all constraints are satisfied, since most violated security constraints were added in (d).

To further reduce the computational time involved in each iteration (d), we do a pre-screening of the SOC terms based on the LODF matrix. Even though most LF_{ij}^{kl} are non-zero, most are very small $< 1e - 3$. By evaluating the post-contingency SOC constraints only for those pairs ij and kl where LF_{ij}^{kl} exceeds a certain threshold, the number of evaluations can be significantly reduced while maintaining acceptable accuracy.

Finally, we check whether the current solution violates any of the pre-contingency constraints. If yes, we restart from (b). Otherwise, the al-

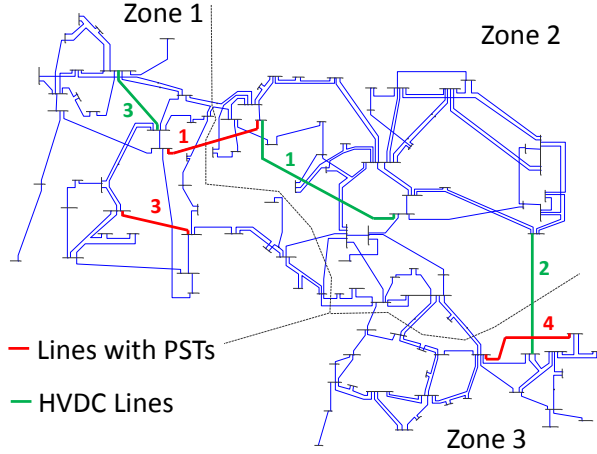


Figure 4.1: IEEE 118 bus system with 3 Zones. The lines with PSTs are marked in red, while HVDC connections are drawn in green.

gorithm is terminated and a globally optimal solution which satisfies all the constraints of the full problem has been found. In all encountered cases, only one pass of the algorithm was required to find an optimal solution without requiring to return to (b).

4.6 Case study - Impact of Corrective Control

In this section, we analyze the benefits of corrective control based on a case study for the IEEE 118 bus system. The scalability of the proposed method and the sequential SOCP algorithm is demonstrated in the next section.

4.6.1 Test System

We use the IEEE 118 bus test system as defined in [73], with the following modifications. Both load and maximum generation capacity are scaled by a factor of 1.25, and the minimum generation capacity is set

Table 4.1: IEEE 118 Bus System - HVDC Connections

	HVDC 1	HVDC 2	HVDC 3
From - To Bus	38 - 65	104 - 62	12 - 17
Replaced Line	96	-	20
Capacity [MW]	500	200	175

to zero. The system loads are interpreted as a mix between load and renewable energy generation, and we assume that all 99 loads fluctuate around their forecasted consumption with standard deviation σ_W equal to 10% of the forecasted consumption. As shown in Fig. 4.1, the system is divided into three zones. We assume that fluctuations within a zone are correlated with correlation coefficient $\rho = 0.3$, that fluctuations in different zones are uncorrelated. For the chance constraints, we apply $\epsilon_L = \epsilon_{DC} = \epsilon_{\vartheta} = 0.01$ for transmission line, HVDC and PST constraints. For the generator constraints, we use $\epsilon_G = 0.001$, which corresponds to the acceptable probability of insufficient reserves in the Swiss power system.

We assume that there are 3 PSTs and 3 HVDC connections installed in the system, as shown in Fig. 4.1. Details for each HVDC connection are listed in Table 4.1. The PSTs are installed on lines 41, 167 and 54, with maximum tap positions $\pm 30^\circ$. The upper bounds on the post-contingency control for HVDC and PSTs are set to $\bar{\delta}_{DC}^{ij} = 0.25 p_{DC}^{max}$ and $\bar{\delta}_{\vartheta}^{ij} = 0.25 \vartheta^{max}$.

The total required down-reserves is set to $R^- = \Phi^{-1}(1 - \epsilon_G)\sigma_\Omega$, which ensures that the total fluctuation Ω will be covered with probability $1 - \epsilon_G$. The total amount of up-reserves R^+ is required to cover the total fluctuation with the same probability or the maximum generation outage, whichever is larger, such that $R^+ = \max\{\bar{p}_g, R^-\}$. This reserve dimensioning ensures that a similar amount of reserves are procured for the chance-constrained and deterministic formulations we will compare. The reserve provision from individual generators is restricted to 20% of total generation capacity, i.e. $\bar{r}^+ = \bar{r}^- = 0.2 p_G^{max}$ in 4.17.

To analyze the performance of the method, we run two Monte Carlo simulations with 2000 samples of ω each:

First, we use uncertainty data which correspond to our assumptions, and draw samples from a multivariate normal distribution with mean

$u = 0$ and covariance matrix Σ_W .

Second, we run out-of-sample tests based on 1 year of historical data from the Austrian Power Grid (APG). The deviations are defined based on the difference between the the DACF and the snapshot for all hours and buses with available data (8492 data points for 28 buses). Splitting the data into four three-month periods, we obtain 2000 data samples for each of the 99 buses. The samples are then scaled to fit the assumed covariance matrix Σ_W and mean $u = 0$. This corresponds to the case where we have a good estimate of the mean and covariance, but do not know the full underlying distribution.

We implement the sequential SOCP algorithm in Julia with JuMP [89], and solve the problem using MOSEK [90]. With this set-up, the solution is obtained within a few minutes on a laptop.

4.6.2 Numerical Results

To demonstrate the benefits of corrective control, we compare five different solutions:

- (a) The *OPF* does not consider uncertainty (i.e., $\Sigma_W = 0$) or constraints related to outages.
- (b) The *SCOPF* considers security constraints, but does not account for uncertainty (i.e., $\Sigma_W = 0$) or post-contingency corrective control ($\delta_{DC}^{ij} = \delta_{\vartheta}^{ij} = 0$).
- (c) The *post-contingency corrective SCOPF* considers security constraints and post-contingency control, but does not account for uncertainty (i.e., $\Sigma_W = 0$).
- (d) The *post-contingency corrective CC-SCOPF* considers uncertainty, security constraints and corrective control for contingencies, but does not include corrective control in reaction to fluctuations (i.e., $\alpha_{DC} = \alpha_{\vartheta} = 0$).
- (e) The *full CC-SCOPF* considers the full problem, including uncertainty, security constraints and corrective control for both fluctuations and contingencies.

We first compare the cost of the generation dispatch and reserves (as obtained directly from the optimization), and analyze the differences.

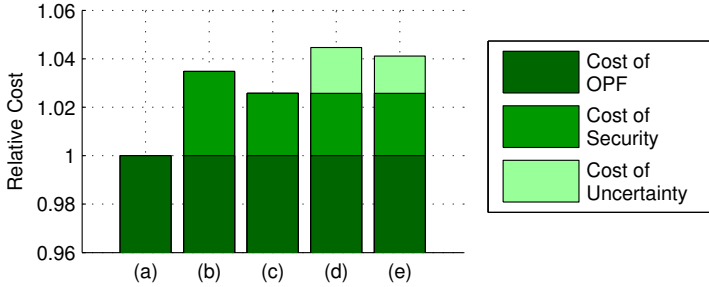


Figure 4.2: Breakdown of the cost for the five OPF formulations: The OPF (a), the SCOPF (b), the post-contingency corrective SCOPF (c), the post-contingency corrective CC-SCOPF (d), and the full CC-SCOPF (e). The costs are normalized by the cost of the OPF (a).

Second, we look into the empirical number of violations observed in the two Monte Carlo simulations for the normally distributed and out-of-sample data. Finally, we investigate the impact of corrective control by comparing the solution of (d) and (e) for different acceptable violation probabilities ϵ and varying uncertainty levels σ_W .

Comparison of Operation Cost

The costs of the problems (a)-(e) are listed in Table 4.2 and are shown graphically in Fig. 4.2, where the costs are normalized by the cost of the standard OPF. The cost increase is analyzed with respect to the criteria defined in Section 2.7.1, *CoS* and *CoU*, repeated here for convenience:

Cost of Security (CoU) is defined as the cost increase between the OPF and the SCOPF, due to the cost of enforcing N-1 constraints.

Cost of Uncertainty (CoS) is defined as the cost increase between the SCOPF and the CC-SCOPF, due to the cost of enforcing chance constraints instead of deterministic constraints.

We observe that the *CoS* can be reduced by 0.9% from (b) to (c), by introducing post-contingency corrective control. Similarly, the *CoU* is reduced by 0.35% from (d) to (e), by introducing uncertainty corrective control.

The reduction in *CoU* can be explained by the better ability to react to the fluctuations in the power injections. In Fig. 4.3, the power flow on

Table 4.2: Cost of the OPF Solutions (a) - (e)

	(a)	(b)	(c)	(d)	(e)
Cost [\$]	84924	87880	87108	88713	88418

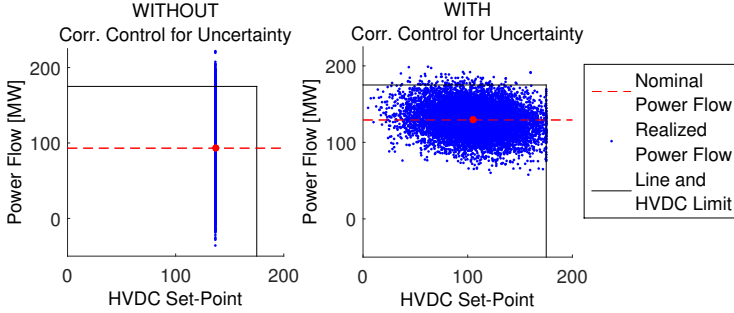


Figure 4.3: Power Flow on Line 119 after outage of Line 126, plotted against the set-point of HVDC 3 for the CC-SCOPF without (left) and with (right) corrective control for uncertainty. The red marker shows the forecasted operating point, the blue points are the realized power flows obtained through the Monte Carlo simulation and the black lines show the line and HVDC limits.

line 119 after the outage of line 126 is plotted against the set-point of HVDC 3 for the CC-SCOPF problems without (left) and with (right) corrective control. The red marker shows the forecasted operating point, and the blue points correspond to the actual, realized conditions as simulated for 2000 samples of ω . Without corrective control (left), the HVDC set-point remains constant for all uncertainty samples. With corrective control (right), the set-point is changing depending on the sample. The post-contingency constraint on line 119 is one of the active constraints in the CC-SCOPF problems (d) and (e), and it is clearly seen how the line flow limit is violated in some cases. However, we also observe that by introducing the corrective control of HVDC and PSTs, it is possible to reduce the standard deviation of the line flow, in this case from 35.2 MW to 19.7 MW. With reduced variance, we need a lower uncertainty margin. Therefore, a higher nominal power flow can be accepted, which leads to a reduction in cost.

The response of the HVDC and PSTs to the uncertainty realizations ω is determined by the corrective control parameters α_{DC} , α_{β} . The

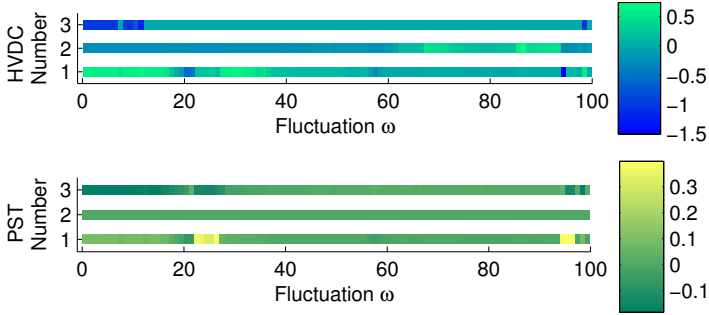


Figure 4.4: Values of the corrective control parameters α_{DC} (top) and α_{θ} (bottom).

values of those variables as obtained from (e) are shown in Fig. 4.4. We observe that the PST and HVDC react strongly to fluctuations that are geographically close to where they are located, and have close to zero entries for most other fluctuations.

Number of Violations and Out-of-Sample Testing

We run two Monte Carlo simulations based on the normally distributed data and the APG data, respectively. For each data set, we compute the number of constraint violations.

We first assess how well the solution of the full CC-SCOPF (e) satisfies the chance constraints by comparing the empirical violation probability ϵ_{emp} for each separate constraint with the pre-defined acceptable violation probability ϵ_L . Fig. 4.5 shows the empirical violation probability per constraint for the CC-SCOPF (e), with normally distributed (green bar) and APG (yellow bar) samples. Only the 19 constraints with empirical violation probabilities $\epsilon_{emp} > 0$ are shown. We observe that there are four tight constraints (3, 7, 8 and 10) for which $\epsilon_{emp} \approx 0.01$ with normally distributed samples. For these constraints, the APG samples lead to $\epsilon_{emp} > 0.01$, indicating that the chance constraint is violated. However, the empirical violation probability remains below 2% for all constraints, and below the acceptable 1% value for the majority of constraints. While the empirical violation probability will vary depending on the uncertainty data, this result indicates that a chance constraint

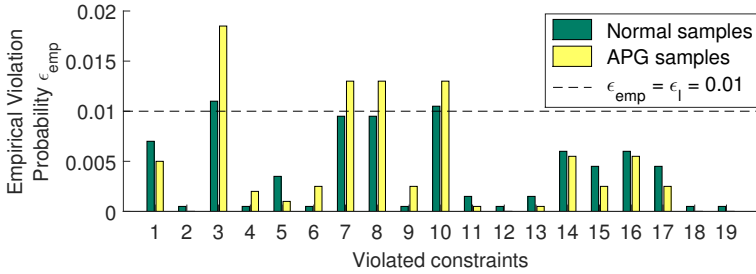


Figure 4.5: Empirical violation probability ϵ_{emp} per constraint with normal samples (green) and APG samples (yellow) for the 19 violated constraints, based on the solution for the full CC-SCOPF (e). The black dotted line corresponds to $\epsilon_{emp} = \epsilon_L = 0.01$.

Table 4.3: Number of Violated Samples for the OPF Solutions (a) - (e)

	(a)	(b)	(c)	(d)	(e)
Normal	2000	1729	1712	98	87
Samples	(100%)	(86.5%)	(85.6%)	(4.9%)	(4.4%)
APG	2000	1697	1655	114	95
Samples	(100%)	(84.9%)	(82.8%)	(5.7%)	(4.8%)

approach based on the normal distribution still significantly reduces the risk of violations.

In a second investigation, we calculate the joint violation probability, i.e., the probability that a sample will exhibit at least one constraint violation. Although each chance constraint in our problem is enforced individually with a violation probability of $1 - \epsilon$, the wind deviations have a different impact on each constraint, and the joint violation probability ϵ_J might be significantly higher than ϵ . It is therefore interesting to understand how many samples lead to at least one constraint violation.

In Table 4.3, the number of Monte Carlo samples which lead to either pre- or post-contingency constraint violations are listed for both the normally distributed and the APG samples. The OPF (a) violates all scenarios, as it neither accounts for outages nor uncertainty. The two SCOPF formulations without (b) and with (c) corrective control

have violations due to uncertainty in around 1700 of the samples. This corresponds to a joint violation probability of around 85%, which is unacceptably high and clearly shows the need to account for uncertainty. Both CC-SCOPF formulations (d) and (e) have violations in around 100 samples, indicating a joint violation probability of around 5%. As expected, this is above the violation probability $\epsilon_L = 0.01$ allowed for each separate constraint, but is much lower than the deterministic formulations. We hence conclude that the CC-SCOPF based on separate chance constraints also effectively reduces the joint violation probability.

The Monte Carlo simulation based on APG data increases number of violated scenarios by less than 1 % compared to the normally distributed samples for the CC-SCOPF cases (a) and (e), indicating that the method performs well also in out-of-sample tests. We further conclude that the CC-SCOPF with corrective control (e) is able to provide a similarly low violation probability as the CC-SCOPF without corrective control (d), but at lower cost.

Influence of Confidence Level

Table 4.4 shows the cost of the generation dispatch without (d) and with (e) corrective control for uncertainty, for different acceptable violation probabilities ϵ and different standard deviations σ_W (in % of forecasted load). We observe that the possibility to react to uncertainty becomes increasingly important (i.e., the reduction in cost is larger) if we want to enforce smaller violation probabilities or if the level of uncertainty increases. With standard deviations of 14.25%, we even observe that the CC-SCOPF with corrective control (e) is feasible, whereas the CC-SCOPF without (d) is not.

4.7 Case study - Scalability

In this case study, we discuss the scalability of the CC-SCOPF and proposed solution algorithm based on the run times for three different test cases, the IEEE 118 bus system, the IEEE 300 bus system and the Polish test case with 2383 buses.

Table 4.4: Cost of CC-SCOPF without (d) and with (e) corrective control for uncertainty, for different values of ϵ with $\sigma = 10\%$ of forecasted load (top) and different standard deviations σ with $\epsilon = 0.05$ (bottom).

	$\epsilon=0.1$	$\epsilon=0.05$	$\epsilon=0.02$	$\epsilon=0.01$	$\epsilon=0.001$
(d)	87898	88136	88466	88713	89483
(e)	87720	87941	88201	88418	89070

	$\sigma_W=5\%$	$\sigma_W=7.5\%$	$\sigma_W=10\%$	$\sigma_W=12.5\%$	$\sigma_W=14.25\%$
(d)	84171	86334	88136	91513	Infeasible
(e)	83991	86119	87941	91022	93063

4.7.1 Test Systems

IEEE 118 bus test system

For the IEEE 118 bus system, we use the same data as described in Section 4.6.1 above.

IEEE 300 bus test system

We use a modified version of the IEEE 300 bus test system found in the NICTA Energy System Test Archive [91]. To obtain a feasible SCOPF case, we increase the line limits by a factor of 5.0 and the generation limits by a factor of 2.0. To obtain a reasonable set of uncertainty sources and an interesting case, we assume that all loads above 50 MW, corresponding to 109 loads and 68% of the system load, are uncertain. Their standard deviations are set to 5% of the forecasted consumption, and the correlation to zero. We assume that there are 3 HVDC connections installed in the system, with corresponding system data listed in Table 4.5. The PSTs are installed on lines 91, 140 and 174, with a control range of $\pm 30^\circ$.

Polish Test System

We base our case study on the Polish Winter Peak test case with 2383 buses as provided with Matpower 5.1 [73]. To obtain a feasible case

Table 4.5: IEEE 300 Bus System - HVDC Connections

	HVDC 1	HVDC 2	HVDC 3
From - To Bus	68 - 198	8 - 18	126 - 145
Replaced Line	-	-	-
Capacity [MW]	900	600	600

Table 4.6: Polish Test Case - HVDC Connections

	HVDC 1	HVDC 2	HVDC 3
From - To Bus	32 - 18	184 - 105	67 - 138
Replaced Line	-	-	169
Capacity [MW]	500	500	1000

with active transmission constraints, the maximum generation capacity is scaled by a factor of 2.0 and the transmission line limits by a factor of 2.5. All loads above 25 MW, corresponding to 157 loads and 29% of the system load, are assumed to be uncertain. Their standard deviations are set to 20% of the forecasted consumption, and the correlation between loads to zero. We assume that there are 3 HVDC connections installed in the system, with corresponding system data listed in Table 4.6. The PSTs are installed on lines 130, 240 and 1381, with a control range of $\pm 30^\circ$.

For both of the above test systems, we choose similar settings as for the IEEE 118 bus test case described in Section 4.6.1. We apply $\epsilon_L = \epsilon_{DC} = \epsilon_\vartheta = 0.01$ for transmission line, HVDC and PST constraints and use $\epsilon_G = 0.001$ for generation constraints. We determine the total required reserve capacities R^- , R^+ and the reserve provision bounds $\bar{r}^+$, $\bar{r}^-$ as described above. We do not include post-contingency corrective control in either of the test cases, but solve the CC-SCOPF with corrective control for uncertainties. The sequential SOCP algorithm is implemented in Julia with JuMP [89], and solved using Ipopt [92].

Table 4.7: Size of the CC-SCOPF for the different test cases

	IEEE 118	IEEE 300	Polish
Variables	54 000	173 000	>8 mill.
Linear Constraints	70 000	343 000	>16 mill.
SOC Constraints	34 000	169 000	>8 mill.

4.7.2 Results

The size of each test case in terms of number of constraints and variables is listed in Table 4.7. Due to the large number of security constraints, already the IEEE 118 bus test case has more SOC constraints than the Polish test case solved in [18]. The problem sizes for the IEEE 300 bus and Polish grid test cases are one and two orders of magnitude larger than the IEEE 118 bus system, as measured in number of constraints, with the Polish test case featuring more than 8 million variables and 8 million SOC constraints.

Table 4.8 shows the run times of the algorithm based on the current implementation. We have included both the total solution time, as well as an overview of the number of SOC evaluations and the total time spent evaluating the SOC constraints. We observe that a significant part of the solution time is spent checking the SOC constraints. The pre-screening of the security constraints based on the LODF matrix reduces the number of security constraints that we need to check by a factor of 3. However, for the largest Polish test case, we still need to evaluate close to 1.4 million SOC constraints.

By using the sequential SOCP algorithm, we are able to keep the number of SOC evaluations very low, with only 2-3 evaluations. While the cutting plane algorithm [18] would solve the problem faster and more reliably in each iteration, it would require a much larger number of evaluations, leading to prohibitively large run times. However, the time required for the SOC check can be significantly reduced by more efficient coding and parallelization, and is left as a topic for future work. This would improve the solution times for both our solution algorithm and the cutting plane algorithm in [18].

Table 4.8: Run times for CC-SCOPF for the different test cases

	IEEE 118	IEEE 300	Polish
Total Run Time	2min 15s	4 min 44s	4h 20min
Number of SOC Evaluations	2	3	2
Time per SOC Evaluation	8s	27 s	1h 36min

4.8 Summary and Conclusions

In this chapter, we have proposed a framework for corrective control which involves corrective actions in response to both contingencies and different uncertainty realizations. The combined corrective control was incorporated in the CC-SCOPF and reformulated using the analytical approach, which leads to an optimization problem with SOC constraints. To be able to handle the problem computationally, we developed a sequential SOCP algorithm.

With this solution algorithm, we are able to solve the IEEE 118 bus case including contingency constraints and 99 uncertain loads within a few minutes on a laptop. The solutions obtained for the IEEE 118 bus system show that the use of corrective control actions in reaction to uncertainty reduces operational cost, without reducing the level of security in the system. The cost reduction is more significant for cases where we want to achieve a low violation probability, or where the level of uncertainty is higher.

We also demonstrate the scalability of the method by presenting results for the IEEE 300 bus test system and the Polish grid with 2383 buses. One main bottleneck of the current implementation is the evaluation of the SOC terms, but a better implementation and parallelization should allow for significant speed up of this part.

Chapter 5

AC Optimal Power Flow with Approximate Chance Constraints

In this chapter, we extend the CC-OPF to a more accurate model involving the full AC power flow equations. Due to the non-linearities and non-convexities of the AC power flow equations, the AC OPF is a hard problem to solve, and trying to incorporate uncertainty makes it even harder. Particularly, modelling how uncertainties propagate throughout the system becomes more challenging when the relations are non-linear. In the following, we propose a method based on partial linearization of the AC power flow equations, which allows for a reasonably accurate, yet computationally tractable representation of the system with uncertainty. First, we use the full AC power flow equations to obtain an accurate solution for the forecasted power injections. Second, assuming that the fluctuations are small compared to the total injections, we linearize the system around the forecasted operating point and model the impact of uncertainty based on the linearized system model. we then formulate a Chance-Constrained AC Optimal Power Flow (AC CC-OPF) with chance constraints for active and reactive power generation, voltage and transmission limits. With the linearized uncertainty representation, we are able to apply the analytical chance constraint reformulation and obtain closed-form expressions for the chance constraints. We further discuss different possibilities for solving the problem, including

both a one-shot and an iterative solution approach. The iterative solution approach solves the chance-constrained problem as an outer iteration, which allows for a straightforward adaption of the method based on any existing AC OPF implementation. We demonstrate the probabilistic performance and scalability of the method in a case study, including simulations for four test systems ranging from the 24-bus IEEE RTS96 system to the Polish test case with 2383 buses. We further compare the proposed method based on linearization and analytical chance constraint reformulation with two alternative reformulations based on (i) a Monte Carlo simulation and (ii) the scenario approach for joint chance constraints. We find that the proposed method efficiently enforces the limit on constraint violations, while allowing for fast computation and a good cost-security trade-off.

5.1 Motivation and Related Work

In this chapter, we propose a stochastic AC OPF formulation with chance constraints to account for the impact of uncertainty. The AC power flow equations entail a more accurate model of the system, including losses and reactive power. This is of particular importance if we are interested in, e.g., distribution networks, where active and reactive power are strongly coupled, or system security, where AC feasibility is of key importance. The AC modelling allows us to include and probabilistically enforce a range of new constraints, such as voltage limits, reactive power constraints and power flow constraints based on apparent power or current magnitude.

While chance constraints offer a comprehensive way of handling forecast uncertainty, it is generally challenging to reformulate the resulting OPF into a tractable optimization problem. In previous chapters and in the literature in general, CC-OPF was typically based on DC power flow, which is easier to handle due to the linear relation between power flows and power injections. Different approximate algorithms for the AC CC-OPF have been suggested in, e.g, [93], where an iterative solution based on numerical integration and linearization of the AC power flow constraints is proposed, or [94], which employs an iterative approach based on the cumulant method and Cornish-Fischer expansion. Generally, many papers on AC CC-OPF utilize linearized versions of the AC power flow equations to obtain linear relationships between power injections, voltages and power flows. Examples include [95], which

solves a linearized AC CC-OPF to obtain optimal redispatch schedules, or [96], which uses linearized equations to solve a voltage-constrained AC CC-OPF for a distribution network with PV in-feeds. To the best of our knowledge, the most comprehensive AC CC-OPF formulation in terms of modelling the actual non-linear equations is provided in [97]. In [97], the chance-constrained problem is formulated using a convex relaxation of the full AC power flow equations and a sample-based approach to handle the chance constraint. While this method guarantees feasibility of the chance constraints, computational tractability for large systems and ensuring that the resulting solution is actually physically meaningful (i.e., that the relaxation is exact) are still challenges to be addressed. All of the above approaches have only been applied to small test cases, signalling a need for approaches that are tractable for utility scale systems [98].

Here, we propose a method which is more accurate than a method based only on linearized equations. The proposed formulation of the AC CC-OPF is based on a partial linearization of the AC power flow formulation, where the forecasted operating point is represented through the full non-linear equations, and the impact of uncertainty is modelled by a linearization around this point. By keeping the full AC equations in the formulation, we ensure AC feasibility for the forecasted point. This is an improvement on previous linearization approaches, e.g. [95, 96], where a change in scheduled generation and voltages away from the linearization point is only accounted for through the linearized equations, and is hence not guaranteed to be feasible. Further, assuming that the deviations are small compared to the overall power generation, the linearization around the expected point can be assumed to be a reasonably accurate representation. Similar ideas for linearization were applied to stochastic load flow in the 1970's, see [99] and discussion therein, and were also adopted in the risk-based OPF in [26].

With the suggested linearization technique, we keep the full non-linear dependence on generation and voltages at the expected operating point, but obtain a representation which is linear in the uncertain variables. The linear dependence allows us to apply the analytical chance constraint reformulation from Section 2.5, leading to closed-form expressions of the constraints. The accuracy of this chance constraint reformulation for apparent power flow constraints was first investigated in [34], although without application in an optimization problem. Preliminary results for the application of the linearization approach to the AC CC-OPF problem were presented in [32], including only parts of

the model and limited testing on a small system.

Here, we present a more comprehensive model, including uncertainty of generator active and reactive power outputs. We further propose different solution algorithms for the problem, including 1) a one-shot optimization, which allows us to co-optimize reactions to the uncertainty realization, or 2) an iterative approach, which allows for integration with any existing AC OPF solver and hence has significant potential for scalability and implementation in existing operating practices. The two solution methods are compared in terms of cost and computation time. We further demonstrate scalability by solving the AC CC-OPF problem for systems with up to 2383 buses.

Solving the AC CC-OPF using a linearization and analytical chance-constraint reformulation introduces two sources of inaccuracy, namely the linearization of the AC power flow equations and the assumption of the distributional parameters. Specifically, our case study assumes that the currents, voltages and generator outputs are normally distributed, as this was found to be a reasonable assumption for line flows in previous chapters. To assess the accuracy of the method, we run both in-sample Monte Carlo simulations to investigate approximation errors from the linearization, and out-of-sample tests to assess the overall performance.

We further assess how well the analytical reformulation compares with two sample based reformulations. The first approach uses a Monte Carlo simulation to define empirical quantiles of the power flows. The second approach is based on the scenario approach [54, 58, 100], and uses a pre-determined number of samples to define a worst-case margin that guarantees a maximum joint violation probability. We compare the three methods in terms of chance constraint feasibility, operational cost and computational time.

The remainder of the chapter is structured as follows. We first describe the AC OPF formulation with chance constraints and the approximate reformulation, before moving on to a description of the different solution approaches. Then a case study to assess the performance of the method is presented, before we summarize and conclude.

The work presented in this chapter is extending the work performed in collaboration with Haoyuan Qu and Jeremias Schmidli during their semester theses work, which was documented in [101, 34, 102, 32].

5.2 Modelling Active and Reactive Power Uncertainty

In this section, we present the modelling of active and reactive power deviations, as well as the reaction from conventional generators in order to maintain system balance and keep the voltage magnitudes at the chosen set-points. As in the above chapters, we consider a power system where $\mathcal{N}$, $\mathcal{L}$ denote the set of nodes and lines. The number of nodes and lines are given by $|\mathcal{N}| = m$ and $|\mathcal{L}| = l$. The set of nodes with uncertain demand or production of energy is given by $\mathcal{U} \subseteq \mathcal{N}$, and the set of conventional generators by $\mathcal{G} \subseteq \mathcal{N}$. To simplify notation, we assume that there is one conventional generator, one composite uncertainty source and one demand per node, such that $|\mathcal{G}| = |\mathcal{U}| = |\mathcal{N}| = m$. Nodes without generation or load can be handled by setting the respective entries to zero, and nodes with multiple entries can be handled through a summation.

To model generation and control at different types of buses, we distinguish between PQ, PV and θV (reference) buses. We denote the sets of PQ, PV and θV buses by $\mathcal{N}_{PQ}$, $\mathcal{N}_{PV}$, $\mathcal{N}_{\theta V} \subset \mathcal{N}$, respectively. The number of PQ and PV buses is given by $|\mathcal{N}_{PQ}| = m_{PQ}$, $|\mathcal{N}_{PV}| = m_{PV}$. In our formulation, we assume only one θV bus, although the method could be extended to account for a distributed slack bus. The sets $\mathcal{G}_{PQ}$, $\mathcal{G}_{PV}$, $\mathcal{G}_{\theta V} \subset \mathcal{G}$ represent generators connected at a PQ, PV and θV buses. We denote variables at PQ buses by subscript $_{PQ}$, variables at PV buses by subscript $_{PV}$ and variables at the reference bus by subscript $_{\theta V}$.

Uncertainty Modelling

As explained in Section 2.2, we can model the deviations in active power production as the sum of the forecasted production, here denoted by p_U , and a zero mean fluctuation ω at each bus. We repeat the definition (2.3) here for convenience:

$$\tilde{p}_U = p_U + \omega .$$

When the active power production or consumption changes, it typically also impacts the reactive power injections. While the previously considered DC power flow formulations neglect reactive power, the AC power

flow model requires a model for changes in the reactive power injections. Different models for reactive power might be applicable, depending on the type of uncertainty source (e.g., load or renewable generation). Here we choose to assume that the fluctuations have a *constant power factor* $\cos \phi$, which might be specified for each uncertainty source. Hence, for each uncertain power injection, the reactive power is given by

$$\tilde{q}_{U,i} = q_{U,i} + \gamma_i \omega_i, \quad \text{where } \gamma_i = \sqrt{\frac{1 - \cos^2 \phi_i}{\cos^2 \phi_i}} \quad \forall i \in \mathcal{U}.$$

The vector of variables $\gamma \in \mathbb{R}^m$ will be referred to as the power ratio in the following. Other possibilities to model reactive power deviations could assume, e.g., that the reactive power injections remain constant, that the reactive power can be dispatched (at least partially) independent of the active power production or that some uncertainty sources (e.g., large wind farms) participate in controlling the voltage at their point of connection. These types of control can be included in the formulation without any conceptual change.

Generation and Voltage Control

In case of deviations from the scheduled reactive and active power injections, the controllable generators must control the produced reactive and active power in order to ensure power balance, and maintain the desired voltage profile. For active power, we assume a similar set-up as in the DC case, where the power mismatch Ω is balanced through a centralized AGC signal as in (2.6). In addition to a direct mismatch in the power injections Ω , the changes in power flows will also influence the system losses. We assume that the change in the active power losses, which is a-priori unknown and denoted by δp , is balanced out by the generator at the reference bus.

$$\tilde{p}_{G,i}(\omega) = p_{G,i} - \alpha_i \Omega, \quad \forall i \in \mathcal{G}_{PQ}, \mathcal{G}_{PV}, \quad (5.1)$$

$$\tilde{p}_{G,i}(\omega) = p_{G,i} - \alpha_i \Omega + \delta p_i, \quad \forall i \in \mathcal{G}_{\theta V}, \quad (5.2)$$

$$\sum_{i \in \mathcal{G}} \alpha_i = 1. \quad (5.3)$$

For reactive power, we assume a more localized control. All generators connected at PV and θV buses change their outputs by δq in order

to keep voltage magnitudes constant. The generators connected at PQ buses do not change their output.

$$\tilde{q}_{G,i}(\omega) = q_{G,i}, \quad \forall i \in \mathcal{G}_{PQ} \quad (5.4)$$

$$\tilde{q}_{G,i}(\omega) = q_{G,i} + \delta q_i, \quad \forall i \in \mathcal{G}_{PV}, \mathcal{G}_{\theta V}. \quad (5.5)$$

The change in voltage magnitudes can be expressed in a similar way. The voltage magnitude is fixed at the reference and PV buses, and changing in reaction to the uncertainty at PQ buses:

$$\tilde{v}_j(\omega) = v_j + \delta v_j, \quad \forall j \in \mathcal{N}_{PQ}, \quad (5.6)$$

$$\tilde{v}_j(\omega) = v_j, \quad \forall j \in \mathcal{N}_{PV}, \mathcal{N}_{\theta V}. \quad (5.7)$$

Note that this type of local reactive power control only applies to how the generators react to the fluctuations ω . The nominal reactive power dispatch of the generators, as well as the voltage set-points at PV and θV buses, are optimized as part of the OPF. We further remark that a centralized Automatic Voltage Regulator (AVR) scheme as in [97], with control variables similar to the participation factors α of the AGC, could also be implemented within the presented framework.

Power flows

With deviations in the active and reactive power injections, the power flows throughout the system also change. Assuming a thermally constrained system, we model the power flows in terms of currents, with i_{ij} representing the current magnitude on line ij . All currents are influenced by the uncertainty which induces a change δi_{ij} in the current magnitude,

$$\tilde{i}_{ij}(\omega) = i_{ij} + \delta i_{ij}, \quad \forall ij \in \mathcal{L}. \quad (5.8)$$

5.3 Chance-Constrained AC Optimal Power Flow

This section first presents the AC CC-OPF, discusses the interpretation of constraints, and points out why the problem in its original form is intractable. We then continue to explain how a tractable, approximate reformulation can be obtained, based on the full AC power flow for the nominal operating point and a linearization around that point for the deviations.

5.3.1 Original Problem

We state the full AC CC-OPF as

$$\min_{p_G, q_G, v} \sum_{i \in \mathcal{G}} (c_{2,i} p_{G,i}^2 + c_{1,i} p_{G,i} + c_{0,i}) \quad (5.9)$$

$$\text{s.t. } f(\tilde{\theta}(\omega), \tilde{v}(\omega), \tilde{p}(\omega), \tilde{q}(\omega)) = 0 \quad \forall \omega \in \mathcal{D} \quad (5.10)$$

$$\mathbb{P}(\tilde{p}_{G,i}(\omega) \leq p_{G,i}^{max}) \geq 1 - \epsilon_P, \quad \forall i \in \mathcal{G} \quad (5.11)$$

$$\mathbb{P}(\tilde{p}_{G,i}(\omega) \geq p_{G,i}^{min}) \geq 1 - \epsilon_P, \quad \forall i \in \mathcal{G} \quad (5.12)$$

$$\mathbb{P}(\tilde{q}_{G,i}(\omega) \leq q_{G,i}^{max}) \geq 1 - \epsilon_Q, \quad \forall i \in \mathcal{G} \quad (5.13)$$

$$\mathbb{P}(\tilde{q}_{G,i}(\omega) \geq q_{G,i}^{min}) \geq 1 - \epsilon_Q, \quad \forall i \in \mathcal{G} \quad (5.14)$$

$$\mathbb{P}(\tilde{v}_j(\omega) \leq v_j^{max}) \geq 1 - \epsilon_V, \quad \forall j \in \mathcal{N} \quad (5.15)$$

$$\mathbb{P}(\tilde{v}_j(\omega) \geq v_j^{min}) \geq 1 - \epsilon_V, \quad \forall j \in \mathcal{N} \quad (5.16)$$

$$\mathbb{P}(\tilde{i}_{ij}(\omega) \leq i_{ij}^{max}) \geq 1 - \epsilon_I, \quad \forall ij \in \mathcal{L} \quad (5.17)$$

$$\theta_{\theta V} = 0 \quad (5.18)$$

The objective (5.9) is to minimize the cost of active power generation, which is modelled through a quadratic cost objective where c_2 , c_1 and c_0 are the quadratic, linear and constant cost coefficients¹. Eq. (5.10) are the nodal power balance constraints based on the full non-linear AC power flow constraints. These equations are functions of the nodal voltage angles $\tilde{\theta}(\omega)$ and magnitudes $\tilde{v}(\omega)$, as well as the nodal injections of active $\tilde{p}(\omega)$ and reactive power $\tilde{q}(\omega)$. The nodal power injections are the sum of the power injections from generator, loads and uncertainty sources,

$$\tilde{p}(\omega) = \tilde{p}_G(\omega) + p_D + p_U + \omega, \quad (5.19)$$

$$\tilde{q}(\omega) = \tilde{q}_G(\omega) + q_D + q_U + \text{diag}(\gamma)\omega. \quad (5.20)$$

Both the power injections and the nodal voltages depend on the realization of ω , and the power balance constraints (5.10) must be maintained for all possible ω within the uncertainty set $\mathcal{D}$.

¹We use a quadratic objective, which is a generalization compared to the linear objectives used in the previous chapters. The quadratic objective is a more accurate model of the true cost of generation, and is particularly of interest when minimizing actual generation cost, as in a vertically integrated utility. A (piecewise) linear cost function is more representative for the bids from generators to a market operator, which must typically be provided in \$/MW.

The remaining constraints are generation constraints for active and reactive power (5.11) - (5.14), constraints on the voltage magnitudes at each bus (5.15), (5.16) and transmission constraints in the form of limits on the current magnitudes $\tilde{i}(\omega)$ (5.17). All these constraints are formulated as chance constraints with acceptable violation probabilities of ϵ_P , ϵ_Q , ϵ_V and ϵ_I , respectively. For the generation constraints, that are in reality hard constraints, a constraint violation would indicate a situation in which regular control actions would leave the system imbalanced and manual intervention would be needed, as explained in Section 2.4. For the current and voltage constraints, a constraint violation would indicate an actual violation of the limit, which can either be tolerated or removed through additional control actions, analogous to the transmission limits on active power discussed in Section 2.4.

Finally, the voltage angle at the reference bus is set to zero by (5.18).

5.3.2 Approximation of Uncertainty Impact

The problem (5.9) - (5.18) is not tractable in its current form. First, it is not possible to directly enforce (5.10) for all ω , as the set of fluctuations $\mathcal{D}$ is uncountable. Further, the chance constraints (5.11)-(5.17) must be reformulated into tractable constraints. In the following, we present an approximate version of (5.9) - (5.18), based on the following main idea:

- For the nominal operating point, given by $\omega = 0$ and the scheduled nodal power injections p , q , we solve the full AC power flow equations. This allows us to get an accurate representation of the forecasted system state, and guarantees AC feasibility for the case without forecast errors.
- The impact of the uncertainty ω is accounted for using a linearized representation of the system, based on Jacobian matrices evaluated at the nominal operating point. This modelling represents a first-order Taylor expansion around the nominal operating point, and can be assumed to be reasonably accurate when the fluctuations ω are small compared to the overall scheduled power injections $|\omega| < |\sum_{i \in \mathcal{N}} p_i|$.

Following the above ideas, we replace (5.10) by a single set of deterministic equations for the forecasted operating point,

$$f(\theta, v, p, q) = 0, \quad (5.21)$$

where θ , v are the voltage angles and magnitudes corresponding to the scheduled injections. Based on the operating point obtained from (5.21), we define sensitivity factors with respect to the fluctuations ω , i.e.,

$$\begin{aligned}\mathbf{\Gamma}_P &= \left. \frac{\partial p_G}{\partial \omega} \right|_{(\theta, v, p, q)}, & \mathbf{\Gamma}_V &= \left. \frac{\partial v}{\partial \omega} \right|_{(\theta, v, p, q)}, \\ \mathbf{\Gamma}_Q &= \left. \frac{\partial q_G}{\partial \omega} \right|_{(\theta, v, p, q)}, & \mathbf{\Gamma}_I &= \left. \frac{\partial i}{\partial \omega} \right|_{(\theta, v, p, q)}.\end{aligned}$$

Here, $\mathbf{\Gamma}_P$, $\mathbf{\Gamma}_Q$ denote the sensitivity factors for the active and reactive power injections, respectively, while $\mathbf{\Gamma}_V$, $\mathbf{\Gamma}_I$ are the sensitivity factors for the voltage and current magnitudes. The sensitivity factors allow us to approximate the changes in the power injections δp , δq and the voltage and current magnitudes δv , δi as linear functions of ω ,

$$\begin{aligned}\tilde{p}_{G,i}(\omega) &= p_{G,i} + \delta p_i \approx p_{G,i} - \alpha_i \Omega + \mathbf{\Gamma}_{P(i, \cdot)} \omega, & \forall i \in \mathcal{G}_{\theta V} \\ \tilde{q}_{G,i}(\omega) &= q_{G,i} + \delta q_i \approx q_{G,i} + \mathbf{\Gamma}_{Q(i, \cdot)} \omega, & \forall i \in \mathcal{G}_{\theta V, \mathcal{G}_{PV}} \\ \tilde{v}_j(\omega) &= v_j + \delta v_j \approx v_j + \mathbf{\Gamma}_{V(j, \cdot)} \omega, & \forall j \in \mathcal{G}_{PQ} \\ \tilde{i}_{ij}(\omega) &= i_{ij} + \delta i_{ij} \approx i_{ij} + \mathbf{\Gamma}_{I(ij, \cdot)} \omega. & \forall ij \in \mathcal{L}\end{aligned}$$

The derivation of the sensitivity factors $\mathbf{\Gamma}_P$, $\mathbf{\Gamma}_Q$, $\mathbf{\Gamma}_V$ and $\mathbf{\Gamma}_I$ can be found in Appendix A. Here, we only note that they depend non-linearly on the forecasted operating point p , q , v and θ , and linearly on the participation factors α and the power ratio γ .

5.3.3 Approximate AC CC-OPF

With the above considerations, we can rewrite (5.9) - (5.18) in the following way:

$$\min_{p_G, q_G, v} \sum_{i \in \mathcal{G}} (c_{2,i} p_{G,i}^2 + c_{1,i} p_{G,i} + c_{0,i}) \quad (5.22)$$

$$\text{s.t. } f(\theta, v, p, q) = 0 \quad (5.23)$$

$$\mathbb{P}(p_{G,i} - \alpha_i \Omega \leq p_{G,i}^{max}) \geq 1 - \epsilon_P, \quad \forall i \in \mathcal{G}_{PQ}, \mathcal{G}_{PV} \quad (5.24)$$

$$\mathbb{P}(p_{G,i} - \alpha_i \Omega \geq p_{G,i}^{min}) \geq 1 - \epsilon_P, \quad \forall i \in \mathcal{G}_{PQ}, \mathcal{G}_{PV} \quad (5.25)$$

$$\mathbb{P}(p_{G,i} - \alpha_i \Omega + \mathbf{\Gamma}_{P(i, \cdot)} \omega \leq p_{G,i}^{max}) \geq 1 - \epsilon_P, \quad \forall i \in \mathcal{G}_{\theta V} \quad (5.26)$$

$$\mathbb{P}(p_{G,i} - \alpha_i \Omega + \mathbf{\Gamma}_{P(i, \cdot)} \omega \geq p_{G,i}^{min}) \geq 1 - \epsilon_P, \quad \forall i \in \mathcal{G}_{\theta V} \quad (5.27)$$

$$q_{G,i}^{min} \leq q_{G,i} \leq q_{G,i}^{max} \quad \forall i \in \mathcal{G}_{PQ} \quad (5.28)$$

$$\mathbb{P}(q_{G,i} + \mathbf{\Gamma}_Q(i, \cdot)\omega \leq q_{G,i}^{max}) \geq 1 - \epsilon_Q, \quad \forall i \in \mathcal{G}_{PV}, \mathcal{G}_{\theta V} \quad (5.29)$$

$$\mathbb{P}(q_{G,i} + \mathbf{\Gamma}_Q(i, \cdot)\omega \geq q_{G,i}^{min}) \geq 1 - \epsilon_Q, \quad \forall i \in \mathcal{G}_{PV}, \mathcal{G}_{\theta V} \quad (5.30)$$

$$\mathbb{P}(v_j + \mathbf{\Gamma}_V(j, \cdot)\omega \leq v_j^{max}) \geq 1 - \epsilon_V, \quad \forall j \in \mathcal{N}_{PQ} \quad (5.31)$$

$$\mathbb{P}(v_j + \mathbf{\Gamma}_V(j, \cdot)\omega \geq v_j^{min}) \geq 1 - \epsilon_V, \quad \forall j \in \mathcal{N}_{PQ} \quad (5.32)$$

$$v_j^{min} \leq v_j \leq v_j^{max} \quad \forall i \in \mathcal{N}_{PV}, \mathcal{N}_{\theta V} \quad (5.33)$$

$$\mathbb{P}(i_{ij} + \mathbf{\Gamma}_I(ij, \cdot)\omega \leq i_{ij}^{max}) \geq 1 - \epsilon_I, \quad \forall ij \in \mathcal{L} \quad (5.34)$$

$$\theta_{\theta V} = 0 \quad (5.35)$$

Note that the active power chance constraints (5.26), (5.27) for the θV bus depends on $\mathbf{\Gamma}_P$, while the reactive power chance constraint for the PV and θV buses depend on $\mathbf{\Gamma}_Q$. The reactive power constraints for the PQ buses (5.28) are deterministic, as the power injections are fixed. Similarly, the chance constraints on the voltage magnitudes at PQ buses (5.31), (5.32) are chance constraints which depend on $\mathbf{\Gamma}_V$, whereas the voltage magnitude constraints for the reference and PV buses are deterministic. All current constraints (5.34) are chance constraints, with the dependency on ω expressed through $\mathbf{\Gamma}_I$. Note that the forecasted current magnitude i_{ij} is a function of the full non-linear AC power flow equations.

5.3.4 Chance Constraint Reformulation

In the above formulation, the chance constraints are linear functions of ω . This implies that we can use the analytical reformulation as in section 2.5. This reformulation is applicable whenever we have a linear dependency on the uncertain variables, even though the sensitivity factors $\mathbf{\Gamma}_P$, $\mathbf{\Gamma}_Q$, $\mathbf{\Gamma}_V$, $\mathbf{\Gamma}_I$ are or the current magnitudes i_{ij} are non-linear functions of the decision variables.

Using this reformulation, we can express (5.22) - (5.35) as

$$\min_{p_G, q_G, v} \sum_{i \in \mathcal{G}} (c_{2,i} p_{G,i}^2 + c_{1,i} p_{G,i} + c_{0,i}) \quad (5.36)$$

$$\text{s.t. } f(\theta, v, p, q) = 0 \quad \forall \omega \in \mathcal{D} \quad (5.37)$$

$$p_{G,i} + \alpha_i f_{\mathcal{P}}^{-1}(1 - \epsilon_P) \sigma_{\Omega} \leq p_{G,i}^{max}, \quad \forall i \in \mathcal{G}_{PQ}, \mathcal{G}_{PV} \quad (5.38)$$

$$p_{G,i} - \alpha_i f_{\mathcal{P}}^{-1}(1 - \epsilon_P) \sigma_{\Omega} \geq p_{G,i}^{min}, \quad \forall i \in \mathcal{G}_{PQ}, \mathcal{G}_{PV} \quad (5.39)$$

$$p_{G,i} + f_{\mathcal{P}}^{-1}(1 - \epsilon_P) \|\Sigma_W^{1/2}(-\alpha_i \mathbf{1}_{1,m} + \mathbf{\Gamma}_P(i, \cdot))^T\|_2 \leq p_{G,i}^{max}, \quad \forall i \in \mathcal{G}_{\theta V} \quad (5.40)$$

$$p_{G,i} - f_{\mathcal{P}}^{-1}(1 - \epsilon_P) \|\Sigma_W^{1/2}(-\alpha_i \mathbf{1}_{1,m} + \mathbf{\Gamma}_{P(i,\cdot)})^T\|_2 \geq p_{G,i}^{min}, \quad \forall i \in \mathcal{G}_{\theta V} \quad (5.41)$$

$$q_{G,i}^{min} \leq q_{G,i} \leq q_{G,i}^{max} \quad \forall i \in \mathcal{G}_{PQ} \quad (5.42)$$

$$q_{G,i} + f_{\mathcal{P}}^{-1}(1 - \epsilon_Q) \|\Sigma_W^{1/2} \mathbf{\Gamma}_{Q(i,\cdot)}^T\|_2 \leq q_{G,i}^{max}, \quad \forall i \in \mathcal{G}_{PV}, \mathcal{G}_{\theta V} \quad (5.43)$$

$$q_{G,i} - f_{\mathcal{P}}^{-1}(1 - \epsilon_Q) \|\Sigma_W^{1/2} \mathbf{\Gamma}_{Q(i,\cdot)}^T\|_2 \geq q_{G,i}^{min}, \quad \forall i \in \mathcal{G}_{PV}, \mathcal{G}_{\theta V} \quad (5.44)$$

$$v_j + f_{\mathcal{P}}^{-1}(1 - \epsilon_V) \|\Sigma_W^{1/2} \mathbf{\Gamma}_{V(j,\cdot)}^T\|_2 \leq v_j^{max}, \quad \forall j \in \mathcal{G}_{PQ} \quad (5.45)$$

$$v_j - f_{\mathcal{P}}^{-1}(1 - \epsilon_V) \|\Sigma_W^{1/2} \mathbf{\Gamma}_{V(j,\cdot)}^T\|_2 \geq v_j^{min}, \quad \forall j \in \mathcal{G}_{PQ} \quad (5.46)$$

$$v_j^{min} \leq v_j \leq v_j^{max} \quad \forall j \in \mathcal{N}_{PV}, \mathcal{N}_{\theta V} \quad (5.47)$$

$$i_{ij} + f_{\mathcal{P}}^{-1}(1 - \epsilon_I) \|\Sigma_W^{1/2} \mathbf{\Gamma}_{I(ij,\cdot)}^T\|_2 \geq i_{ij}^{max}, \quad \forall ij \in \mathcal{L} \quad (5.48)$$

$$\theta_{\theta V} = 0 \quad (5.49)$$

As for the previous chance-constrained formulations, we observe that the consideration of uncertainty introduce a similar uncertainty margin, i.e., a tightening of the nominal constraints. The uncertainty margins for active and reactive power are defined by

$$\lambda_{P,i} = \alpha_i f_{\mathcal{P}}^{-1}(1 - \epsilon_P) \sigma_{\Omega}, \quad \forall i \in \mathcal{G}_{PQ}, \mathcal{G}_{PV}, \quad (5.50)$$

$$\lambda_{P,i} = f_{\mathcal{P}}^{-1}(1 - \epsilon_P) \|\Sigma_W^{1/2}(-\alpha_i \mathbf{1}_{1,m} + \mathbf{\Gamma}_{P(i,\cdot)})^T\|_2, \quad \forall i \in \mathcal{G}_{\theta V}, \quad (5.51)$$

$$\lambda_{Q,i} = f_{\mathcal{P}}^{-1}(1 - \epsilon_Q) \|\Sigma_W^{1/2} \mathbf{\Gamma}_{Q(i,\cdot)}^T\|_2, \quad \forall i \in \mathcal{G}_{PV}, \mathcal{G}_{\theta V}, \quad (5.52)$$

while the voltage and current margins are given by

$$\lambda_{V,j} = f_{\mathcal{P}}^{-1}(1 - \epsilon_V) \|\Sigma_W^{1/2} \mathbf{\Gamma}_{V(j,\cdot)}^T\|_2 \quad \forall j \in \mathcal{N}_{PQ}, \quad (5.53)$$

$$\lambda_{I,ij} = f_{\mathcal{P}}^{-1}(1 - \epsilon_I) \|\Sigma_W^{1/2} \mathbf{\Gamma}_{I(ij,\cdot)}^T\|_2 \quad \forall ij \in \mathcal{L}. \quad (5.54)$$

The general function $f_{\mathcal{P}}$ is used to emphasize that we can apply either of the five analytical reformulations in Table 2.1.

5.4 Solution Algorithms

In the following, we discuss two ways of solving the approximate AC CC-OPF formulated above.

5.4.1 One-Shot Optimization

The reformulated AC CC-OPF given by (5.36) - (5.49) is a continuous, non-convex optimization problem. The most obvious way to find a solution is therefore to pass the entire problem to a suitable solver. The solver will optimize the generation dispatch, while inherently accounting for the dependency of the sensitivity factors on the current nominal point p , q , v , θ and the participation factors α , γ . By including α and γ as optimization variables, it is thus possible to optimize not only the nominal dispatch, but also the reaction to fluctuations. Note that the problem is not guaranteed to converge to a global optimum, as it is non-convex.

The drawback of attempting a one-shot solution is the problem complexity, which might lead to long solution times. The deterministic AC OPF is already a non-convex problem, and adding additional terms with complex dependencies on the decision variables only increases the difficulty of solving the problem. This can be a bottleneck for adoption in more realistic settings, where scalability and robustness of the solutions are important criteria.

5.4.2 Iterative Solution Algorithm

To address the problem of increased computational complexity, we propose an iterative solution algorithm which allows us to solve the AC CC-OPF using any existing AC OPF tool. For example, the iterative algorithm can utilize convex relaxations of the deterministic AC OPF to find globally optimal solutions for the nominal operating point, or be added as an outer loop on AC OPF implementations already available in industry. This facilitates adoption into existing processes, and allows the algorithm to take advantage of previous efforts to increase robustness and scalability of the underlying AC OPF tool.

The iterative solution algorithm is based on the observation that the uncertainty only occurs in the uncertainty margins defined by (5.50) - (5.54). Assume that the participation factors α for the AGC and the power ratio γ of the uncertain power injections are known. Then, for a given operating point, the uncertainty margins can be pre-calculated, and the AC CC-OPF can be solved as a standard AC OPF with limits

adjusted by the uncertainty margin:

$$p_G^{min} + \lambda_P \leq p_G \leq p_G^{max} - \lambda_P \quad (5.55)$$

$$q_G^{min} + \lambda_Q \leq q_G \leq q_G^{max} - \lambda_Q \quad (5.56)$$

$$v^{min} + \lambda_V \leq v \leq v^{max} - \lambda_V \quad (5.57)$$

$$i \leq i^{max} - \lambda_I \quad (5.58)$$

However, the linearization point (θ, v, p, q) is not known a-priori. Therefore, we solve the problem using an iterative approach, which uses the maximum change of the uncertainty margins (between subsequent iterations) as a measure of convergence. The approach consist of the following steps:

-
1. *Initialization:* Set uncertainty margins $\lambda_P^0 = \lambda_Q^0 = \lambda_V^0 = \lambda_I^0 = 0$, and iteration count $\kappa = 0$.
 2. *Solve AC OPF:* Solve the AC OPF defined by (5.36) - (5.37) and (5.55) - (5.58), and increase iteration count $\kappa = \kappa + 1$.
 3. *Evaluate uncertainty margins:* Compute the uncertainty margins of the current iteration λ_P^κ , λ_Q^κ , λ_V^κ and λ_I^κ by evaluating (5.50) - (5.54). Then, compute the maximum difference to the last iteration for the uncertainty margins of the power injections, voltages and currents,

$$\begin{aligned} \eta_P^\kappa &= \max\{|\lambda_P^\kappa - \lambda_P^{\kappa-1}|\} , & \eta_Q^\kappa &= \max\{|\lambda_Q^\kappa - \lambda_Q^{\kappa-1}|\} , \\ \eta_V^\kappa &= \max\{|\lambda_V^\kappa - \lambda_V^{\kappa-1}|\} , & \eta_I^\kappa &= \max\{|\lambda_I^\kappa - \lambda_I^{\kappa-1}|\} . \end{aligned}$$

4. *Check convergence:* If the maximum difference is small enough, stop.

$$\eta_P^\kappa \leq \hat{\eta}_P , \quad \eta_Q^\kappa \leq \hat{\eta}_Q , \quad \eta_V^\kappa \leq \hat{\eta}_V , \quad \eta_I^\kappa \leq \hat{\eta}_I .$$

If not, move back to step 2.

The accuracy of the solution is defined through the choice of the stopping criteria $\hat{\eta}_P$, $\hat{\eta}_Q$, $\hat{\eta}_V$, and $\hat{\eta}_I$.

While the above algorithm is straight forward to implement, it has some drawbacks.

First, the uncertainty margins can only be computed as part of the outer loop, which takes the forecasted operating point and values for α and γ as inputs. Since the uncertainty margins are not part of the inner loop optimization, the iterative AC CC-OPF does not define the operating point, α or γ in a way that minimizes the uncertainty margins on congested lines or on buses with tight voltage constraints. It is however possible to include the influence of α on the generation dispatch by keeping α as a variable in the generation constraints (5.38) - (5.41).

Second, the solution is not guaranteed to converge. In our simulations, the algorithm converged within a few iterations when $\hat{\eta}_P$, $\hat{\eta}_Q$, $\hat{\eta}_V$ and $\hat{\eta}_I$ are chosen not smaller than 0.001 MVA for active and reactive power, and 10^{-5} p.u. for voltage and current magnitudes. However, we observed that when the stopping criteria $\hat{\eta}_P$, $\hat{\eta}_Q$, $\hat{\eta}_V$, and $\hat{\eta}_I$ are chosen to be even smaller, the solution does not settle. Instead, each iteration finds a slightly different solution in the vicinity of the optimum.

However, the main change in the solution occurs between the first and second iterations, where the uncertainty margins increase from zero to the value corresponding to the optimal point of the deterministic AC OPF. Even in case of non-convergence, stopping the algorithm after a given number of iterations and choosing the solution corresponding to the smallest change η_P^κ , η_Q^κ , η_V^κ , η_I^κ between iterations should, in most cases, provide a reasonable approximation of the true solution to the chance-constrained problem.

5.4.3 Alternative Iterative Solution Algorithms

When solving the problem using the iterative approach, the uncertainty margins λ_P^κ , λ_Q^κ , λ_V^κ and λ_I^κ are calculated only in the outer iteration. This opens for the possibility of using other methods than the analytical reformulation to calculate the uncertainty margins.

Empirical Uncertainty Margins from Monte Carlo Simulation

One possibility is to define empirical uncertainty margins based on a Monte Carlo simulation. The uncertainty margins (5.50) - (5.54) represent an approximation of the quantile of the distribution of the currents, voltages and power generation. By sampling the uncertainty vector ω and calculating the resulting power flows for a large number of samples, we can compute empirical uncertainty margins $\lambda_P^{u,l}$, $\lambda_Q^{u,l}$, $\lambda_V^{u,l}$ and $\lambda_I^{u,l}$.

For example, assume we want to ensure a voltage magnitude constraint with $\epsilon_V = 0.01$ and use 1000 samples of ω . To compute the upper and lower quantiles, we run the power flow for each sample. The upper quantile corresponds to the 10^{th} largest magnitude v_j^{10} , and the lower quantile to the 10^{th} lowest magnitude v_j^{990} . The upper and lower uncertainty margins are defined by $\lambda_V^u = v_j^{10} - v_j$ and $\lambda_V^l = v_j - v_j^{990}$, respectively. Note that the empirical uncertainty margins are not necessarily symmetric due to the non-linearity of the AC power flow equations.

A similar Monte Carlo approach to define the uncertainty margins was implemented as part of the prototype toolbox in the UMBRELLA project [41], where it was shown to significantly reduce the probability of constraint violations. However, the implemented approach calculated the uncertainty margins based only on the forecasted operating point, and did not include any subsequent updates. Since the outcome of the Monte Carlo simulation and hence the uncertainty margins depend on the solution to the AC OPF for the forecasted operating point, we implement the Monte Carlo based approach as part of the iterative framework.

Enforcing Joint Chance Constraints

While the uncertainty margins (5.50) - (5.54) represent the quantiles of the respective constraints, they can more generally be understood as a tightening which is necessary to secure the system against uncertainty. A similar constraint tightening can be observed in other stochastic and robust formulations of the OPF problem. One example is the CC-OPF formulations based on joint chance constraints, in e.g. [14, 97].

In [97], the stochastic AC OPF is solved as a joint chance-constrained problem. The AC power flow equations are represented using a convex SDP relaxation [103], and solved as a robust problem. The robust uncertainty set is defined by the method in [55], and represented in the AC OPF using the set vertices, defined as the maximum and minimum realizations of the observed variables.

Another way of guaranteeing chance constraint feasibility is using the scenario approach [58, 54], which defines a number of randomly drawn samples N_S for which the system must remain within bounds. The number of samples is related to the decision variables N_X and the acceptable

joint violation probability ϵ_J , and is given by [100]

$$N_S \geq \frac{2}{\epsilon_J} \left(\ln \frac{1}{\beta} + N_X \right). \quad (5.59)$$

The value $1 - \beta$ represents the confidence level for satisfying the chance constraint, and can be chosen to be very small. By enforcing the system constraints for all samples, it is guaranteed that the system remains jointly secure with the desired probability.

Representing uncertainty sets based on samples, or representing a robust set through the vertices as in [97], only applies when the underlying problem is convex, since convexity guarantees that the linear combination of feasible points remain within the convex, feasible set. The convex relaxation is hence necessary for the approach in [97] to guarantee chance constraint feasibility. However, the relaxation itself is not guaranteed to be AC feasible, but might provide solutions that are not physical solutions to the power flow problem.

Here, we take a slightly different approach. We solve the full, non-convex AC OPF for the forecasted operating point, ensuring that we find a physically meaningful solution². We then draw a finite number of samples for which the solution should remain secure, as prescribed by the scenario approach [100]. While the scenario approach is not directly applicable to non-convex problems, we justify our approach by the fact that we only evaluate the samples in the vicinity of a local optimum, where the problem can be assumed to be locally convex.

Assuming that the fluctuations ω are small enough for us to stay within this locally convex region, we run a power flow simulation for each of the N_S scenarios. Based on the results, we find the two limiting scenarios for each constraint,

$$v_j^u = \max\{v_j^m, m = 1, \dots, n_S\} \quad \text{and} \quad v_j^l = \min\{v_j^m, m = 1, \dots, n_S\}$$

i.e., the scenarios that lead to the highest and lowest flows. We then define the upper and lower uncertainty margins as $\lambda_V^u = v_j^u - v_j$ and $\lambda_V^l = v_j - v_j^l$, and solve the problem again.

As for the other iterative approaches, the computation of v_j^u and v_j^l depends on the solution obtained with the AC OPF for the forecasted

²The AC OPF for the forecasted operating point can be solved using a convex relaxation, as long as we ensure feasibility of the resulting solution and apply a tighter relaxations, such as the higher-order moment relaxation in [104, 105], if the initial solution is not exact.

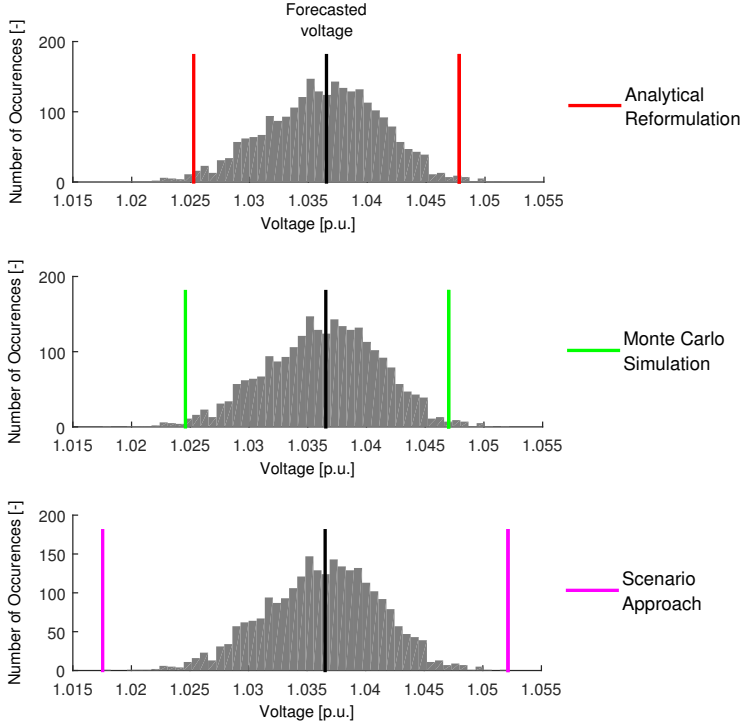


Figure 5.1: Uncertainty margins $\lambda_{V,j}^u$, $\lambda_{V,j}^l$ for an example voltage constraint, as obtained with a given set of samples. The black line represents the forecasted voltage, and the histogram shows the empirical voltage distribution for the given sample set. From top to bottom, we see the uncertainty margins obtained with the analytical reformulation, with a Monte Carlo simulation and with the scenario approach.

operating point. The iterations hence continue until the uncertainty margins have converged. With the final solution, where none of the samples are violated, the problem is guaranteed to have a joint violation probability below the prescribed value.

A comparison of the three approaches to define uncertainty margins is shown for an example voltage constraint in Fig. 5.1. The histogram represents the empirical distribution of the voltage magnitudes, as calculated based on the samples used in the Monte Carlo simulation and the scenario approach. The analytical margins (top) are calculated based on (5.53) with a normal distribution assumption, the Monte Carlo margins (middle) correspond to the empirical margin of the given set of samples, and the scenario approach (bottom) uses the extreme scenarios to define the margins. We observe that the analytical margins are symmetric, which in this case leads to a slight overestimation of the upper margin and a slight underestimation of the lower margin compared with the empirical Monte Carlo margins. The difference is however very small, although the underlying samples are not normally distributed. Further, we see that the scenario approach has much larger uncertainty margins than either of the two other approaches, as it only depends on the worst among the drawn scenarios. However, it should also be noted that the uncertainty margins for the scenario approach are not directly comparable to the uncertainty margins for the analytical and Monte Carlo approach. In this example, the number of samples in the scenario approach was chosen to guarantee a joint violation probability of $\epsilon_J \leq 0.1$, whereas the uncertainty margins of the analytical and Monte Carlo reformulations aim at limiting the violation probability of each constraint to $\epsilon \leq 0.01$.

5.5 Case Study

In the following, we study the performance of the approximate AC CC-OPF. We first investigate the different solution algorithms in terms of nominal operating cost and computational complexity, and demonstrate scalability of the iterative approach. We continue with an assessment of the performance of the approximate chance constraint reformulation. We perform both in-sample and out-of-sample testing, and compare the iterative approach based on analytic uncertainty margins with the iterative approaches based on Monte Carlo simulations and the scenario approach.

5.5.1 Test systems

We run simulations for four different test systems, as described below. As a base case, we enforce all chance constraints with violation probabilities $\epsilon_P = \epsilon_Q = \epsilon_V = \epsilon_I = 0.01$. We assume that uncertainty from load and renewable generation is observed as fluctuations in the net load. The number of uncertain loads and their respective standard deviations are chosen to obtain congested, but feasible test cases. For the analytical reformulation, we assume that ω follows a multivariate normal distribution. For the iterative solution algorithms we set $\hat{\eta}_P = \hat{\eta}_Q = 0.001MV A$, $\hat{\eta}_V = 10^{-5}p.u.$ and $\hat{\eta}_I = 0.001kA$.

IEEE RTS96 One Area Test Case

We base our test case on the data for the RTS96 system provided with Matpower 5.1 [73]. We do not consider unit commitment, and hence set the lower limit on the generator output to zero. To obtain an interesting case, the generation limits are increased by a factor of 1.5. All 17 loads are assumed to be uncertain, with standard deviations equal to 10% of the forecasted consumption and zero correlation between loads.

IEEE 118 Bus Test System

We use a modified version of the 118 bus system included in the NICTA Energy System Test Archive [91]. For similar reasons as above, we set the lower generation limit to zero and increase the upper limit by a factor of 1.5. The 99 loads are all assumed to be uncertain, with standard deviations equal to 5% of their forecasted consumption. The 118 bus system is split into three zones as described in Section 4.6.1, and we assume a correlation coefficient $\rho = 0.3$ between loads within one zone, and zero correlation between loads in different zones.

IEEE 300 Bus Test System

We use IEEE 300 bus test system included in the NICTA Energy System Test Archive [91], with the lower generation limit to zero. The loads with consumption between 0 and 100 MW are assumed to be uncertain, with standard deviations equal to 5% of their forecasted consumption and zero correlation between loads. This corresponds to 131 uncertain loads, and 21% of the total system demand.

Polish 2383 Bus, Winter Peak Test Case

We use the Polish Winter Peak Test Case with 2383 buses from Matpower 5.1 [73], with the following modifications. We set the lower generation limit to zero and increase the upper limit by a factor of 2. Many of the generators in the test case have very low (or even zero) reactive power capabilities. These generators are connected at PV buses where the voltage magnitude should be kept constant, but due to their lack of reactive power capability, they are not able to affect the voltage at their point of connection. Hence, these generators cannot keep the bus voltage constant during fluctuations, even for very small amounts of uncertainty, and the AC CC-OPF is always infeasible. To relieve this situation, we increase the upper and lower reactive power limits by ± 10 MVar for all generators. All loads with consumption between 10 and 50 MW are assumed to be uncertain, with standard deviations equal to 10% of the forecast and zero correlation between loads. This corresponds to 941 uncertain loads, and 67% of the total system demand.

5.5.2 Investigated Approaches

We compare several different versions of the AC CC-OPF, as well as different methods to solve them:

A Standard AC OPF without consideration of uncertainty.

B Iterative AC CC-OPF is the iterative solution algorithm presented in Section 5.4.2. We assume that α is pre-defined according to (2.7), and that γ is determined by the nominal (pre-defined) power factor of the uncertainty sources. We compare three different approaches to define the uncertainty margins:

B1 Analytical with uncertainty margins defined by the closed-form expressions (5.50) - (5.54).

B2 Monte Carlo with empirical uncertainty margins obtained from a Monte Carlo simulation, as described in Section 5.4.3.

B3 Scenario Approach with empirical uncertainty margins obtained from the limiting scenarios, as described in Section 5.4.3.

C One-Shot AC CC-OPF is used to denote the solution obtained when solving (5.36) - (5.49) as a one shot problem. We

compare both with pre-determined and optimized uncertainty response α , γ .

C1 Fixed α, γ where α and γ are pre-defined in the same way as for the iterative approach.

C2 Optimized α, γ where α and γ are co-optimized along with the generation dispatch.

5.5.3 Comparison of Solution Algorithms

In the following, we compare the solutions obtained with the different solution approaches based on the analytical expressions for the uncertainty margins. Specifically, we use the deterministic AC OPF (A) as a benchmark, and solve the AC CC-OPF using the iterative approach (B1), the one-shot optimization with fixed participation factors α, γ (C1) and the one-shot optimization with α, γ as decision variables (C2). We compare the cost, the solution times and the number of iterations for the RTS96 and the 118 bus test cases. In order to fairly compare the computational times, the listed computational times are based on our own AC OPF implementations, which are based on functions available in Matpower and solved using KNITRO. The results are listed in Table 5.1. The costs are shown relative to the deterministic AC OPF solution.

As expected, the use of chance constraints in the AC CC-OPF formulations increases operational cost compared to the deterministic solution. When co-optimizing the participation factors α, γ , the increase in cost is lower, as seen in previous chapters. While co-optimizing α, γ reduces cost, it also significantly increases computational time. For the 118 bus system, the optimization (C2) does not converge, even after running for 13h on a desktop computer. To keep the computational time low and at least partially co-optimize α , we can keep α as an optimization variable in the generation constraints (which are anyways linear in α). Another option would be to pre-optimize α using, e.g., the DC CC-OPF from previous chapter.

We would like to note that the solution obtained with the iterative and one-shot solution approaches not only have the same cost, but also find the same optimal point (except for some very minor differences).

Table 5.1: Cost of generation, solution time and number of iterations with different solution approaches. The numbers are for the IEEE RTS96 and IEEE 118 bus test systems.

	Standard (A)	Iterative (B1)	O-S Fixed (C1)	O-S Opt (C2)
RTS96				
Cost	\$36'771	+7.7%	+7.6%	+3.2%
Time	1.5s	4.3s	4.3s	12.5s
Iterations		4		
118 Bus				
Cost	\$3'504.1	+1.9%	+1.9%	-
Time	2min 5s	7min 46s	10min 51s	-
Iterations		4		

5.5.4 Scalability

To demonstrate how the iterative solution approach can be used with existing tools and scaled to large systems, we implement the iterative AC CC-OPF as an outer iteration on the standard Matpower “runopf” function [73] and solve the problem for all four test systems. The resulting times and the number of iterations are shown in Table 5.2. We also show the evolution of the generation cost between first, second and last iterations.

The problems converge within 4-5 iterations. The overall problem solves very fast, and the solution is obtained within a few seconds for the smaller test systems and within half a minute for the Polish test case. Further, we observe that the main change in cost happens between the first and the second iteration, with only minor adjustments until final convergence. This implies that even if the problem should not converge, the intermediate iterations are typically also good approximations of the solution to the chance constrained problem. Finally, we would like to note that Matpower with the default solver MIPS finds a slightly different optimal point than KNITRO, although the differences are quite minor.

Table 5.2: Solution times, number of iteration and generation costs in the first, second and final iterations for different test cases. The problems are solved using the iterative approach (B1) with Matpower 5.1 and the default MIPS Solver.

	RTS96	118 Bus	300 Bus	Polish
Time	0.54s	1.15s	3.37s	31.89s
Iterations	5	4	5	4
Cost (1st)	36 771	3504.1	16 779	787 987
Cost (2nd)	40 249	3575.7	17 173	802 529
Cost (final)	40 127	3575.3	17 143	802 238

5.5.5 Evaluation of Chance Constraint Reformulation

The approximate AC CC-OPF has two main sources of inaccuracy that might lead to frequent overloads and violation of the chance constraint probability. First, the impact of uncertainty is approximated by a linear function, while it in reality depends on the full, non-linear AC power flow equations. Second, the analytical reformulation applied in this case study assumes a normal distribution, which might be an inaccurate description of the true uncertainty distribution. In the following case study, we investigate the performance of the analytical reformulation (B1) in terms of observed violation probabilities. We further compare the analytical reformulation (B1) with the sample-based methods to define the uncertainty margins, based on the empirical quantiles of the Monte Carlo approach (B2) and the uncertainty margins based on the scenario approach (B2). The comparisons are based on the RTS96 system, and the AC OPF within each iteration is solved using Matpower with MIPS.

In-Sample Test of the Analytical Reformulation

To assess how well the linearized model represents the actual impact of the uncertainty, we draw 10'000 samples from a multivariate normal distribution which matches the assumed distribution (i.e., we assume

Table 5.3: In-sample testing of the performance of the analytical chance constraint reformulation (B1). The maximum empirical violation probability ϵ_{emp} observed for any individual chance constraint, as well as the number of samples with at least one constraint violation, corresponding to the joint violation probability. In the upper table, where σ_W is varied, $\epsilon_I = \epsilon_V = 0.01$. In the lower table, where $\epsilon = \epsilon_I = \epsilon_V$ is varied, $\sigma_W = 0.1$.

	$\sigma_W = 0.075$	$\sigma_W = 0.1$	$\sigma_W = 0.125$
Max. ϵ_{emp}	0.011	0.013	0.017
# of Violated	647	650	812
Samples	(6.5%)	(6.5%)	(8.1%)
	$\epsilon = 0.01$	$\epsilon = 0.05$	$\epsilon = 0.1$
Max. ϵ_{emp}	0.013	0.044	0.092
# of Violated	650	1370	2194
Samples	(6.5%)	(13.7%)	(21.9%)

perfect knowledge of the distribution). Based on these samples, we assess the empirical violation probabilities through a Monte Carlo simulation. We perform the assessment for different standard deviations $\sigma_W = \{0.075, 0.1, 0.125\}$ and different acceptable violation probabilities $\epsilon_V = \epsilon_I = \{0.01, 0.05, 0.1\}$. Different values of the standard deviation σ_W allow us to assess the performance of the method with fluctuations of different size, where larger standard deviations imply fluctuations further away from the linearization point. With different violation probabilities ϵ , we assess the accuracy of predictions for different parts of the distribution (smaller ϵ imply that we are further into the tail), and for quantiles that are closer (small ϵ) or further (large ϵ) away from the operating point. The acceptable violation probabilities of the generator constraints on active and reactive power are kept constant at $\epsilon_P = \epsilon_Q = 0.01$, as these constraints are not influenced by the nonlinearities of the power flow equations.

The results of the in-sample testing are shown in Table 5.3. If the analytical reformulation is accurate, the maximum empirical violation probability for a single chance constraint, denoted by ϵ_{emp} should be close to

the desired violation probability $\epsilon = 0.01$. The method performs quite well with the smallest standard deviation $\sigma_W = 0.075$, but gets worse as the standard deviation increases. This is as expected, as the linearization becomes less accurate for larger deviations. We further observe that the method performs better for larger accepted violation probabilities $\epsilon = 0.05$ and 0.1 . On the one hand, this might be because the quantiles corresponding to $\epsilon = 0.05$ and 0.1 are closer to the nominal operating point, in a region where the linearization is more accurate. On the other hand, a normal distribution might be a better approximation for the mid-part of the distribution than for the tails, hence providing better results when ϵ is not too small.

We further observe that the AC CC-OPF limits the total number of violated samples to a relatively small percentage, even though it only aims at enforcing the separate chance constraints.

Out-Of-Sample Test of the Analytical Reformulation

The analytical reformulation assumes a normal distribution, while the true distribution of the forecast errors might be different. Therefore, we also include an out-of-sample test with uncertainty samples based on the historical data from APG described in Section 2.8. We assign one set of historical samples, in total 8492 data points, to each uncertain load. The samples are then rescaled to match the assumed standard deviation σ_W and assumed correlation coefficient $\rho = 0$. Since the number of available samples are limited, we base the Monte Carlo simulation on 5000 samples.

The results of the out-of-sample testing based on APG historical data is shown in Table 5.4. Again, the maximum empirical violation probability for a single chance constraint, denoted by ϵ_{emp} should be close to the desired violation probability $\epsilon = 0.01$. We observe that the analytical reformulation based on the assumption of a normal distribution is still performing reasonably well, although the highest observed violation probability is 0.02 instead of 0.01 . Different from the in-sample testing, the magnitude of the standard deviation does not determine how well the method performs. However, we again observe that the method is better at enforcing the chance constraints with larger acceptable violation probabilities $\epsilon = 0.05, 0.1$. The total number of violated samples is slightly higher than in the in-sample test, but remain in the same range.

Table 5.4: Out-of-sample testing of the performance of the analytical chance constraint reformulation (B1). The maximum empirical violation probability ϵ_{emp} observed for any individual chance constraint, as well as the number of samples with at least one constraint violation, corresponding to the joint violation probability. In the upper table, where σ_W is varied, $\epsilon_I = \epsilon_V = 0.01$. In the lower table, where $\epsilon = \epsilon_I = \epsilon_V$ is varied, $\sigma_W = 0.1$.

	$\sigma_W = 0.075$	$\sigma_W = 0.1$	$\sigma_W = 0.125$
Max. ϵ_{emp}	0.017	0.014	0.020
# of Violated	374	369	461
Samples	(7.5%)	(7.4%)	(9.2%)
	$\epsilon = 0.01$	$\epsilon = 0.05$	$\epsilon = 0.1$
Max. ϵ_{emp}	0.014	0.054	0.093
# of Violated	369	724	1165
Samples	(7.4%)	(14.5%)	(23.3%)

Comparison with Other Reformulations

Finally, we compare the different iterative solution approaches (B1) with analytical uncertainty margins and $\epsilon = 0.01$, (B2) with empirical uncertainty margins based on a Monte Carlo simulation and $\epsilon = 0.01$, and (B3) with uncertainty margins based on the scenario approach and an acceptable joint violation probability of $\epsilon = 0.1$. Within the optimization algorithm, we use 1000 samples to calculate the Monte Carlo margins, while the scenario approach prescribes the use of at least $N_S = 2465$ samples. For the a-posteriori Monte Carlo simulation to evaluate the violation probabilities, we do an out-of-sample test using 5000 different samples from the historical data.

In Table 5.5, the generation cost, solution time and number of iterations are listed. We also show the maximum empirical violation probability ϵ_{emp} and the total number of samples with violations. When comparing the cost of the solutions, as well as the empirical violation probabilities, it is important to remember that the sample-based solutions (B2) and (B3) depend on the samples that are drawn. If we would resolve the

Table 5.5: Comparison of the iterative solution approach with (B1) analytical uncertainty margins and $\epsilon = 0.01$, (B2) empirical uncertainty margins based on a Monte Carlo simulation and $\epsilon = 0.01$, and (B3) uncertainty margins based on the scenario approach and an acceptable joint violation probability of $\epsilon = 0.1$.

	Analytical (B1)	Monte Carlo (B2)	Scen. Approach (B3)
Cost	40 127 (+9.1%)	39 922 (+8.6%)	41 455 (+12.7%)
Max. ϵ_{emp}	0.014	0.019	0.003
# of Violated Samples	369 (7.4%)	502 (10.0%)	35 (0.7%)
Time	0.54s	2min 7s	6min 12s
Iterations	5	5	6

problem with a different set of samples, we would obtain a different solution. While we would like to check the solution by resolving it several times, we refrain from further simulations due to the limited amounts of available data and the large number of samples necessary for the scenario approach.

We observe that the analytical and Monte Carlo approaches lead to relatively similar solutions, although the Monte Carlo solution is lower. This decrease in cost does however come at the expense of higher violation probabilities and larger number of violated samples. It is interesting to see that the Monte Carlo approach, despite being calculated based on the full AC power flow equations and without any assumptions on the uncertainty distribution, actually performs worse than the analytical reformulation in terms of satisfying the upper limit on the violation probability ϵ_{emp} of the chance constraint. A larger number of samples could be used to obtain a better solution, but since we run a power flow for each sample in each iteration, a larger sample size would also increase computational time. The Monte Carlo already requires a lot more computational resources than the analytical reformulation, with a solution time of 2min 7s compared with 0.5s.

The scenario approach has a higher cost than the other two solutions, but also significantly lower violation probabilities. Actually, it oversatisfies the joint chance constraint by an order of magnitude. The limit on the acceptable, joint violation probability was set to $\epsilon_J = 0.1$, whereas the empirical, joint violation probability is less than 1 %. As a matter of fact, none of the solutions violates the limit on the joint probability, although the analytical and Monte Carlo approaches have lower cost. This shows one of the drawbacks of the scenario approach: While the solution guarantees feasibility of the chance constraint, the solution can be far from cost optimal. The sample bound for the scenario approach given in [100] is for general, convex problems. By using information about the specific structure of the problem, it might be possible to obtain tighter bounds that provide tighter bounds with fewer samples. However, as we have seen in Chapter 2, it is hard to obtain a tight reformulation without assuming at least some properties of the underlying distribution.

5.6 Conclusions

In this chapter, we described an approximate formulation for the AC CC-OPF. The nominal operating point is represented using the full AC power flow equations, and the impact of uncertainty is accounted for through a linearization around the operating point. With the linearization, we can express the changes in voltage and current magnitudes, as well as active and reactive power production as linear functions of the fluctuations. This allows us to apply the previously presented analytical reformulation of the chance constraints, hence providing a reasonably accurate and tractable formulation of the AC CC-OPF.

Different methods to solve the problem were proposed and discussed in the case study. The one-shot solution approach allow us to co-optimize the reaction to the fluctuations, which reduces cost, but also increases computational complexity. With the current implementation of the one-shot approach, the optimization did not converge for systems larger than the RTS96. However, improvements in the implementation and development of better solution algorithms can be expected to significantly improve computation time and scalability.

As an alternative to the one-shot approach, we suggest an iterative solution method, where the chance constraints are enforced through an outer iteration on the deterministic AC OPF. The iterative solution

method has several advantages. On the one hand, we are able to leverage the solution speed of existing, deterministic AC OPF solvers, which makes the problem easily scalable to large scale systems with a significant number of uncertainty sources. On the other hand, the iterative approach opens up for alternative definitions of the uncertainty margins, such as empirical evaluation based on a Monte Carlo simulation or the definition of uncertainty margins that allow us jointly enforce the chance constraints.

In the case study, we assessed the performance of the method based on four test systems, the IEEE RTS96 system, the IEEE 118 bus and 300 bus systems, as well as the Polish Winter Peak test case with 2383 buses. For all test cases, a solution could be obtained within seconds to half a minute using the iterative solution approach with Matpower. The analytical reformulation based on the assumption of normally distributed uncertainty was shown to perform reasonably well in both in-sample and out-of-sample tests. In a comparison between the analytical reformulation, a Monte Carlo based reformulation and a reformulation based on the scenario approach, the analytical reformulation outperformed the Monte Carlo reformulation in terms of meeting the violation probability of the individual chance constraints, and the scenario approach in terms of cost.

Part II

Risk-Based Optimal Power Flow With Uncertainty

Chapter 6

Limiting the Probability of High Risk Operation

In power systems operation, not only the probability of constraint violations, but also the impact of the violation on system security matters in evaluating risk. In the following, we extend our risk modeling from the chance-constrained OPF with N-1 security constraints in the previous part of the thesis towards a more comprehensive risk model, where we consider risk as the product between probability of an event and the a measure of its impact. We propose a risk model for risks related to outages, which accounts for available remedial measures and the impact of cascading events. The new risk model is used to formulate risk-based constraints for the post-contingency line flows, which are included in an OPF formulation. Forecast uncertainty is accounted for by enforcing the relevant constraints as a joint chance constraint, which is reformulated using a sampling-based technique. In a case study of the IEEE 30 bus system, we demonstrate how the proposed risk-based, probabilistic OPF allows us to control the risk level, even in presence of uncertainty. We investigate the trade-off between generation cost and risk level in the system, and show how accounting for uncertainty leads to a more expensive, but more secure dispatch.

6.1 Motivation and Related Work

In our modelling, we distinguish between two types of disturbances in power system operations, *random outages* and *forecast uncertainty*, which have inherently different characteristics. Whereas outages can be characterized as discrete events with a (usually) low probability, forecast uncertainty (deviations in power in-feeds arising from load, RES or short-term trading) is characterized by a continuous probability distribution. The method proposed in this chapter addresses both types of disturbances. Random outages are handled by a risk-based extension of the N-1 criterion. The risk-based N-1 security constraints utilize additional information about the probability of outages, the extent of post-contingency violations and the cost and availability of remedial actions, and thus provide a more quantitative measure of power system security. Forecast uncertainty is accounted for using a chance-constrained formulation of the OPF. The resulting formulation allows us to limit the risk of outages, even in presence of forecast uncertainty.

There exist two main approaches to model risk in power system operation.

On the one hand, risk can be modeled through overall reliability parameters like Expected Energy Not Served (EENS). These parameters incorporate the effect of cascading events and reflect the risk faced by the customers in the system. However, computing the risk requires extensive calculations (i.e., Monte-Carlo simulations), and these types of risk measures are therefore typically used to analyze the risk for a given operating condition [24], [25], as opposed to inclusion in an optimization problem.

On the other hand, risk can be modeled in terms of violation of technical limits, e.g., dependent on the power flow of a line or on the voltage magnitude. Such risk measures typically consider the situation after an N-1 outage, and do not simulate how a potential cascade would develop further. Thus, these risk measures do not reflect the full risk of cascading events, but are much easier to compute. Further, when a risk measure is related to specific components, it is easier for the system operator to identify actions to influence the risk. This type of risk measures have therefore often been proposed for incorporation in OPF formulations. The OPF formulations in [106], [107] describe risk as violations and near-violations of voltage limits and line transfer capacities, modeled as linear functions of the voltage magnitude and the line flow. In [28], risk

is expressed as a quadratic function of the line flow, whereas [26] models risk as the cost of equipment aging in function of, e.g., the line flow or voltage magnitude. Here, risk is expressed as a piecewise affine function of the line flow. The parameters of the risk function are computed based on the cost and availability of remedial measures, and also reflects the risk of initiating a cascading event.

The proposed OPF is formulated as a central dispatch problem minimizing overall generation cost, similar to the OPF problems solved in markets with Locational Marginal Pricing (LMP). Recently, [108] investigated how a risk-based OPF impacts the LMPs, showing that the risk-based OPF introduces a new cost component which reflects the system risk level. While [108] also uses piecewise affine risk function, no systematic argumentation for the choice of the risk function parameters is provided. In contrast, the parameters introduced in this chapter are based on actual system properties, providing a less arbitrary definition of the risk function.

Although the OPF is formulated as a central dispatch problem, the proposed method does not focus on the market clearing aspect, but rather on ensuring that the dispatch has a sufficiently low level of risk. Therefore, it can also be applied in self-dispatch markets by changing the objective to minimize changes to the market outcome instead of minimizing overall generation cost. For security considerations, the relation between the risk function and the cost and availability of remedial measures is particularly interesting. All risk-based formulations [106, 107, 28, 26, 108] allow for post-contingency line overloading under some circumstances, but do not provide remedial actions to bring the system back to normal operation. Since the parameters of the risk function proposed here are calculated based on the cost and availability of remedial actions, the method proposes effective post-contingency remedial actions, ensuring that the system can be brought back within operational limits when a contingency occurs.

Although several risk-based OPF formulations exist, few of them account for forecast uncertainty in a comprehensive way. The OPF formulation in [26] considers normally distributed load uncertainty, and limits the expected value of the risk. In contrast, the method proposed here guarantees that the risk limit will hold with a chosen probability. This is achieved by formulating a chance-constrained optimization problem, following along the lines of [14, 44]. The problem is solved using the randomized optimization technique proposed in [55], based

on the scenario approach from [54]. This technique requires no assumptions on the distribution of the forecast errors, beyond the availability of samples.

The remainder of this chapter is organized as follows. First, we define a risk function based on system parameters such as available remedial measures. We then introduce risk-based constraints for post-contingency line flows, and show how to choose appropriate upper bounds on the risk by comparing the risk-based constraints to the deterministic N-1 constraints. Further, we formulate a chance-constrained DC OPF with risk-based constraints. In a case study for the IEEE 30-bus system, we analyze the proposed formulation and compare it with other OPF formulations with regards to cost, risk level, and number of post-contingency overloads. An additional sensitivity study investigates the relationship between remedial actions and accepted post-contingency overloads, as well as the proposed remedial actions.

The material presented here is based on the publications [19, 31].

6.2 Risk Modelling for Post-Contingency Overloads

A risk measure for use in within the OPF problem must provide a clear relation between the risk and the decision variables, and must be inexpensive to compute to maintain computational tractability. In the following, we present a risk model which focuses on risk as a function of post-contingency transmission line loading. Previous risk formulations (e.g., [107], [28]) use the same severity function for all transmission lines independent of which contingency has taken place. Here, we improve this formulation in two ways. First, we explicitly account for different types of risk (such as moderate overloads that can be mitigated through redispatch, or high overloads that might lead to cascading events) by formulating a piecewise linear severity function. Second, we formulate the severity function separately for each line and contingency, which allows us to account for the effect of available remedial measures on each line in each post-contingency situation. The proposed risk formulation sets post-contingency line flow limits based on the available remedial measures and potential impacts of cascading events. This allows for less restrictive limits than the traditional N-1 criterion, but ensures that

effective remedial measures are available in the cases where we allow for post-contingency overloads.

6.2.1 Definition of the risk measures

A risk measure for post-contingency overloads should reflect both the probability of an outage and the severity of the resulting operating condition. The risk related to a specific line ij and outage $k \in \mathcal{K}$, where $\mathcal{K}$ is the set of considered outages, is expressed as

$$\mathcal{R}_{(k,ij)}^{spec} := \mathcal{P}_{(k)} \cdot \mathcal{S}_{(ij|k)}, \quad (6.1)$$

where $\mathcal{P}_{(k)}$ is the probability of outage k and $\mathcal{S}_{(ij|k)}$ is the severity of the operating condition on line ij given outage k . This expression can be seen as the risk-based counterpart of the N-1 criterion, as it describes the risk for a specific line in one specific post-contingency state. Using $\mathcal{R}_{(k,ij)}^{spec}$ as a basis, we define:

$$\mathcal{R}_{(k)}^{out} := \sum_{ij \in \mathcal{L}} \mathcal{P}_{(k)} \cdot \mathcal{S}_{(ij|k)} \quad (6.2)$$

$$\mathcal{R}_{(ij)}^{line} := \sum_{k \in \mathcal{K}} \mathcal{P}_{(k)} \cdot \mathcal{S}_{(ij|k)} \quad (6.3)$$

$$\mathcal{R}^{tot} := \sum_{k \in \mathcal{K}} \sum_{ij \in \mathcal{L}} \mathcal{P}_{(k)} \cdot \mathcal{S}_{(ij|k)} \quad (6.4)$$

$\mathcal{R}_{(k)}^{out}$ expresses the risk after an outage k , and is obtained by summing the risk of all lines ij in this post-contingency state. $\mathcal{R}_{(ij)}^{line}$ is the risk related to line ij , summed over all outages k . $\mathcal{R}^{tot}$ is the total risk in the system, summed over all outages k and all lines ij .

In order to evaluate (6.1), the outage probabilities $\mathcal{P}_{(k)}$ must be estimated, and the severity $\mathcal{S}_{(ij|k)}$ has to be defined. We assume that the outage probabilities are calculated a-priori (e.g., based on historical data and current weather conditions [109], [27]) and given as an input to the optimization. The severity modeling is described in the next section.

6.2.2 Severity modeling

To capture different types of risk arising from different levels of post-contingency line loading, we define the severity $\mathcal{S}_{(ij|k)}$ as a piecewise

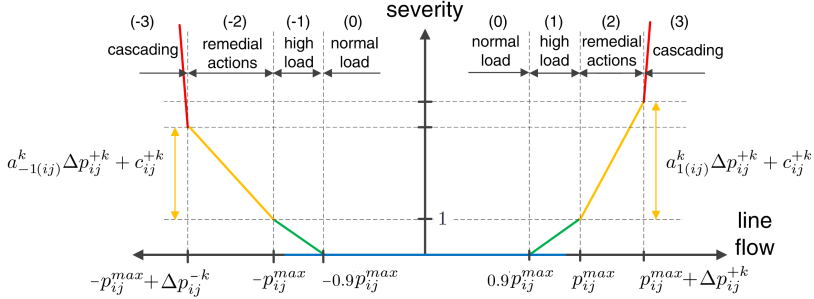


Figure 6.1: Piecewise linear severity function for line ij after outage k . Notice that the function is not symmetric. In particular, it has different slope in the third (yellow) segment corresponding to flows that require remedial actions.

linear function of the line flow. We define four different segments for the severity function. The four segments are shown in Fig. 6.1, and correspond to 0) normal load, 1) high load, 2) moderate overload which requires remedial actions and 3) cascading overload which might lead to a cascading event.

For the derivation of the severity function parameters, we consider the outage of any of the l lines or m generators, such that the set of contingencies $\mathcal{K} = \mathcal{L} \cup \mathcal{G}$. Although we restrict ourselves to those N-1 outages, any other outage situations, e.g. N-2 or common mode outages, can be included without any change to the method. In the following, the post-contingency line flows on line ij after outage k is denoted by p_{ij}^k . Mathematically, we define the piecewise linear severity function as the pointwise maximum over a set of affine functions of the line flow,

$$\mathcal{S}_{(ij|k)}(p_{ij}^k) := \max_j \{a_{n(ij)}^k p_{ij}^k + b_{n(ij)}^k\}, \quad n \in \{-3, \dots, 3\} \quad (6.5)$$

where $a_{n(ij)}^k, b_{n(ij)}^k$ are parameters for the n^{th} segment belonging to line ij and outage k . Formulating the severity function like this ensures convexity of the optimization problem. We will now give a physical interpretation of the different severity zones, and explain how to compute the parameters $a_{n(ij)}^k$ and $b_{n(ij)}^k$ with $n = 0, \dots, 3$ for positive line flows. The severity function is not symmetric in general, but the parameters for negative line flows with $n = -3, \dots, 0$ can be calculated in an analogous way.

Normal load

At low load ($p_{ij}^k < 0.9 \cdot p_{ij}^{max}$), the severity is assumed to be zero. This segment of the severity function, marked blue in Fig. 6.1, connects the two points

$$\mathcal{S}_{(ij|k)}(0) = 0 \quad (6.6)$$

$$\mathcal{S}_{(ij|k)}(0.9 \cdot p_{ij}^{max}) = 0, \quad (6.7)$$

with parameters $a_{0(ij)}^k = b_{0(ij)}^k = 0$.

High load

High loading ($0.9 \cdot p_{ij}^{max} < p_{ij}^k < p_{ij}^{max}$) might lead to a failure of the line (e.g., due to relay malfunction), and the severity is therefore non-zero. We choose $a_{1(ij)}^k$ and $b_{1(ij)}^k$ such that the "high load" segment of the severity function connects the two points given by Eq. (6.7) and (6.8). This segment is marked green in Fig. 6.1.

$$\mathcal{S}_{(ij|k)}(p_{ij}^{max}) = 1. \quad (6.8)$$

Moderate overload

Many system operators allow a transmission line overload for a short time (e.g., for a short period after a contingency). However, if the operating limit is violated, removing the overload requires remedial actions like transmission switching or redispatch. Here, we define moderate overload as one which can be relieved within a given time frame (e.g., 5 minutes). The corresponding severity zone goes from $p_{ij}^{max} < p_{ij}^k < (p_{ij}^{max} + \Delta p_{ij}^{+k})$, where we define Δp_{ij}^{+k} as the largest achievable line flow reduction for line k after outage i . Δp_{ij}^{+k} is a function of the available remedial measures (e.g., available redispatch). The effectiveness of a remedial measure depends on the location of a line in the system, topology changes (e.g., after a line outage), or changes in availability of remedial measures (e.g., after the outage of a generator offering redispatch). Therefore, Δp_{ij}^{+k} must be calculated separately for each line and each outage.

The increase in risk from p_{ij}^{max} to $(p_{ij}^{max} + \Delta p_{ij}^{+k})$ depends on both the inherent risk of overloading the line and the cost of removing this overload. Allowing for a temporary overload always involves an inherent

risk, because the probability of line trip increases due to increased line sag and higher probability of relay malfunction, and because the remedial actions might not work as planned such that the line will stay overloaded longer than expected. This inherent risk of overloading is modelled in a similar way as for the high load case, with the same slope $a_{1,k}^i$. In addition, allowing for overloads incurs an additional cost, since we need to implement remedial actions. The cost of reducing the line flow by Δp_{ij}^{+k} is denoted by c_{ij}^{+k} , and can be a function of, e.g., the cost of a redispatch measure. Some remedial measures, such as switching measures, does not incur any additional cost (i.e., $c_{ij}^{+k} = 0$).

If both Δp_{ij}^{+k} and c_{ij}^{+k} are known, the severity function parameters $a_{2(ij)}^i$ and $b_{2(ij)}^i$ are chosen such that the severity function connects the two points given by (6.8) and (6.9). This segment is marked yellow in Fig. 6.1.

$$\mathcal{S}_{(ij|k)}(p_{ij}^{max} + \Delta p_{ij}^{+k}) = 1 + a_{1(ij)}^k \Delta p_{ij}^{+k} + c_{ij}^{+k} . \quad (6.9)$$

The term $a_{1(ij)}^i \Delta p_{ij}^{+k}$ in (6.9) represent the inherent risk of overloading, and c_{ij}^{+k} the cost of the remedial action. Note that the severity function (6.5) is convex for all remedial actions, since $a_{2(ij)}^i \geq a_{1(ij)}^i$.

In general, there are many ways to consider remedial measures in this formulation. We now suggest one way of defining Δp_{ij}^{+k} and c_{ij}^{+k} with the following choice: (i) We only consider generation redispatch. (ii) We only allow one redispatch measure (i.e., one shift between two generators) per line and contingency to ensure fast and easy implementation in real time. (iii) We choose the redispatch measure that leads to the largest flow reduction, without consideration of cost.

We further assume that the available positive and negative redispatch capacity at each generator is known and given by the vectors $p_R^+, p_R^- \in \mathbb{R}_{+0}^m$, with $p_{R,i}^+$ denoting the positive redispatch capability of generator i . The corresponding redispatch costs are also assumed to be known and are given by $c_R^+, c_R^- \in \mathbb{R}^m$. To compute the largest possible reduction of the line flow Δp_{ij}^{+k} , we first need to compute the largest possible shift in generation $\Delta p_{g \rightarrow h}$ from any generator g to any other generator h . This is given by

$$\Delta p_{g \rightarrow h} = \min\{p_{R,g}^+, p_{R,h}^-\} . \quad (6.10)$$

To describe how much a generation shift from g to h decreases the power flow on line ij , we use the power transfer distribution factor $PTDF_{ij,g \rightarrow h}$ [86]. The largest power flow decrease Δp_{ij}^{+k} can then be

calculated as

$$\Delta p_{ij}^{+k} = -\min_{g,h}\{PTDF_{ij,g \rightarrow h}^k \Delta p_{g \rightarrow h}^k\}, \quad (6.11)$$

since Δp_{ij}^{+k} is defined as a positive value. Note that both $PTDF_{k,g \rightarrow h}^k$ and $\Delta p_{g \rightarrow h}^k$ depend on the outage k . For a generator outage g , the topology remains unchanged, but the redispatch availability of generator g is lost. Thus, $\Delta p_{g \rightarrow h}^k$ might be different from normal operation since $p_{R,g}^{+k} = p_{R,g}^{-k} = 0$. For a line outage k , the redispatch availability remains unchanged, but the topology changes. Thus, we need to recalculate $PTDF_{k,g \rightarrow h}^i$ for line outages.

When we know which generation shift $\Delta p_{g \rightarrow h}^k$ that lead to the largest line flow decrease Δp_{ij}^{+k} , the cost c_{ij}^{+k} of the redispatch measure is given by

$$c_{ij}^{+k} = (c_{R,g}^{+} + c_{R,h}^{-}) \Delta p_{g \rightarrow h}. \quad (6.12)$$

With this result, we can calculate $a_{2(ij)}^k$ and $b_{2(ij)}^k$ as the affine function connecting (6.8) and (6.9).

Cascading overload

Short term overloads might be acceptable up to a certain point, but should not be allowed to evolve into cascading events. Here, we consider two types of cascade initiation. First, any overload that cannot be removed by remedial actions might eventually lead to a line trip (delayed trip due to time settings in the relays or flash-over due to sag). We therefore define line loadings higher than $p_{ij}^{max} + \Delta p_{ij}^{+k}$ as cascading overload. Second, overloads above a certain threshold (e.g., $p_{ij}^k > 1.2 \cdot p_{ij}^{max}$) might lead to immediate trip of the line. To avoid this, we introduce a limit for Δp_{ij}^{+k} (e.g., $\Delta p_{ij}^{+k} \leq 0.2 \cdot p_{ij}^{max}$).

Since the potential impacts of a cascading event can be very high, the severity increases very rapidly as we enter the cascading zone. The increase in severity per additional MW loading is denoted by ρ_C . The severity function parameters $a_{3(k)}^i$ and $b_{3(k)}^i$ are chosen such that the severity function connects the two points given by (6.9) and (6.13). This part of the severity function is marked red in Fig. 6.1.

$$\mathcal{S}_{(ij|k)}(p_{ij}^{max} + \Delta p_{ij}^{+k} + 1) = 1 + a_{2(ij)}^k \Delta p_{ij}^{+k} + \rho_C \quad (6.13)$$

There are many possible ways of defining the value of ρ_C . On the one hand, if $\bar{\rho}_C \rightarrow \infty$, no violation of the limit $p_{ij}^{max} + \Delta p_{ij}^{+k}$ is allowed. On the other hand, if contingency analysis show that an outage of line ij following the outage k would not have an adverse effect on the system as a whole, we can choose $\underline{\rho}_C = a_{2(ij)}$. This lower bound ensures that $\rho_C \geq a_{2(k)}$, which is required for the convexity of the severity function (6.5). Thus, we get that ρ_C can be chosen anywhere in the range $\underline{\rho}_C \leq \rho_C \leq \bar{\rho}_C$. Here, we choose $\rho_C = \max\{a_C, a_{2(k)}\}$, where choosing a high number a_C guarantees that the risk increases rapidly when the line loading exceeds $p_{ij}^{max} + \Delta p_{ij}^{+k}$.

6.2.3 Risk Constraints

Since the risk is modelled based on the post-contingency line flows, we can formulate risk-based constraints for these line flows. For other quantities, i.e. line flows in normal operation or generator outputs, we keep the standard constraints in which the line flows are limited by the steady-state value and the generators by their technical generation maximum.

Formulation of risk constraints

Based on the risk measures defined by (6.1) to (6.4), we can formulate constraints to limit the risk:

$$\mathcal{R}^{spec} : \mathcal{P}_{(k)} \cdot \mathcal{S}_{(ij|k)}(p_{ij}^k) \leq \bar{\mathcal{R}}_{(k)} \quad (6.14)$$

$$\mathcal{R}^{out} : \sum_{i=1}^{N_{out}} \mathcal{P}_{(k)} \cdot \mathcal{S}_{(ij|k)}(p_{ij}^k) \leq \bar{\mathcal{R}}^{out} \quad (6.15)$$

$$\mathcal{R}^{line} : \sum_{k=1}^{N_l} \mathcal{P}_{(k)} \cdot \mathcal{S}_{(ij|k)}(p_{ij}^k) \leq \bar{\mathcal{R}}^{line} \quad (6.16)$$

$$\mathcal{R}^{tot} : \sum_{i=1}^{N_{out}} \sum_{k=1}^{N_l} \mathcal{P}_{(k)} \cdot \mathcal{S}_{(ij|k)}(p_{ij}^k) \leq \bar{\mathcal{R}}^{tot} \quad (6.17)$$

Eq. (6.14) constrains the risk for each line ij to stay below a constant limit $\bar{\mathcal{R}}_{(k)}$ after the outage k . Eq. (6.15) limits the risk of outage k , while (6.16) limits the risk of line ij and (6.17) limits the total risk in the system.

Relation to N-1 constraints

Including the constraints (6.14) - (6.17) in the optimization problem allows us to control the risk level in the system according to our preferences, but also requires us to define the risk limits $\overline{\mathcal{R}}$ in a reasonable way. To provide a physical interpretation of the risk limits, we relate the risk-based constraints (6.14) to the traditional N-1 constraints. This relation is then used to propose reasonable values for the risk limits, and to show how the choice of the risk limits influences the accepted post-contingency flows.

Assuming that the outage probability $\mathcal{P}_{(k)}$ is given as an input to the optimization, (6.14) can be reformulated as an upper bound on the severity

$$\mathcal{S}_{(ij|k)}(p_{ij}^k) \leq \frac{\overline{\mathcal{R}}_{(k)}}{\mathcal{P}_{(k)}}. \quad (6.18)$$

If the risk limit is chosen equal to the probability of the outage, $\overline{\mathcal{R}}_{(k)} = \mathcal{P}_{(k)}$, the risk constraint (6.18) reduces to

$$\mathcal{S}_{(ij|k)}(p_{ij}^k) \leq 1. \quad (6.19)$$

The severity function, as defined in Section II.B, takes the value $\mathcal{S}_{(ij|k)}(p_{ij}^{max}) = 1$ at the line capacity limit p_{ij}^{max} , and $\mathcal{S}_{(ij|k)}(p_{ij}) \leq 1$ when $p_{ij}^k \leq p_{ij}^{max}$. Thus, (6.19) is equivalent to the traditional N-1 constraint $p_{ij}^k \leq p_{ij}^{max}$.

To enforce the traditional N-1 criterion for the overall system, the upper bound on the severity must be equal to 1 for all risk constraints (6.14). To achieve this, the risk limit must be set to $\overline{\mathcal{R}}_{(k)} = \mathcal{P}_{(k)}$ for all outages i . This choice implies that the accepted risk level $\overline{\mathcal{R}}_{(k)}$ is different for outages with different outage probabilities $\mathcal{P}_{(k)}$.

This discussion highlights two important characteristics of the traditional N-1 criterion. First, the N-1 criterion corresponds to an implicit risk level, which might be different from the desired limit on risk. Second, the N-1 criterion leads to a situation where the accepted risk level differs between outages, because the outage probabilities are not accounted for. These drawbacks of the N-1 criterion can be mitigated by enforcing risk-based constraints with an appropriately chosen, consistent risk limit. The value of this risk limit should be in the same order of magnitude as the outage probabilities, to obtain a risk level which is not too different from the traditional N-1 situation.

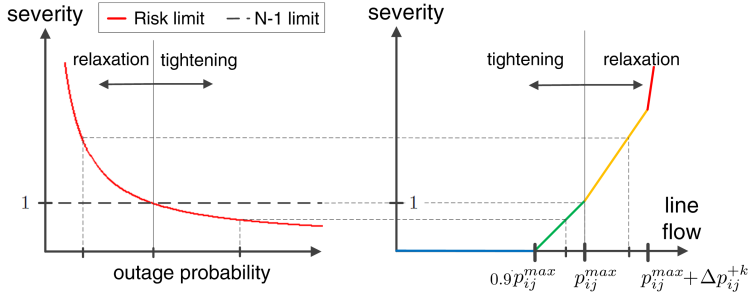


Figure 6.2: Left: Comparison between the risk-based severity limit (red line), and the N-1 severity limit (black line). Right: Tightening and relaxation of the line flow constraint as a function of the severity limit and the line and outage specific severity function.

If the risk limit $\overline{\mathcal{R}}_{(k)}$ is chosen to be the same for all outages i , the upper bound on the severity depends on the outage probability $\mathcal{P}_{(k)}$ and a higher severity $\mathcal{S}_{(ij|k)}$ is accepted for outages with low probability. This is illustrated in Fig. 6.2 (left), where the red line is the severity limit for a given value of $\overline{\mathcal{R}}_{(k)}$ and the black dashed line is the severity limit for the N-1 criterion (which is constant equal to 1, independent of the outage probability). Since the severity is a function of the post-contingency line flow, accepting a higher (or lower) severity relaxes (or tightens) the constraint on the post-contingency line flow. The amount of relaxation (or tightening) depends on the severity function, as shown in Fig. 6.2 (right).

6.3 Risk-based Optimal Power Flow with Probabilistic Guarantees

This section introduces the mathematical formulation of the RB-CC-SCOPF, with risk-based constraints for the post-contingency line flows and a joint chance constraint to account for the forecast uncertainty. The objective is to find the minimal cost dispatch that satisfies the tolerated risk level for all outages as well as the desired violation level for the chance constraint. The setup is similar to the probabilistic SCOPF described in [14], using a joint chance constraint to handle uncertainty. However, the security constraints for the post-contingency

line flows are substituted with the proposed risk-based constraints. The power flow modelling is the same as in Chapter 2. To balance the system, we assume an affine balancing policy with a predefined α according to (2.7), (2.8).

6.3.1 Objective and Constraints

The resulting optimization problem is given by

$$\min_{p_G} \sum_{i \in \mathcal{G}} (c_{2,i} p_{G,i}^2 + c_{1,i} p_{G,i}) \quad (6.20)$$

s.t.

$$\sum_{i \in \mathcal{N}} p_{G,i} - d_i + u_i = 0 \quad (6.21)$$

$$\mathbb{P} \left(\begin{array}{c} -p_{ij}^{max} \leq \mathbf{M}_{(ij, \cdot)}(p_G - \alpha\omega - d + u + \omega) \leq p_{ij}^{max} \\ p_G^{min} \leq p_G - \alpha\Omega \leq p_G^{max} \\ p_G^{min} \leq p_G^k - \alpha^k(\Omega + p_{G,k}) \leq p_G^{max}, \forall k \in \mathcal{K} \\ \mathcal{R} \leq \bar{\mathcal{R}} \end{array} \right) \geq 1 - \epsilon_J \quad (6.22)$$

Eq. (6.20) and (6.21) define the objective function and the power balance constraints, with $c_1, c_2 \in \mathbb{R}^{N_G}$ being the linear and quadratic cost coefficients. Eq. (6.22) describes a probabilistic constraint, stating that all inequalities within the brackets must hold jointly with a probability of at least $1 - \epsilon_J$, where ϵ_J is the joint violation probability, which we will refer to as the violation level. The first inequality is the line flow constraints for normal operating conditions, enforced using the standard line flow limits. The second and third inequalities enforce capacity limits for the generators during normal operation and after generator outages. The last inequality describes one or more risk limitations for the post-contingency line flows, which can be chosen from the options proposed in Section 6.2.3.

Note that the problem remains convex after introduction of the risk constraints. Inserting the expressions for the power flows in (6.5), $\mathcal{S}_{(ij|k)}(p_{ij}^k)$ can be expressed as a the pointwise maximum over a set of affine functions with p_G as an argument. $\mathcal{P}_{(i)} \cdot \mathcal{S}_{(ij|k)}(p_{ij}^k)$ is hence convex with respect to p_G . This means that all the risk measures are sums of convex functions and hence convex themselves.

6.3.2 Chance Constraint Reformulation

To achieve a tractable optimization problem, we need to reformulate the joint chance constraint (6.22) to an equivalent deterministic constraint. One possibility is to use the so-called scenario approach [54], also discussed in Chapter 5, to deal with the chance constraint. According to this method we replace the chance constraint with a finite number of hard constraints, each corresponding to a different scenario for our uncertain variables. No assumptions or knowledge regarding the underlying probability distribution are required. In [54], [110] implicit and explicit sample size bounds are provided so that the resulting solution satisfies the chance constraint with a chosen confidence. The number of necessary samples are related to the number of decision variables, in this case the number of generators m , as well as the chosen violation level ϵ_J and the confidence level β . The confidence level β is a secondary violation probability, defining the probability that the chance constraint might still be violated, despite using the given number of samples. In this chapter, we follow the probabilistically robust approach, a randomized optimization technique proposed in [55]. The method includes two steps.

In the first step, we solve an optimization problem to determine, with a confidence of at least $1 - \beta$, a set $\mathcal{D}$ that contains at least $1 - \epsilon_J$ probability mass of the distribution of the uncertain variables ω . This set is defined using the scenario approach, where the number of required scenarios is related to the number of uncertain variables, in our case the number of uncertain in-feeds m . The number of scenarios is given by [54]

$$N_S \geq \frac{1}{\epsilon_J} \frac{e}{e-1} \left(\ln \frac{1}{\beta} + 2m - 1 \right), \quad (6.23)$$

where e is the base of the natural logarithm. Further details on how to compute $\mathcal{D}$ is given in [55], [15].

In the second step, we use the probabilistically computed set $\mathcal{D}$ to solve a robust problem for all uncertainty realizations within this set. The chance constraint (6.22) is substituted by the robust constraints

$$\begin{aligned} -p_{ij}^{max} &\leq \mathbf{M}_{(ij, \cdot)}(p_G - \alpha\omega - d + u + \omega) \leq p_{ij}^{max}, & \forall \omega \in \mathcal{D} \\ p_G^{min} &\leq p_G - \alpha\Omega \leq p_G^{max}, & \forall \omega \in \mathcal{D} \\ p_G^{min} &\leq p_G^k - \alpha^k(\Omega + p_{G,k}) \leq p_G^{max}, & \forall k \in \mathcal{K}, \omega \in \mathcal{D} \\ \mathcal{R} &\leq \bar{\mathcal{R}} & \forall \omega \in \mathcal{D} \end{aligned}$$

Since all our constraints are convex with respect to ω (independent of the value of p_G), the robust constraints hold for all $\omega \in \mathcal{D}$ if they hold for ω at the vertices of the set, i.e., all possible combinations of minimum and maximum values of the vector ω . With one wind power plant, we need to consider only 2 scenarios, the maximum and minimum ω in $\mathcal{D}$. With m uncertain in-feeds, we must consider 2^m scenarios. When m is a large number, enumeration of the vertices should be avoided to solve the problem more efficiently. This can be achieved by applying techniques proposed in [111].

6.4 Case Study - Risk and Uncertainty

In the following we assess how our model of risk-based security impacts both the nominal cost and the risk level in the system. Further, we assess how the choice of the risk limit and acceptable violation level ϵ_J influences system operation, and look at the number of cases in which we experience overloads. Finally, we assess how the availability and cost of remedial actions influences the solution.

6.4.1 Considered Formulations

The RB-CC-SCOPF, given by (6.20) - (6.22) includes risk-based limits for the post-contingency line flows and accounts for wind in-feed uncertainty. However, some small changes allow us to change the OPF type. By choosing $\overline{\mathcal{R}}_{(k)} = \mathcal{P}_{(k)}$ for each outage k and $\overline{\mathcal{R}}^{out} = \overline{\mathcal{R}}^{line} = \overline{\mathcal{R}}^{tot} = \infty$, the risk-based constraints become equivalent to traditional N-1 constraints. If the forecast is perfect, i.e. there exist no forecast uncertainty, we set $\omega = 0$ and the probabilistic constraint (6.22) reduces to a deterministic constraint. Thus, we can define four different SCOPF formulations:

- 1) *Standard SCOPF* (SCOPF) considers traditional N-1 constraints (by setting $\overline{\mathcal{R}}_{(k)} = \mathcal{P}_{(k)}$), and no forecast errors for the wind in-feed ($\omega = 0$).
- 2) *Chance-Constrained SCOPF* (CC-SCOPF) considers traditional N-1 constraints (by setting $\overline{\mathcal{R}}_{(k)} = \mathcal{P}_{(k)}$), and a joint chance constraint with violation level ϵ_J .
- 3) *Risk-Based SCOPF* (RB-SCOPF) considers risk-based post-contingency line flow constraints, but no forecast errors for the wind

in-feed ($\omega = 0$).

4) *Risk-Based, Chance-Constrained SCOPF* (RB-CC-SCOPF) considers risk-based post-contingency line flow constraints, and a probabilistic constraint with violation level ϵ_J .

6.4.2 Test System

The OPF formulations presented above are applied to the IEEE 30-bus network [73], which is modified to include two wind power generators at bus 7 and 12. The forecasted wind power at the two buses is 50 and 40 MW, respectively. To generate wind power scenarios, we use the Markov Chain based model described in [112], [14]. For the chance-constrained optimization, we account for up to 2000 scenarios (depending on ϵ_J). In addition, we have 8000 scenarios which are used only for evaluation of the risk. The line outage probabilities were estimated based on the line length (estimated from the line reactance) and the average outage probability for transmission lines in Germany [109]. The generator outages were estimated based on reliability data from the IEEE RTS-96 system [69]. We assume that all generators are able to provide both positive and negative redispatch and that the up and down redispatch capability are the same, i.e., $p_R^+ = p_R^- = p_R$. The redispatch capability is defined as 10% of the maximum output for each generator, $p_R = 0.1 \cdot p_G^{max}$, and all generators are paid the same price for redispatch [\$/MW], with $c_{R,g}^+ = 1.1 \max(c_1)$ and $c_{R,g}^- = 0$. The slope of the risk function for cascading overloads was set to $a_C = 120$, which is a large value compared with the cost of remedial actions in the system. We also assume that a_C is the same for all lines, assuming that cascading events should be generally avoided.

To set the risk limits, we introduce a new base quantity $\mathcal{R}^{base}$. This quantity is defined as the median probability of the line outages,

$$\mathcal{R}^{base} = \text{median}(\mathcal{P}_{(k)}) .$$

When we define risk limits $\overline{\mathcal{R}}_{(k)}$ and $\overline{\mathcal{R}}^{out}$ or evaluate the risks $\mathcal{R}_{(k,ij)}^{spec}$ and $\mathcal{R}^{tot}$ in the case study, we normalize the values by $\mathcal{R}^{base}$ to obtain numerical values that are easier to interpret.

All optimization problems were implemented in MATLAB and solved using CPLEX via the MATLAB interface TOMLAB.

Table 6.1: Generation cost with the different SCOPF formulations

	Generation cost (% of SCOPF)
SCOPF	100.00
CC-SCOPF	100.48
RB-SCOPF	99.88
RB-CC-SCOPF	100.35

6.4.3 Results

Generation Cost and Risk Level for SCOPF Formulations

We first compare the generation cost obtained with the four SCOPF formulations shown in section 6.4.1. The contingency and line specific risk constraint $\mathcal{R}_{(k,ij)}^{spec} \leq \overline{\mathcal{R}}$ is used for the risk-based formulations, with $\overline{\mathcal{R}} = \mathcal{R}^{base}$. For the probabilistic formulations, the maximum violation level was set to $\epsilon_J = 0.05$.

Table 6.1 lists the generation cost for each SCOPF formulation. The variation in generation cost is not very large. However, using the risk-based criterion in this case reduces the cost by about 0.1% (RB-SCOPF compared to SCOPF), whereas accounting for wind uncertainty increases cost by about 0.5% (RB-CC-SCOPF compared to RB-SCOPF).

To explain the differences in cost, we compare the risk level for the different formulations. Considering the solutions of the optimization problems above, we compute $\mathcal{R}_{(k,ij)}^{spec}$ according to (6.1). We first discuss the deterministic SCOPF and RB-SCOPF solutions with $\omega = 0$, and then move on to the probabilistic RB-CC-SCOPF result.

In Fig. 6.3, the severity $\mathcal{S}_{(ij|k)}$ is plotted against $\mathcal{P}_{(k)}$ for each line ij and each outage k . The black diamonds and the blue dots are the results from the SCOPF and the RB-SCOPF, respectively. Note that since we calculate the severity for all lines after each outage, there are several dots and circles for each outage. The red line is the risk limit $\overline{\mathcal{R}}_{(k)} = \mathcal{R}^{base}$, and dashed black line is the N-1 limit $\mathcal{S}_{(ij|k)} = 1$.

For most outages, $\mathcal{S}_{(ij|k)} = 0$ for all lines ij because all post-contingency line flows are below $0.9p_{ij}^{max}$. For other outages, $\mathcal{S}_{(ij|k)} > 0$, but is

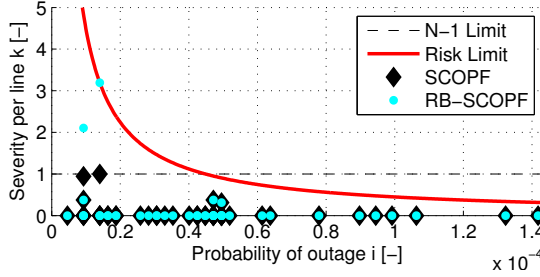


Figure 6.3: Comparison of solutions obtained with the traditional SCOPF and the RB-SCOPF. The black diamonds (SCOPF) and the blue dots (RB-SCOPF) show the evaluation of $\mathcal{R}_{(k,ij)}^{spec}$ for the case where $\omega = 0$.

similar for both the RB-SCOPF and the SCOPF. However, the solution of the RB-SCOPF leads to two outages with $\mathcal{S}_{(ij|k)} > 1$, implying that the traditional N-1 limit is violated. For one of the binding constraints, the severity is allowed to increase from $\mathcal{S}_{(ij|k)} = 1$ with the SCOPF to $\mathcal{S}_{(ij|k)} = 3.2$ with the RB-SCOPF. Because of the relaxation of this binding constraint, the RB-SCOPF has a lower cost than the SCOPF. In other cases, the RB-SCOPF could lead to a more expensive solution than the SCOPF, since the RB-SCOPF leads to a constraint tightening for outages with high probability.

The above discussion of the SCOPF and the RB-SCOPF only considers the risk level for the forecasted wind $\omega = 0$. We now assess how the risk level changes with forecast uncertainty ω . We compute $\mathcal{R}_{(k,ij)}^{spec}$ for the case where no wind deviation occurs, $\omega = 0$, and for the cases with forecast deviation equal to the worst case scenarios in the robust set $\mathcal{D}$, denoted by ω^{max} . Fig. 6.4 shows the risk computed for the solution of the RB-CC-SCOPF for different wind in-feed scenarios. The green dots corresponds to no wind deviation $\omega = 0$, while the blue crosses denote the risk for the worst case scenarios $\omega = \omega^{max}$. Comparing Fig. 6.4 to Fig. 6.3, we observe that the RB-CC-SCOPF has $\mathcal{S}_{(ij|k)}$ well below the acceptable limit when $\omega = 0$. However, for the worst-case scenario ω^{max} , the risk reaches the limit. Thus, by keeping a margin and being conservative for the case with $\omega = 0$, the RB-CC-SCOPF avoids constraint violations for all $\omega \in \mathcal{D}$. This conservativeness affects the generation cost, meaning that the RB-CC-SCOPF will always be more expensive than the RB-SCOPF.

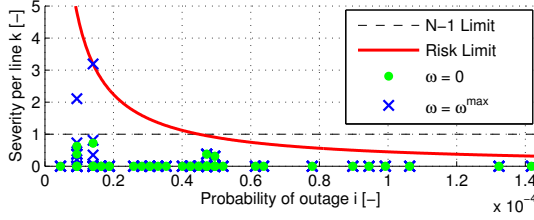


Figure 6.4: Evaluation of $\mathcal{R}_{(k,ij)}^{spec}$ for the solution obtained with the RB-CC-SCOPF. The green dots is $\mathcal{R}_{(k,ij)}^{spec}$ evaluated for $\omega = 0$. The blue crosses are $\mathcal{R}_{(k,ij)}^{spec}$ for the worst-case scenarios $\omega = \omega^{max}$.

Influence of Risk Limit and Acceptable Violation Probability on the RB-CC-SCOPF

We now compare generation cost and risk level for the RB-CC-SCOPF for different upper bounds on the risk and violation level. The acceptable violation level ϵ_J is varied in the range between 0.01 - 0.08. The scenarios are first drawn for $\epsilon_J = 0.08$, and additional samples are added as ϵ_J is decreased. The upper bound in the contingency specific risk constraint $\mathcal{R}_{(k)}^{out} \leq \overline{\mathcal{R}}^{out}$ is varied in the range $(0.5 - 5) \mathcal{R}^{base}$. The generation cost is obtained directly as an output from the optimization, while the risk level $\mathcal{R}^{tot}$ is computed as the average total risk (6.4) for 8000 wind scenarios.

The generation cost of the RB-CC-SCOPF for different risk levels $\overline{\mathcal{R}}^{out}$ and violation probabilities ϵ_J is depicted in Fig. 6.5. The cost increases as the acceptable violation level ϵ_J or the risk limit $\overline{\mathcal{R}}^{out}$ decreases. This is as expected, since a lower violation level means that the robust set $\mathcal{D}$ becomes larger and a lower risk limit implies stricter limits on the severity. The generation cost is not influenced very much by the choice of ϵ_J , except for a cost decrease as ϵ_J increases from 0.03 to 0.04. Note that the changes in cost between different levels of ϵ_J depends on the specific set of samples. The cost might have decreased more smoothly if we had drawn another set of samples for the uncertainty realizations. The relaxation of the risk limit $\overline{\mathcal{R}}^{out}$ from 0.5 to 5 appears to decrease the cost linearly.

Fig. 6.6 shows the average total risk $\mathcal{R}^{tot}$ over 8000 wind scenarios. The risk decreases as the acceptable violation level ϵ_J or the risk limit $\overline{\mathcal{R}}^{out}$ decreases. The total risk is not very sensitive to the choice of ϵ_J ,

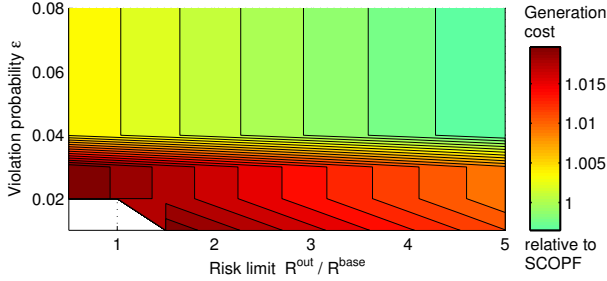


Figure 6.5: Generation cost obtained with the RB-CC-SCOPF, relative to the cost of the SCOPF, for different limits on the violation level ϵ_J and the risk $\mathcal{R}_{(k)}^{out} \leq \overline{\mathcal{R}}^{out}$. In the white region, there is no feasible solution to the problem.

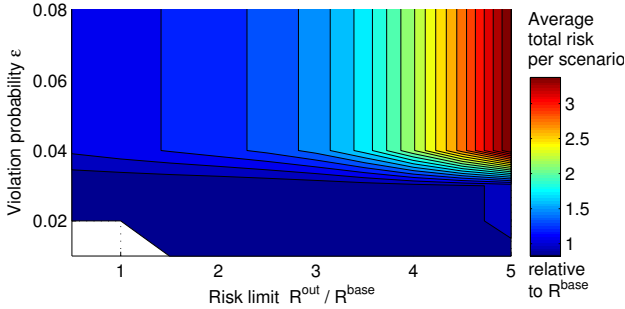


Figure 6.6: Average total risk across 8000 wind realizations, normalized by $\mathcal{R}^{base}$. The risk is evaluated for the solutions of the RB-CC-SCOPF for different limits on the violation level ϵ_J and the risk $\mathcal{R}_{(k)}^{out} \leq \overline{\mathcal{R}}^{out}$. In the white region, there is no feasible solution to the problem.

except when ϵ_J increases from 0.03 to 0.04 (which coincides with the cost increase observed in Fig. 6.5). The choice of the risk limit $\overline{\mathcal{R}}^{out}$ has a more significant influence. For higher risk limits $\overline{\mathcal{R}}^{out} > 2.5 \cdot \mathcal{R}^{base}$, the risk increases very fast, and seems to increase faster than the generation cost decreases.

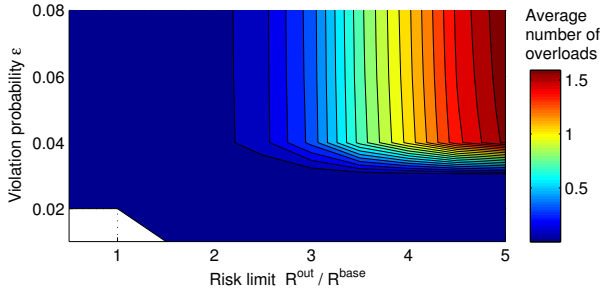


Figure 6.7: Average number of N-1 violations across 8000 wind realizations, as obtained with the RB-CC-SCOPF for different limits on the violation level ϵ_J and the risk $\mathcal{R}_{(k)}^{out} \leq \overline{\mathcal{R}}^{out}$. In the white region, there is no feasible solution to the problem.

Number of cases with overloads

The risk-based formulation allows post-contingency overload (i.e., $\mathcal{S}_{(ij|k)} > 1$) for contingencies with low probability. In such cases, post-contingency actions to reduce the line flow are required, which introduce additional cost and require manual activation by the operator. Since it is not desirable that this happens too often, we investigate how many cases there are with $\mathcal{S}_{(ij|k)} > 1$. Fig. 6.7 shows the average number of post-contingency constraints with $\mathcal{S}_{(ij|k)} > 1$ for the above RB-CC-SCOPF solutions with varying acceptable violation probabilities and different risk levels. This average was calculated based on the 8000 wind in-feed scenarios. Although the number of constraints with $\mathcal{S}_{(ij|k)} > 1$ increases when either the violation level or the risk limit increases, the average always remains below two violations per scenario. As this is a relatively small number, we believe that these situations can be handled by the operator, particularly since the RB-CC-SCOPF proposes effective remedial actions to relieve these overloads if the outage should happen.

As an example of the proposed remedial actions, we look at the redispatch measures for the RB-CC-SCOPF case in Fig. 6.4. The two lines that risk a post-contingency overload (i.e., where the severity is larger than 1) are the lines from bus 6-10 and 6-28. Both would be overloaded if the other is outaged. However, this overload is tolerated since a generation shift of 5.5 MW between generator 2 and 4 would reduce the line flow by 3.5 MW, which is sufficient to remove the overload.

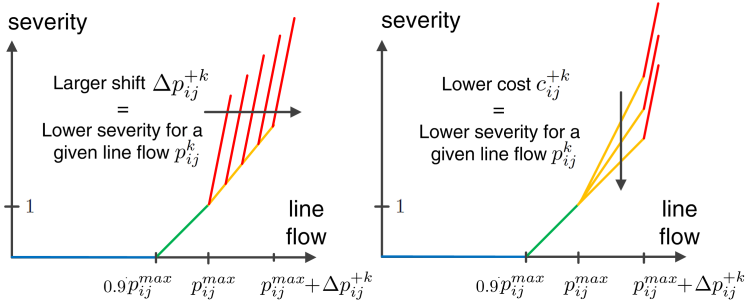


Figure 6.8: Left: Change in severity function with higher amount of available redispatch p_R , leading to a higher Δp_{ij} . Right: Change in severity function with a lower redispatch cost c_{ij}^+ , leading to a less steep severity function.

Influence of Cost and Availability of Remedial Measures

If the availability or cost of redispatch change, the severity function changes. Fig. 6.8 shows the influence of changes in available redispatch power (left) and changes on redispatch cost (right) on the severity function. As more redispatch becomes available or the cost decreases, the severity related to overloads decreases. This means that for some lines, a higher post-contingency line flow will be acceptable. In the following, the influence of the amount and cost of redispatch on the accepted line overloads and objective cost is investigated.

The RB-CC-SCOPF is run with $\epsilon_J = 0.05$ and the contingency and line specific risk constraint $\mathcal{R}_{(k,ij)}^{spec} \leq \mathcal{R}^{base}$. The amount of available redispatch p_R is varied in the range $(0 - 0.25) p_G^{max}$, and the cost of the redispatch c_R^+ is varied from $(0 - 1.2) c_1$ (with $c_R^- = 0$). Fig. 6.9 shows the average accepted overloads for all lines after all contingencies with $\mathcal{P}_{(i)} \geq \text{median}(\mathcal{P}_{(k)})$ (i.e., for all contingencies where the acceptable severity limits is $\mathcal{S}_{(ij|k)} \geq 1$) for different values of p_R and c_R^+ . The values are given as a percentage of the line capacity.

If the cost c_R^+ is high, higher amounts of available redispatch p_R does not lead to an significant increase in the accepted overloads. This is because of the steep slope of the severity curve, which means that the accepted overload for most lines is below Δp_{ij} . If the cost c_R^+ is low, higher amounts of available redispatch p_R leads to an increase in the accepted overloads. In this case the slope of the severity curve is lower,

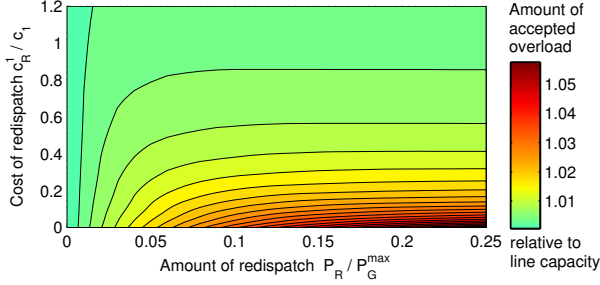


Figure 6.9: Average accepted overload for all lines after all contingencies with $\mathcal{P}_{(k)} \geq \text{median}(\mathcal{P}_{(k)})$ (i.e., all contingencies with $\mathcal{S}_{(ij|k)} \geq 1$) for different values of p_R and c_R^+ . The values are given as a percentage of the line capacity.

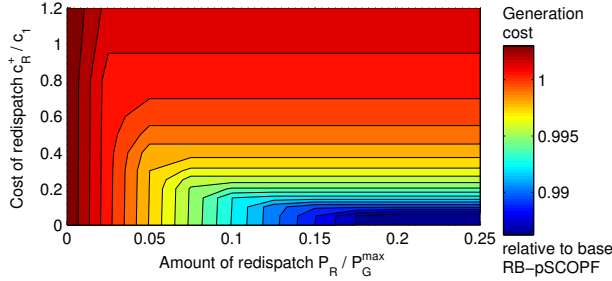


Figure 6.10: Cost of the RB-CC-SCOPF with $\epsilon_J = 0.05$ and the contingency and line specific risk constraint $\mathcal{R}_{(k,ij)}^{spec} \leq \mathcal{R}^{base}$ for different values of p_R and c_R^+ .

meaning that the accepted overload is higher than Δp_{ij} . Similarly, if p_R is low, the cost c_R^+ is of little importance, whereas if p_R is high, the accepted overloads are more sensitive to the cost c_R^+ .

With more and cheaper remedial actions, it is thus possible to relax the post-contingency line constraints more, which has an influence on the overall cost of the optimization problem. Fig. 6.10 shows the generation cost from the RB-CC-SCOPF for different values of p_R and c_R^+ . The pattern is similar to Fig. 6.9, confirming that the relaxation of the post-contingency line constraints leads to overall lower cost.

6.5 Summary and Conclusions

This chapter proposes a new way of modeling risk of post-contingency overloads, accounting for system properties like the effect and availability of redispatch. The resulting risk measure is used to formulate risk-based constraints for the post-contingency line flows. By discussing how the risk-based constraints compares with traditional N-1 constraints, we provide some guidance on how risk limits can be chosen. The new constraints are included in an SCOPF formulation which also accounts for forecast uncertainty. This risk-based, probabilistic SCOPF (RB-CC-SCOPF) is formulated such that we guarantee that the risk level and the rest of the system constraints will be enforced with a high probability $1 - \epsilon_J$.

The OPF formulation was applied to a case study of the IEEE 30 bus system. We show that we are able to limit the risk even when the in-feeds deviate from the forecast. Further, the risk-based formulation allows us to choose the desired risk level, as opposed to the N-1 criterion which only deems the system as secure or insecure. As expected, enforcing a lower level of risk or a lower violation level ϵ_J both increases generation cost, but leads to lower average risk and fewer N-1 violations.

The proposed method allows us not only to control the system risk level, but also to account for the effect of available remedial measures during the operational planning process. Through the use of risk-based constraints, the post-contingency line flow limits are set based on which measures are available. The case study demonstrates how the cost and amount of available redispatch influences the severity function, and how less costly and larger amounts of redispatch allow us to relax the post-contingency line flow constraints, leading to lower generation cost.

The method and particularly the severity model can be further developed. Here, the amount of available redispatch is assumed to be known, which allows us to pre-compute the severity function before the optimization starts. The severity function computation could be defined as part of the optimization, which would allow for co-optimization of the generation dispatch and available redispatch. Further, we could include more than one remedial action for each line, and consider the possibility of incorporating other remedial actions with lower cost than redispatch, such as switching actions. Finally, a complete model should account for the influence of the remedial actions not only on the overloaded line, but also on other lines in the system.

Chapter 7

Limiting the Expected Operational Risk

In the previous chapter, we proposed a risk-based SCOPF where chance constraints were used to limit the probability that a given risk limit is exceeded. The risk constraints considered only the post-contingency line flows, whereas the line flows in normal operations were enforced using standard chance constraints. In this chapter, we focus on modelling risk in the pre-contingency operating condition, and express our risk-based constraints in terms of the expected risk of overload, formalized through the concept of Weighted Chance Constraints (WCC). The WCC account for the full probability distribution of the line flows and generator outputs, and express the severity of constraint violations through the use of a weight function. We focus on generic weight functions such as a linear or quadratic function, although the weight function could be tailored to represent any specific type of risk. Further, we prove that the WCC remain convex for any convex weighting function, and for very general generation control policies. We comment on the relation to the standard chance constraints, and formulate an OPF with WCC and general generation control policies. Finally, we provide details on the evaluations of the WCC constraints for example weight functions and generation control policies, and assess the performance in a case study.

7.1 Motivation and Related Work

One way to handle uncertainty within the OPF problem is to limit the probability of constraint violation by formulating a chance-constrained optimization problem, as explained in previous chapters. While previous versions of the chance-constrained OPF, including the RB-CC-SCOPF in the previous chapter, have shown the ability to limit the probability of violations, they have some drawbacks. Standard chance constraints do not distinguish between large and small violations, although those might carry significantly different risk. Further, the previous CC-OPF formulations only consider affine policies for generation control, which is not necessarily realistic for large forecast errors and may lead to sub-optimal results. The method proposed in the following addresses these drawbacks.

Here, we suggest to account for both the probability and the size of constraint violations, and calculate the expected risk of constraint violations. To this end, we define a *weighting function* for the overload, and calculate the expected risk as the product between the weighting function and the probability distribution of the overload. As this can be interpreted as a generalization of the standard chance constraint, we call this type of constraint Weighted Chance Constraints. While the set-up allows for general weighting functions, the considered examples are chosen to be applicable to the power system problem. The examples are closely related to the severity functions applied in risk-based OPF, e.g. [106, 28, 19, 31], where they are used to account for the severity of post-contingency overloads. Another example of a related problem is the risk-based OPF in [26], where the risk function is defined in terms of expected cost of overloads. The weighted chance constraint suggested here is also somewhat similar to a constraint on the Conditional-Value-at-Risk (CVaR), a risk measure borrowed from the finance sector, which was introduced by [113] and applied to the OPF problem in [61]. Common to all the above examples is that they assign a higher risk to a larger overload, implying that it is reasonable to assume a *convex* risk functions.

One key observation is that the WCC remain convex for any convex, non-decreasing weight function and for very general generation control policies. Extending the generation control from affine to more general control policies allow us to model additional control actions, such as manual activation of reserves or generation redispatch during large deviations. With more flexible policies, we are able to obtain cheaper,

more realistic solutions while maintaining the same level of risk. This was also observed for piecewise affine policies in [84]. With the convexity properties of the WCC, we can optimize over such generation control policies, without compromising convexity of the resulting OPF problem. Convexity of the WCC-OPF is important for a number of reasons. It allows us to find globally optimal solutions, or conversely, to show problem infeasibility. Further, several efficient solution algorithms are available for convex optimization problems, which suggests a path for scaling our methods to large systems and to general uncertainty distributions (which are computationally more involved). One such example is the outer-approximation cutting-plane algorithm from [18], which was found to be very effective for the CC-OPF.

The remainder of the chapter is organized as follows. We first introduce the WCC, along with a proof of convexity. Then, we provide the mathematical formulation for the WCC-OPF, before showing some relevant examples for practical weighting functions with affine and piecewise affine generation control. Finally, the performance of the method is illustrated in a case study based on the IEEE RTS96 test system.

The contents of this chapter is based on the material in [20].

7.2 Weighted Chance Constraints

We propose a generalized weighted chance constraint of the form

$$\int_{-\infty}^{\infty} f(y(\omega))P(\omega)d\omega \leq \epsilon. \quad (7.1)$$

Here, $P(\omega)$ is the multivariate distribution function of the fluctuations. The quantity $y(\omega)$ denotes the magnitude of overload, e.g. the amount of transmission line loading above the limit. Generally, whenever we have $y(\omega) > 0$, it indicates a violation of the limit, while $y(\omega) \leq 0$ implies that we are in a safe operating region. The weight function $f(\cdot)$, which is nonzero only if $y > 0$, translates a given overload $y(\omega)$ into a risk value $f(y(\omega))$.

Overall, the integral on the left hand side of the equation represents the expected value of $f(y(\omega))$. The constraint (7.1) thus limits the expected risk to stay below a given value ϵ .

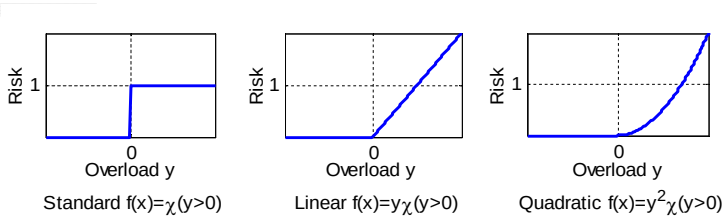


Figure 7.1: Examples of risk functions for chance constraints: Standard (left), linear (middle) and quadratic (right).

Weight Functions

The weight function $f(y)$ can be chosen in many ways. For example, (7.1) is equivalent to a standard chance constraint if $f(y)$ is the unit step function $\chi(y > 0)$ (i.e., 0 for $y < 0$ and 1 for $y \geq 0$). Other possibilities include weight functions that have been proposed in the context of risk-based OPF, such as linear [106] and quadratic [28] severity functions, the piecewise-affine function defined in the previous chapter or the more general risk functions describing the component cost of overload in [26]. For comparison, the linear and quadratic weight functions are plotted in Fig. 7.1 along with the weight function corresponding to a standard chance constraint (i.e., the unit step function). While unit step function assigns equal risk to all magnitudes of overloads, the risk increases with increasing overload for the linear or quadratic weight functions. Note that the risk limit ϵ has different physical interpretations and different units depending on the definition of $f(\cdot)$. For the standard chance constraints, ϵ represents the acceptable violation probability. For the linear weighted chance constraint, ϵ limits the expected overload and is specified in MW . For the quadratic weight function, the unit of ϵ is MW^2 .

General Generation Control Policies

In previous chapters, we assumed that the system was balanced according to an affine control policy. However, when the changes from the schedule are large, the power system operator might use additional controls to balance the system and control power flows. Examples include the use of slower reserves (e.g., spinning or tertiary reserves) or wind

power curtailment. Since these controls have a significant influence on the ability to handle disturbances, they should be included in the OPF to avoid unrealistic or sub-optimal solutions. In the following sections, we therefore discuss the application of more general generation control policies $\tilde{p}_G(\omega)$.

Since the the network must obey total power balance at all times, the generation policy must enforce the total power balance constraint,

$$\sum_i \tilde{p}_{G,i}(\omega) + u_i + \omega_i - d_i = 0, \quad \forall \omega. \quad (7.2)$$

We further note that, assuming a DC power flow approximation, the power flows remain linear functions of the power injections, i.e.,

$$\tilde{p}_{ij}(\omega) = (\mathbf{M}_{(ij, \cdot)})(\tilde{p}_G(\omega) - d + u + \omega). \quad (7.3)$$

We can hence express the overloads $y(\omega)$ for the upper and lower generator limits as

$$y_i^u(\omega) = \tilde{p}_G(\omega) - p_G^{max}, \quad y_i^l(\omega) = p_G^{min} - \tilde{p}_G(\omega). \quad (7.4)$$

For the upper and lower line flow limits, we have

$$y_{ij}^u(\omega) = \tilde{p}_{ij}(\omega) - p_{ij}^{max}, \quad y_{ij}^l(\omega) = p_{ij}^{min} - \tilde{p}_{ij}(\omega). \quad (7.5)$$

Convexity of the Weighted Chance Constraint

Convexity of the OPF problem is important, as it guarantees globally optimal solutions and allows for implementation of efficient solution algorithms. Solution algorithms based on outer approximation with cutting planes has been shown to make the CC-OPF problem tractable for large systems with thousands of nodes [18]. The weighted chance constraint (7.1) is a convex constraint for general generation control policies and general probability distributions for ω whenever the weight function $f(x, \omega)$ is a convex function of the decision variables x for any possible realization of ω . We state our result regarding convexity in Theorem 1.

Theorem 1. *Let $\tilde{p}(\omega)$ be any general generator control policy, and convex with respect to the decision variables for every ω . Consider a general weighted chance constraint as defined in (7.1),*

$$\int_{-\infty}^{\infty} f(y(\omega)) P(\omega) d\omega \leq \epsilon. \quad (7.6)$$

Assume that the weighting function $f(\cdot)$ is a convex function of its argument, and non-decreasing. Then (7.6) is a convex constraint in the control policy $\tilde{p}(\omega)$.

Proof. Referring to (7.4), (7.5) and (7.3), we see that the overload $y(\omega)$ is always a convex function, as it depends linearly on the convex generator control policy $\tilde{p}$. Combined with a convex and non-decreasing weight function $f(\cdot)$, this implies that the function $f(y(\omega))$ is a convex function for every ω . The proof then follows from the fact that the expectation of a convex function is also convex. $\square$

Theorem 1 shows that the WCC are convex under general control policies if the weighting function $f(\cdot)$ is convex. For the standard chance constraint, this is not the case, as the unit step function is non-convex. For the linear and quadratic risk functions, and other example risk functions mentioned above, $f(\cdot)$ is convex. Generally, a convex weight function, which assigns a higher risk to a higher overload, is natural from an engineering point of view.

Optimizing the Generation Control Policy

For every ω , the constraints (7.2), (7.3) are simply linear constraints. This means that the control policy $\tilde{p}$ can be a variable within a convex WCC-OPF problem. However, for practical implementation of the WCC within a standard convex optimization program, it is necessary to represent the control policy with a finite number of parameters. It is straight forward to see that the convexity of (7.1) and (7.2) is preserved as long as we use any linear-in-parameter representation of the control policy $\tilde{p}$ given by

$$\tilde{p}_G(\omega) = p_G - \sum_{k=1}^K \alpha_k g_k(\omega), \quad (7.7)$$

along with the constraints

$$\sum_{i \in \mathcal{N}} p_{G,i} + u_i - d_i = 0, \quad (7.8)$$

$$\sum_{i \in \mathcal{G}} \sum_{k=1}^K \alpha_{k,i} g_k(\omega) + \Omega = 0, \quad (7.9)$$

where $\alpha_k \in \mathbb{R}^m$ and $g_k(\omega) \in \mathbb{R}$. The original affine policy (2.6) can be obtained as a special case of (7.7) by setting $K = 1$ and $g_k(\omega) = \Omega$.

In the above, $g_k(\omega)$ can be any general function of ω . The expression for the control policy in (7.7) has extensive representation potential. For example, we can represent most general control policies $\tilde{p}(\omega)$, e.g. continuous or piece-wise continuous, approximately in the form of (7.7) by using a piece-wise constant representation of $\tilde{p}(\omega)$,

$$\tilde{p}_{G,i}(\omega) = p_{G,i} - \sum_{k=1}^K \alpha_{k,i} \chi(\omega \in S_k). \quad (7.10)$$

Here, S_k are sets that form a partition of the domain of ω and α_k is the value of $\tilde{p}$ on S_k . Increasing the number of terms K in the representation allows us to approximate general control policies with higher fidelity, but the number of variables to optimize over also increases. Thus, a trade-off between fidelity and computational effort must be made.

Note that related, piecewise affine policies were investigated in the context of joint chance constraints in [84].

7.3 Optimal Power Flow with Weighted Chance Constraints and General Generation Control Policies

Following our standard notation and assuming a general generation control policy $\tilde{p}_G(\omega)$ for the generator outputs, we state the WCC-OPF as

$$\min_{p_G, \alpha} \sum_{i \in \mathcal{G}} c_i p_{G,i} \quad (7.11)$$

$$\text{s.t.} \quad (7.12)$$

$$\sum_{i \in \mathcal{N}} p_{G,i} - d_i + u_i = 0 \quad (7.13)$$

$$\sum_{i \in \mathcal{G}} \alpha_i = 1, \quad \alpha \geq 0 \quad (7.14)$$

$$WCC [\tilde{p}_{G,i}(\omega) > p_i^{max}] < \epsilon_G \quad \forall i \in \mathcal{G} \quad (7.15)$$

$$WCC [\tilde{p}_{G,i}(\omega) < p_i^{min}] < \epsilon_G \quad \forall i \in \mathcal{G} \quad (7.16)$$

$$WCC [\mathbf{M}_{(ij, \cdot)}(\tilde{p}_{G,i}(\omega) - d + u + \omega) > p_{ij}^{max}] < \epsilon_L \quad \forall \{ij\} \in \mathcal{L} \quad (7.17)$$

$$WCC [\mathbf{M}_{(ij, \cdot)}(\tilde{p}_{G,i}(\omega) - d + u + \omega) < -p_{ij}^{max}] < \epsilon_L \quad \forall_{\{ij\} \in \mathcal{L}} \quad (7.18)$$

As usual, the objective (7.11) is to minimize generation cost, while (7.13), (7.14) enforces power balance during normal operation and with deviations. The constraints (7.15) - (7.18) enforce generation and transmission constraints using WCC. Here, we use a shorthand notation for the WCC (7.1). Specifically, whenever we need to enforce that the risk of the quantity $y(\tilde{\omega})$ is small, we denote it by

$$WCC(y(\omega) > 0) \leq \epsilon. \quad (7.19)$$

For example, the constraint (7.15) limits the risk of a generator limit violation $\tilde{p}_{G,i}(\omega) > p_{G,i}^{max}$, which is equivalent to the overload $y(\omega) = \tilde{p}_{G,i}(\omega) - p_{G,i} > 0$, to a maximum value ϵ .

7.4 Expressions for the Weighted Chance Constraints

In this section, we derive closed form expressions for the WCC with linear and quadratic weight functions, assuming normally distributed fluctuations ω . Note that the WCC can also be computed when the distribution is not normal, although this might require numerical methods for evaluation of the integrals.

7.4.1 With Affine Control Policy

We first provide expressions for the WCC with an affine control policy,

$$\tilde{p}_G(\omega) = p_G - \alpha\Omega. \quad (7.20)$$

For completeness, we also show the expressions for the original chance constraint in our current notation.

Mean and Covariance of the Overloads

With the assumption of a multivariate normal random vector ω , we observe that the overloads y in (7.4), (7.5) are distributed as normal random variables in all cases, since they are linear combinations of ω .

For the violations of the upper generation limit y_i^u , the mean $\mu_{y,i}^u$ and variance $(\sigma_i^u)^2$ of the overload is given by

$$\mu_{y,i}^u = p_{G,i} - p_{G,i}^{max}, \quad (\sigma_i^u)^2 = (\alpha_i \sigma_\Omega)^2. \quad (7.21)$$

Similarly, we get that the line overload y_{ij}^u is Gaussian with

$$\mu_{y,ij}^u = \mathbf{M}_{(ij,\cdot)}(p_G - d + u) - \bar{p}_{ij}, \quad (7.22)$$

$$(\sigma_{ij}^u)^2 = \mathbf{M}_{(ij,\cdot)}(\mathbf{I} - \alpha \mathbf{1}_{1,m}) \Sigma_W(\mathbf{M}_{(ij,\cdot)}(\mathbf{I} - \alpha \mathbf{1}_{1,m}))^T \quad (7.23)$$

Similar expressions can be obtained for the lower limits y_i^l , y_{ij}^l .

Linear Weight Function

With a linear weight function, we substitute $f(y) = y\chi(y > 0)$ in (7.1), and get

$$\int_{-\infty}^{\infty} y\chi(y > 0)P(\omega)d\omega = \int_0^{\infty} y P(y)dy \leq \epsilon. \quad (7.24)$$

The left hand side of (7.24) is the expectation of a truncated Gaussian random variable, and can be expressed as

$$\mu_y \left(1 - \Phi \left(\frac{-\mu_y}{\sigma} \right) \right) + \frac{\sigma}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{-\mu_y}{\sigma} \right)^2} \leq \epsilon. \quad (7.25)$$

The WCC for generator and line overloads are obtained by inserting the corresponding means and variances from (7.21)-(7.23) in (7.25).

Quadratic Weight Function

Substituting the quadratic weight function $f(y) = y^2\chi(y > 0)$ in (7.1), we get

$$\int_{-\infty}^{\infty} y^2\chi(y > 0)P(\omega)d\omega = \int_0^{\infty} y^2 P(y)dy \leq \epsilon. \quad (7.26)$$

The left hand side of (7.26) is the second moment of a truncated Gaussian random variable and can be rewritten as

$$(\mu_y^2 + \sigma^2) \left(1 - \Phi \left(\frac{-\mu_y}{\sigma} \right) \right) + \frac{\mu_y \sigma}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{-\mu_y}{\sigma} \right)^2} \leq \epsilon. \quad (7.27)$$

As for the linear weight function, the WCC for the quadratic weight function can be obtained by substituting the means and variances from (7.21)-(7.23) in (7.27).

Standard chance constraints and affine policy

We include the expressions for the standard chance constraints in our current notation for the sake of completeness. Substituting the unit step function $f(y) = \chi(y > 0)$, we obtain

$$\int_{-\infty}^{\infty} \chi(y > 0) P(\omega) d\omega = \int_0^{\infty} P(y) dy \leq \epsilon \quad (7.28)$$

which can be reformulated to our usual chance constraint

$$\mu_y + \Phi^{-1}(1 - \epsilon)\sigma \leq 0. \quad (7.29)$$

The final form can be obtained by substituting (7.21)-(7.23) in (7.29).

7.4.2 With Piecewise Affine Control Policy

We now consider a piece-wise affine policy which can be expressed as

$$\tilde{p}_G \doteq \begin{cases} p_G - \alpha\Omega, & \Omega^- \leq \Omega \leq \Omega^+ \\ p_G - \alpha\Omega + \xi_i^+, & \Omega \geq \Omega^+ > 0 \\ p_G - \alpha\Omega + \xi_i^-, & \Omega \leq \Omega^- < 0 \end{cases} \quad (7.30)$$

Here, ξ^+ , ξ^- represent additionally deployed reserves in case of large wind fluctuations, when Ω is larger (or smaller) than a given threshold Ω^+ (or Ω^-). These activated reserves can be for example non-spinning or tertiary reserves that require manual intervention by the operator, or a manually initiated redispatch to relieve transmission line overloads. To ensure a balanced system, we additionally enforce $\sum_{i \in \mathcal{G}} \xi^+ = \sum_{i \in \mathcal{G}} \xi^- = 0$.

Mean and Covariance of the Overloads

With the piecewise affine control policy (7.30), the overload y is conditional on the total fluctuation $y|\Omega$. The overload is normally distributed and its mean and variance depends on which of the three regions in (7.30) the total wind deviation Ω belongs to. For generator power, the conditional mean and variance are given by

$$\mu_{y,i}^u(\Omega) = p_{G,i} - p_{G,i}^{max} + \xi_i(\Omega) - \alpha_i\Omega, \quad (7.31)$$

$$(\sigma_i^u)^2 = 0, \quad (7.32)$$

where

$$\xi_i(\Omega) = \begin{cases} \xi_i^-, & \Omega < \Omega^-, \\ 0, & \Omega^- \leq \Omega \leq \Omega^+, \\ \xi_i^+, & \Omega > \Omega^+. \end{cases} \quad (7.33)$$

Similarly, we can calculate the conditional means and variances of the line overloads as

$$\begin{aligned} \mu_{y,ij}^u(\Omega) &= \mathbf{M}_{(ij,\cdot)} [p_{G,i} - d + u + \xi(\Omega)] - p_{ij}^{max} \\ &\quad + 1/\sigma_\Omega^2 (\mathbf{M}_{(ij,\cdot)}(\mathbf{I} - \alpha \mathbf{1}_{1,m}) \Sigma_W \mathbf{1}_{1,m}^T) \Omega \end{aligned} \quad (7.34)$$

$$\begin{aligned} (\sigma_{ij}^u)^2 &= \mathbf{M}_{(ij,\cdot)}(\mathbf{I} - \alpha \mathbf{1}_{1,m}) \Sigma_W (\mathbf{M}_{(ij,\cdot)}(\mathbf{I} - \alpha \mathbf{1}_{1,m}))^T \\ &\quad - 1/\sigma_\Omega^2 (\mathbf{M}_{(ij,\cdot)}(\mathbf{I} - \alpha \mathbf{1}_{1,m}) \Sigma_W \mathbf{1}_{1,m})^2, \end{aligned} \quad (7.35)$$

with $\xi(\Omega)$ defined as in (7.33).

Linear Weight Function

Under a piece-wise affine policy, the linearly weighted chance constraint can be written as

$$\begin{aligned} \int_0^\infty yP(y)dy &= \int_{-\infty}^\infty \int_0^\infty yP(y|\Omega)P(\Omega)dyd\Omega \quad (7.36) \\ &= \int_{-\infty}^{\Omega^-} \int_0^\infty yP(y|\Omega)P(\Omega)dyd\Omega \\ &\quad + \int_{\Omega^-}^{\Omega^+} \int_0^\infty yP(y|\Omega)P(\Omega)dyd\Omega \\ &\quad + \int_{\Omega^+}^\infty \int_0^\infty yP(y|\Omega)P(\Omega)dyd\Omega. \end{aligned} \quad (7.37)$$

The final version of the linear WCC is obtained by substituting the expression for the expectation of a truncated Gaussian from (8.27) in (7.37), and insert the means and variances from (7.31), (7.32) and (7.34), (7.35).

Quadratic Weight Function

Following the same derivation as above, we can obtain an expression analogous to (7.37) by simply replacing y by y^2 . The rest of the deriva-

tion is identical with the only difference that we substitute the expression for the second moment of a truncated Gaussian from (7.27).

7.5 Case Study

In this case study, we first investigate the impact of the different risk functions by solving the WCC-OPF with a standard, linear and quadratic risk function and the standard affine control policy. To assess the performance, we tune the parameters to obtain a similar objective cost, and compare the number and size of the constraint violations. In the second part of the case study, we investigate how a the more flexible piece-wise affine policy for generation control performs compared with the standard affine policy in the case of the linear WCC-OPF.

7.5.1 Test System

The case study is based on the IEEE RTS96 system [69], with some modifications. The line capacities are reduced to 80 % of the steady state capacity listed in [69], since only active power flows are considered. The generation costs used in the OPF are linear costs adopted from [70], and we also assume that the generation capacity is twice larger than those from the original RTS96 model [69]. The minimal generator output is set to 0 for all generators. Aggregations of wind power plants are located at bus 8 and bus 15, with a forecasted in-feed of 125 and 175 MW, respectively. The wind power fluctuations are assumed to follow a multivariate normal distribution with zero mean and standard deviations of 9.4 and 13.1 MW, and are correlated with a correlation coefficient $\rho = 0.2$.

7.5.2 Results

Comparison of Weighting Functions

We compare the standard CC-OPF with the linear and quadratic WCC-OPF assuming an affine policy for generation control. We choose $\epsilon_L = 0.1$ for the transmission line constraints in all settings discussed. For the generation constraints, we choose different ϵ_G for the different problems, with $\epsilon_G = 0.001$ for the CC-OPF and the linear WCC-OPF

Table 7.1: Cost for the CC-OPF and the WCC-OPFs

	Standard CC-OPF	Linear WCC-OPF	Quadratic WCC-OPF
Total cost [\$]	16546	16547 (+0.01%)	16562 (+0.1%)

and $\epsilon_G = 10^{-5}$ for the quadratic WCC-OPF. The choice of ϵ was tuned to obtain similar cost, in order to compare the trade-offs that each formulation makes in terms of number and size of constraint violations. The cost of the three OPF solutions is shown in Table 7.1. The difference is less than 0.1%, with the standard chance constraint being slightly cheaper than the other two, and the quadratic weighting function being the most expensive.

To test the OPF solutions, we generate 10 000 samples from the multivariate distribution of the wind power fluctuations and compute the number and size of the resulting violations. For the generation constraints, the choice of a very small ϵ_G ensures that there are very few significant violations of the generation limits. However, since $\epsilon_G > 0$, the linear and quadratic WCC assign a small, but non-zero α to some generators producing at the limit. These generators thus experience a violation probability of 50%, but with very small violations < 0.1 MW.

For the transmission line constraints, where ϵ is larger, there are more violations. We only consider the active transmission line constraints, since this is where the effect of the weighting function is visible. The active transmission constraints are also the only ones who experience a significant number of violations. In Fig. 7.2, the empirical probability ϵ_{emp} of violations larger than 0, 1, 2, 5 and 10 MW is plotted for three transmission lines, line 12 (top), line 23 (middle) and line 28 (bottom). For the standard chance constraint, the probability of any violation $> 0MW$ is close to 0.1 for all three lines. The linear WCC lead to a higher violation probability for line 12 and 23, but lower violation probability for line 28. The quadratic WCC is more restrictive, with all empirical violation probabilities $\epsilon_{emp} < 0.1$. Considering the size of the different violations, we see that line 23 has some violations $> 5MW$. For this line, we observe how the standard chance constraint allows for more large violations than the linear and quadratic WCC. Particularly the quadratic WCC reduces the number of large overloads to almost zero.

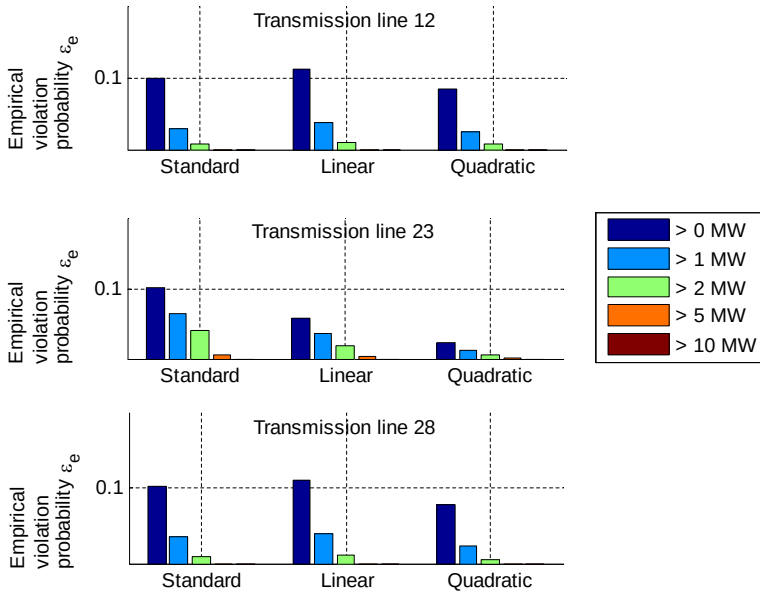
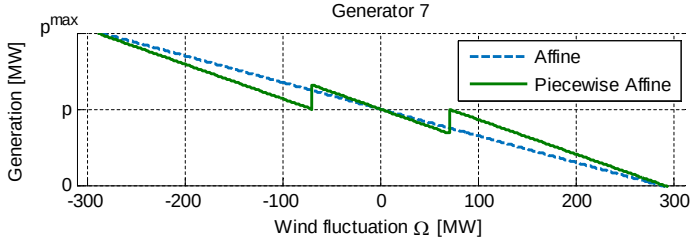


Figure 7.2: Empirical violation probability ϵ_{emp} for transmission line 12 (upper part), line 23 (middle) and line 28 (bottom). The results for all three formulations (standard, linear and quadratic) are shown in each plot, and the color of the bar indicate the empirical probability of exceeding certain violation thresholds (0, 1, 2, 5 and 10 MW).

Table 7.2: Cost for the affine and piecewise affine policies

	Affine	Piecewise Affine
Total cost [\$]	16547	16515 (-0.2%)

**Figure 7.3:** Generation output of generator 7 as a function of the total wind fluctuation Ω for the affine and the piecewise affine policy.

Based on the results, we see that the linear and quadratic WCC are less restrictive than the standard chance constraint for small violations, with higher violation probabilities ϵ_{emp} than the standard chance constraint. However, both the linear and quadratic WCC are more effective in reducing the probability of large overloads, as seen for line 23. This demonstrates the purpose of enforcing a weighted chance constraint as opposed to a standard chance constraint.

Impact of More Flexible Generation Control Policies

We now investigate how a the more flexible piece-wise affine policy for generation control (7.31) performs compared with the standard affine policy (2.6) in the case of the linear WCC-OPF. We use the same test case set-up as above, but choose $\epsilon_L = 0.01$ and $\epsilon_G = 0.01$. For the piecewise affine policy, we define the threshold for deploying additional reserves ξ^+ , ξ^- as $\Omega^+ = -\Omega^- = 70 \text{ MW}$.

The cost of the WCC-OPF solutions are shown in Table 7.2. The cost of the CC-OPF with piecewise affine generation control is lower than the policy with affine control. The reason for this cost decrease is due to more flexible use of generation resources. Fig. 7.3 shows the generation output of one generator as a function of the total wind fluctuation Ω for the affine (straight line) and the piecewise affine policy (straight line

with jumps). We observe that both policies lead to similar set-points for the scheduled generation p , and that they hit the maximum and minimum generation bounds at the same time. However, the piecewise affine policy allows a larger participation factor α which allows for a steeper slope than the affine policy. This is possible due to the use of additional reserves ξ^+ , ξ^- when the total wind fluctuation exceeds the thresholds Ω^+ , Ω^- . For the congested transmission lines 12, 23 and 28, a similar effect is observed. For large fluctuations, redispatching generation by ξ^+ , ξ^- shifts the power flows to lower values, thus lowering risk.

7.6 Summary and Conclusions

In this chapter, we introduced weighted chance constraints for modelling risk of overloads in power system operations, and formulated the WCC-OPF as an extension to previous CC-OPF formulations. The WCC account for the size of constraint violations through a weighting function, which can be chosen to assign a higher risk to larger violations. The framework is very general, and has the beneficial property of maintaining convexity under general generation control policies, and for general probability distributions of the fluctuations ω .

The performance of the new WCC-OPF was demonstrated in a case study for the IEEE RTS96 system. It was shown that the WCC-OPF allows for a larger number of small violations, but reduces the number of severe overloads compared to the CC-OPF. Further, it was shown that a more flexible, piecewise affine policy for generation control allows for lower cost compared with the original affine policy, while maintaining the same risk level.

The possibility to formulate general control policies while maintaining convexity can be exploited to model other power system controls. For example, we can extend the control of HVDC and PSTs beyond the current affine models applied in chapter 4. Another example is given in the subsequent chapter, where piecewise affine control is used to model different ways of controlling wind power output.

In the above case study, both transmission and generation constraints were assigned the same type of weighting function. It might however be reasonable to use different weighting functions for generation and transmission, since the consequences of a violation are different for the two.

Chapter 8

Optimal Power Flow with Wind Power Control

Over the past years, the share of electricity production from wind power plants has increased to significant levels in several power systems across Europe and the United States. In order to cope with the fluctuating and partially unpredictable nature of renewable energy sources, Transmission System Operators (TSOs) have responded by requiring wind power plants to be capable of providing reserves or following active power set-point signals. In this chapter, we address the issue of efficiently incorporating these new types of wind power control in the day-ahead operational planning. We review the technical requirements the wind power plants must fulfill, and propose a mathematical framework for optimizing wind power control. The framework is based on an optimal power flow formulation with weighted chance constraints of the type described in the previous chapter. In a case study based on the IEEE 118 bus system, we use the developed method to assess the effectiveness of different types of wind power control in terms of operational cost, system security and wind power curtailment.

8.1 Motivation and Related Work

Over the last decade, electricity production from wind power plants has reached significant levels in several regions of Europe and the United States. The forecast errors and fluctuations inherent to wind power generation has lead TSOs to reassess their reserve dimensioning policies [114]. In systems with large wind penetrations, such as Denmark [115] or Ireland [116], the grid codes now require wind power plants to be able to provide active power control. These control capabilities include droop control for frequency stabilization, down-regulation of the active power output for provision of spinning reserves, capability of following an active power set-point signal, and possibility to enforce a cap on the total wind power generation [115, 116, 117]. Pilot projects have already shown the technical capability of wind turbines to provide this kind of control, see e.g. [118].

With the above mentioned capabilities, wind power plants are able to participate in ancillary service provision, system balancing and congestion management. While increased controllability should generally improve system performance, wind power plants still differ from conventional generators in that their generation output is fluctuating. Further, their capacity is not fully known in day-ahead operational planning due to forecast uncertainty. Therefore, if ancillary service provision from wind power plants is not planned appropriately, the use of wind power control might significantly increase operational risk. For example, wind power fluctuations can render the wind power plants unable to deliver the required reserve capacities in real-time.

In this chapter, we address the question of how to optimally and securely incorporate wind power control in day-ahead planning. We account for wind power variability through the use of WCC, which were presented in the previous chapter. The WCC allow us to i) account for both the probability and size of constraint violations, and ii) assign higher risk values to larger overloads as in risk-based OPF [27, 28, 19]. Further, the WCC maintains convexity of the OPF formulation while modelling more general reactions to wind power fluctuations. Control actions such as wind power curtailment above a certain threshold is one example of such controls. Convexity is important to design efficient solution algorithms that scale well, and we present one such implementation below.

The remainder of this chapter is organized as follows. We first review the wind power control capabilities based on existing grid codes [115, 116]

and formulate the control policies associated with them. In particular, we model two types of wind control capabilities, a droop control policy that enables the wind power plants to provide reserves, and a control policy that enforces a cap on the maximum wind power output. We then incorporate the control policies as tractable extensions to the WCC-OPF. Leveraging the convexity of the WCC, we implement an efficient successive cutting-plane algorithm, and test our formulation on a modified version of the IEEE 118 bus system with 25 wind power plants. We quantify the benefits of each type of wind power control in terms of operational cost, system security and the amount of wind power curtailed, before summarizing and concluding.

The contents of this chapter are based on [33].

8.2 Active Power Control from Wind Turbines

Current grid-codes in countries with significant penetration of electrical energy from wind power [115, 116] require new installations of wind power plants to provide a variety of active power controls to stabilize the grid frequency and balance the system. Here, we consider the situation where the wind power plants adjust the active power output by changing the amount of energy extracted from the wind (e.g., through pitch control [117]) to follow a reference signal from the TSO. Since each wind power plant typically has a large number of wind turbines, we assume that the power output is continuously controllable between zero and maximum available power. While the grid codes require wind power plants to control their output in many different ways, we only model the two control mechanisms we consider to be the most suitable for system balancing and congestion management:

ΔP control: The wind power plants monitor the maximum available wind power (given by the current local wind condition) and keeps the output ΔMW below the maximum. The TSO can ask the wind power plant to implement this control to curtail excess wind energy, relieve congestion, or to keep wind power capacity available for reserve provision.

Output cap: The output cap is an absolute cap on the active power provided from a single wind power plant. The wind power plant produces at maximum as long as the maximum is below the output cap. If the

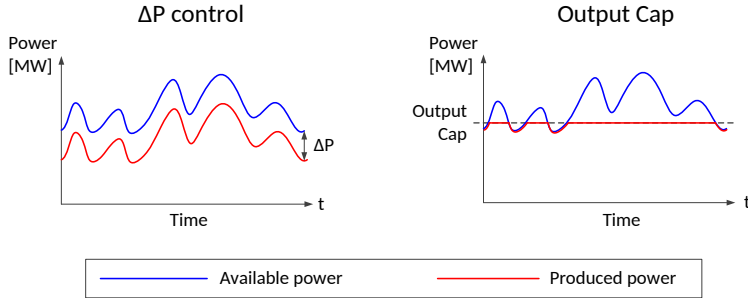


Figure 8.1: Two types of active power control (ΔP control and output caps), with wind power generation as a function of time (left) and the "controlled" wind power curve (right).

maximum available power exceeds the output cap, the wind power output is kept constant at the cap. The TSO can use this policy to reduce the variability of the wind power output (with a low cap, the output will essentially be constant) or to handle local transmission constraints by decreasing the peak production.

The two types of control are shown schematically in Fig. 1. Notice that the power output (red line) is always less or equal to the maximum possible production (blue line) at any given point in time. In case of reserve provision, a nominal curtailment is necessary for the turbine to be able to increase the output power in reaction to a control signal from the TSO. In the case of an output cap, any production above the cap will be curtailed. In the following, we present a mathematical model of the two types of control, and show how they can be incorporated in an OPF formulation.

8.3 Power System Modelling with Wind Power Control

We now provide an extension to our power system model, considering active power control of wind power plants. We follow the notation from previous sections, with the following additions. The set of wind generators is denoted by $\mathcal{W} \subseteq \mathcal{N}$. The available wind power is considered to be a random variable, but we assume that the wind power plants are able to continuously control their power output to any set-point below the

available power. In the following, we consider wind as the only random variable. However, the formulation can be extended to other uncertainty sources such as solar PV or load.

Wind Power Uncertainty

The wind in-feeds w_j of an uncontrolled wind power plant is modeled as the sum of the forecasted electricity production from wind $u_j = \mathbb{E}[w_j]$ and a zero mean fluctuation around the mean ω_j , such that

$$w_j = u_j + \omega_j \quad (8.1)$$

The total wind power fluctuation is denoted by $\Omega \doteq \sum_{j \in \mathcal{W}} \omega_j$. When we introduce wind power control, the output of the wind power plant changes depending on the control policy implemented. Both the mean u and the (distribution of the) uncertain deviation ω can be affected by the control.

Control of Conventional Generators

Conventional generators are scheduled to provide a nominal power output $p_{G,i}$. In addition, they might contribute to balancing through our standard affine control policy (2.6), where the participation factors α are subject to optimization.

Wind Power Plants with ΔP Control

The TSO might use ΔP control to reduce the scheduled mean power v_j from a wind power plant $j \in \mathcal{W}$ to any value below the mean, $v_j \leq u_j$. The resulting unused wind energy can be used to provide reserves, by defining participation factors α in a similar way as for conventional generators. The power output of wind power plants with ΔP control is thus given by

$$\tilde{w}_j = v_j + \omega_j - \alpha_j \Omega \quad \forall j \in \mathcal{R}. \quad (8.2)$$

where $\mathcal{R} \subset \mathcal{W}$ is the set of wind power plants providing reserves.

Wind Power Plants with Output Caps

With output cap control, the TSO can enforce an upper limit on the power output of individual wind power plants. In this case, the power output is given by

$$\tilde{w}_j = v_j + \min\{\omega_j, \bar{\omega}_j\} \quad \forall j \in \mathcal{C}, \quad (8.3)$$

where the cap $\bar{\omega}$ is the upper bound on the wind power fluctuation. The cap $\bar{\omega}$ is defined relative to the nominal output, and can be either positive or negative. The set $\mathcal{C} \subset \mathcal{W}$ denotes all wind power plants with output caps. We assume that these wind power plants do not provide reserves, i.e., $\alpha_j = 0 \quad \forall j \in \mathcal{C}$.

With output cap control, the distribution of ω changes. The new total wind deviation is given by

$$\tilde{\Omega} = \sum_{j \in \mathcal{R}} \omega_j + \sum_{j \in \mathcal{C}} (\min\{\omega_j, \bar{\omega}_j\} - \min\{0, \bar{\omega}_j\}), \quad (8.4)$$

where the last term is a correction term to account for negative output caps $\bar{\omega}_j \leq 0$, which limits the output to stay below the mean output u_j .

Power Balance and Power Flow Modelling

To ensure power balance, we enforce the constraint

$$\sum_{i \in \mathcal{V}} \tilde{p}_{G,i} + \tilde{w}_i - d_i = 0. \quad (8.5)$$

Since quantities $\tilde{p}_i$ and $\tilde{w}_i$ are random quantities whose values depend on the random wind fluctuation ω , the above relation must hold for all possible realizations of ω . This can be enforced by separating the nominal part (when the fluctuation is zero) and the stochastic part of (8.5). The nominal constraint is obtained by substituting $\omega = 0$ in the wind power output equations (8.2),(8.3) and in the generator control policy (2.6), and plugging the resulting expressions into (8.5),

$$\sum_{i \in \mathcal{G}} p_{G,i} - \sum_{i \in \mathcal{D}} d_i + \sum_{j \in \mathcal{W}} v_j + \sum_{j \in \mathcal{C}} \min\{0, \bar{\omega}_j\} = 0. \quad (8.6)$$

When the wind fluctuation ω is non-zero, we must have

$$\begin{aligned}
0 &= \sum_{i \in \mathcal{G}} (p_{G,i} - \alpha_i \tilde{\Omega}) - \sum_{i \in \mathcal{D}} d_i + \sum_{j \in \mathcal{R}} (v_j + \omega_j - \alpha_j \tilde{\Omega}) \\
&\quad + \sum_{j \in \mathcal{C}} (v_j + \min\{\omega_j, \bar{\omega}_j\}) \\
&= \sum_{i \in \mathcal{G}, \mathcal{R}} -\alpha_i \tilde{\Omega} + \sum_{j \in \mathcal{R}} \omega_j + \sum_{j \in \mathcal{C}} (\min\{\omega_j, \bar{\omega}_j\} - \min\{0, \bar{\omega}_j\}) \\
&= (1 - \sum_{i \in \mathcal{G}, \mathcal{R}} \alpha_i) \tilde{\Omega}. \tag{8.7}
\end{aligned}$$

From the last line it follows that to ensure power balance for every value of random wind fluctuation, it is enough to enforce the constraint

$$\sum_i \alpha_i = 1. \tag{8.8}$$

The power flows on each transmission line is computed according to the standard DC approximation,

$$\tilde{p}_{ij} = \mathbf{M}_{(ij, \cdot)} (\tilde{p}_G + \tilde{w} - d) \quad \forall_{ij \in \mathcal{E}}. \tag{8.9}$$

8.4 Optimal Power Flow with Weighted Chance Constraints and Wind Power Control

The objective is to minimize the sum of the generation and reserve cost. This is expressed as

$$\min_{p_G, v, \alpha, r^+, r^-} \sum_{i \in \mathcal{G}} c_i p_{G,i} + \sum_{j \in \mathcal{W}} c_j v_j + \sum_{i \in \mathcal{W}, \mathcal{G}} (c_i^+ r_i^+ + c_i^- r_i^-) \tag{8.10}$$

The decision variables are the scheduled power from generators $p_{G,i}$ and wind power plants v_j , the AGC participation factors α , and the up- and down-reserves r^+ , r^- . The vectors c , c^+ , c^- denote the cost, i.e. bids, from the generators and wind power plants for energy, up- and down reserves. As described above, the power balance constraints are given

by,

$$\sum_{i \in \mathcal{G}} p_{G,i} - \sum_{i \in \mathcal{D}} d_i + \sum_{j \in \mathcal{W}} v_j + \sum_{j \in \mathcal{C}} \min\{0, \bar{\omega}_j\} = 0, \quad (8.11)$$

$$\sum_i \alpha_i = 1. \quad (8.12)$$

What remains is to enforce generation and transmission constraints. For quantities that are functions of fluctuating wind in-feeds, we use WCCs to ensure that the risk remain small in a probabilistic sense. For the sake of readability, we use the same shorthand notation for the WCC as in the previous chapter,

$$WCC(y(\tilde{\omega}) > 0) \leq \epsilon, \quad (8.13)$$

where ϵ represents the risk limit.

Constraints for Conventional Generators

For a conventional generator, we enforce generation limits and constraints on reserve availability in the following way:

$$p_G + r^+ \leq p_G^{max}, \quad (8.14)$$

$$p_G - r^- \geq p_G^{min}, \quad (8.15)$$

$$WCC(-\alpha_i \tilde{\Omega} > r_i^+) \leq \epsilon_G, \quad \forall i \in \mathcal{G}, \quad (8.16)$$

$$WCC(-\alpha_i \tilde{\Omega} < r_i^-) \leq \epsilon_G, \quad \forall i \in \mathcal{G}, \quad (8.17)$$

$$0 \leq r_i^+ \leq r_{max,i}^+, \quad 0 \leq r_i^- \leq r_{max,i}^-, \quad \forall i \in \mathcal{G}. \quad (8.18)$$

As in previous chapters, (8.14), (8.15) enforce generation constraints. Eqs. (8.16), (8.17) describe the activation of reserves in reaction to the wind fluctuation $\tilde{\Omega}$, and use WCCs to limit the risk of not having enough reserves. Eq. (8.18) ensure that r^+ , r^- are non-negative, and r_{max}^+ , r_{max}^- are upper limits on the ability or willingness of the generators to provide reserves.

Constraints for Wind Power Plants

Wind power plants with ΔP control are able to maintain constant capacities r^+ , r^- available for reserves, similar to a conventional generator. However, for the wind power plants, the risk of the wind power

plant not being able to provide the promised reserves must be limited, since the power output of a wind power plant is dependent on the wind realization. This can be achieved by enforcing the following constraints:

$$v_j + r_j^+ \leq u_j, \quad \forall j \in \mathcal{R}, \quad (8.19)$$

$$WCC(v_j + \omega_j - r_j^- \leq 0) \leq \epsilon_W, \quad \forall j \in \mathcal{R}, \quad (8.20)$$

$$WCC(-\alpha_j \tilde{\Omega} > r_j^+) \leq \epsilon_G, \quad \forall j \in \mathcal{R}, \quad (8.21)$$

$$WCC(-\alpha_j \tilde{\Omega} < r_j^-) \leq \epsilon_G, \quad \forall j \in \mathcal{R}. \quad (8.22)$$

$$0 \leq r_j^+ \leq r_{max,j}^+, \quad 0 \leq r_j^- \leq r_{max,j}^-, \quad \forall j \in \mathcal{R}. \quad (8.23)$$

Eq. (8.19) is the nominal constraint that ensures that the scheduled power generation v and the capacity allocated for up-reserves r^+ remain below the forecasted power u . Eq. (8.20) ensures that the actual produced power $v_j + \omega_j$ is high enough to provide the expected down-reserves r^- . This constraint limits the risk ϵ_W that the wind power plant will not be able to provide the allocated reserve capacity. Constraints (8.21) - (8.23) have the same interpretation as the corresponding constraints for the conventional generators (8.16) - (8.18).

For wind power plants with output caps, we simply enforce

$$v_j \leq u_j, \quad \forall j \in \mathcal{C}. \quad (8.24)$$

Constraints for Line Flow Limits

To limit the risk of transmission line overloads, we enforce the power flow constraints using WCCs,

$$WCC(\mathbf{M}_{(ij, \cdot)}(p_G - \alpha \tilde{\Omega} + \tilde{w} - d) > p_{ij}^{max}) \leq \epsilon_L, \quad \forall ij \in \mathcal{E}, \quad (8.25)$$

$$WCC(\mathbf{M}_{(ij, \cdot)}(p_G - \alpha \tilde{\Omega} + \tilde{w} - d) < -p_{ij}^{max}) \leq \epsilon_L, \quad \forall ij \in \mathcal{E}, \quad (8.26)$$

where ϵ_L is the risk limit for line overloads.

8.4.1 Expressions for the Weighted Chance Constraints with Linear Weight Functions

We choose a linear weight function $f(y) = y\chi(y > 0)$ for the WCC. The following expressions assume that the line flows and generation

outputs are normally distributed. The WCC can also be evaluated when the distribution is not normal, although this might require numerical methods and more higher computational requirements.

Without Cap Control

When the wind fluctuation ω is a multivariate Gaussian random variable and there are no wind power plants enforcing cap control, the overload $y(\omega)$ is normally distributed for each of the WCC. In this case, the linear weighted chance constraint can be computed using the expression for the expectation of a truncated normal random variable (8.27),

$$\int_{-\infty}^{\infty} f(y(\omega))P(\omega)d\omega = \int_0^{\infty} yP(y)dy = u_y \left(1 - \Phi \left(\frac{-u_y}{\sigma_y} \right) \right) + \frac{\sigma_y}{\sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{-u_y}{\sigma_y} \right)^2} \leq \epsilon, \quad (8.27)$$

where u_y and σ_y denote the mean and standard deviation of y .

With Cap Control

When some of the wind power plants are enforcing cap control, the distribution of y changes when an output cap is reached. The computation of the linear weighted chance constraint thus becomes more involved. Let $\mathcal{C} = \{i_1, \dots, i_K\}$ denote the indices of the wind power plants implementing cap control. For each of these wind power plants, we split the integral over $\omega_{\mathcal{C}} = (\omega_{i_1}, \dots, \omega_{i_K})$ in (7.1) into two parts, one where the fluctuation in the wind power output is below the cap $\bar{\omega}_{\mathcal{C}}$, and one where the fluctuation is above the cap, by defining the sets $\mathcal{S}_{k,b} \in \mathbb{R}$ for $k = 1, \dots, K$ and $b \in \{0, 1\}$ as

$$\mathcal{S}_{k,b} = \begin{cases} (-\infty, \bar{\omega}_k], & \text{for } b = 0, \\ (\bar{\omega}_k, \infty) & \text{for } b = 1. \end{cases}$$

Starting from the original WCC (7.1), we can split the integral into an integral over each set $\mathcal{S}_{k,b}$. The linear weighted chance constraints can

then be computed as

$$\int_0^\infty \int_{-\infty}^\infty y(\omega) P(y(\omega), \omega_{\mathcal{C}}) d\omega_{\mathcal{C}} dy \quad (8.28)$$

$$= \sum_{b \in \{0,1\}^K} \int_0^\infty \int_{\mathcal{S}_{1,b_1}} \dots \int_{\mathcal{S}_{K,b_K}} y P(y, \omega_{\mathcal{C}}) d\omega_{\mathcal{C}} dy \leq \epsilon. \quad (8.29)$$

In the above summation, whenever $b_k = 1$, the corresponding ω_{i_k} is larger than $\bar{\omega}_{i_k}$ and the power output is fixed at $\bar{\omega}_{i_k}$. The overload y is jointly distributed along with $\omega_{\mathcal{C}}$ according to a multivariate Gaussian distribution. One can evaluate the integrals using efficient numerical schemes for Gaussian integration [119]. For smaller values of K , it is efficient to split (8.28) into the summation in (8.29) and evaluate each of the terms using Gaussian quadrature integration specialized to rectangular domains. When K is large, it is more appropriate to perform numerical integration directly on Eq. (8.28) using Monte-Carlo sampling. Note that such a Monte-Carlo based numerical integration technique allows for extension to general probability distributions, but increases computational complexity.

We remark here that the linear WCC (8.29) are convex with respect to the optimization variables, namely p_G, α and r . The same constraints can be shown to be non-convex for standard chance constraints (7.29).

8.5 Implementation of the WCC-OPF with Wind Power Control

We leverage the convexity of the WCC to devise an efficient outer-approximation-cutting-plane algorithm, similar to the one in [18]. The algorithm is implemented in Julia language [120] using JuMP [89]. At each step, a linear program corresponding to an outer approximation of the feasible set is solved. The outer approximation is progressively tightened by adding tangential cutting-planes to the convex WCC that eliminate the current infeasible solution. Each successive linear program is warm started using the solution of the previous one.

Since the constraints and gradients for the WCC-OPF with cap control must be evaluated using numerical integration methods, it is desirable to reduce the number of function evaluations corresponding to these integrals. The cutting-planes algorithm accomplishes this by selectively

evaluating the gradients only for the violated constraints, and shifting most of the computations over to the linear programming solver.

8.6 Case Study

8.6.1 Test System

We base our case study on the IEEE 118-bus system [72], with a few modifications as follows. For the generator bids for energy and reserves c , c^+ , c^- , we use the linear cost coefficients. For wind power plants, we assume zero marginal cost, and set $c_j = c_j^+ = c_j^- = 0$. Although the formulation could be extended to include unit commitment, it is not considered here. Therefore, the minimum generation output of the conventional generators is set to zero. To obtain a more stressed system state, we increase the load by a factor of 1.25 and decrease the transmission limits by a factor of 0.75. Wind power plants are located at 25 different buses throughout the system. Their locations, forecasted power output and correlation matrix can be found in [121]. The standard deviation of each wind power plant is set to 10% of the forecasted power output. When considering different levels of wind power penetration, both the forecasted power output and the standard deviations are scaled by a factor corresponding to the penetration of wind power relative to total system load. The risk limits for all WCCs were set to $\epsilon = 0.1$ MW. With the cutting-plane algorithm described above, a solution to the WCC-OPF for the 118 bus system is obtained within seconds for the cases with ΔP control and within a couple of minutes for the cases with output caps.

8.7 Results

Impact of Reserve Provision from Wind Power Plants

We compare the cases when the wind power plants using ΔP control can (a) only curtail their mean and (b) the curtailed energy can additionally be used to provide reserves. Figure 8.2 shows the total cost, cost of generation and cost of reserves for the two cases without and with wind power reserves under various levels of wind penetration, and Figure 8.3

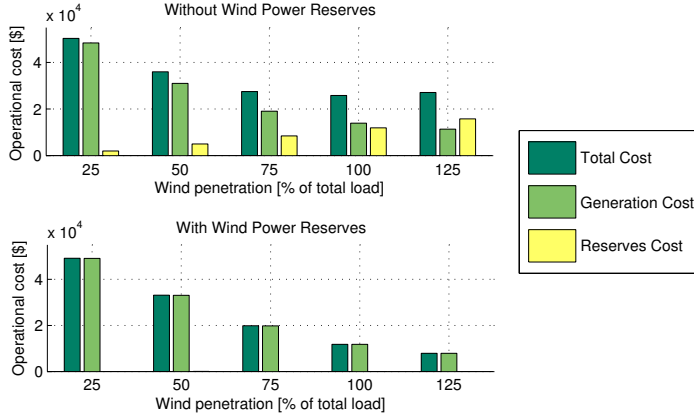


Figure 8.2: Total cost, cost of production and cost of reserves for the two cases with and without reserves from wind power.

shows the total amount of generation and reserves provided by both conventional generators and wind power plants.

As one might expect, the total cost is lower when the wind power plants are providing reserves. This difference is especially amplified when the system has higher wind penetration, and a large fraction of the cost difference is attributed to a higher cost of reserves. With high wind penetration, there is significantly more uncertainty in the system, and without (zero cost) wind power reserves, a large amount of reserves must be bought from conventional generators to balance fluctuations.

Additionally, in the case without wind power reserves, the nominal generation cost is also higher. As observed in Figure 8.3, the generators must run at a higher nominal set-point p in order to be able to provide enough down reserve r^- , particularly with high wind penetrations. Thus, the wind power nominal set points v must be lowered, leading to under-utilization of available cheap wind power.

Effect of Wind Power Output Caps

We now investigate the benefits of output cap control on wind power plants for the case with 75% wind power penetration. We enable output cap control for the wind power plants at bus 85 and bus 117, since these

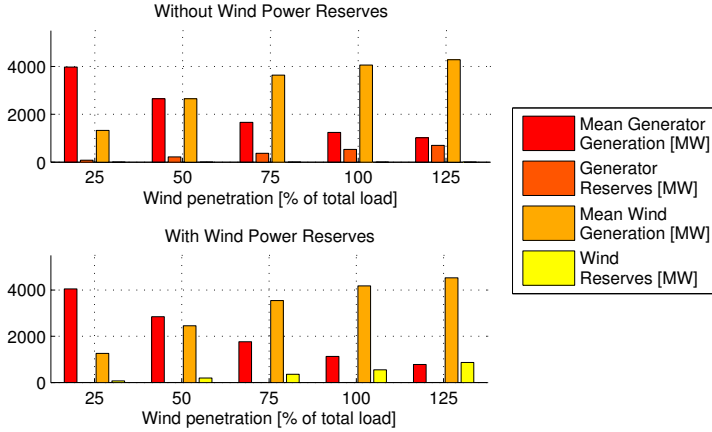


Figure 8.3: Production and reserves from generators and wind power plants for each level of wind penetration.

have particularly large values of means (364 and 188 MW) and standard deviations (36.4 and 18.8 MW). We allow the rest of the wind power plants to provide reserves according to (8.2). We compare this with the case when no wind power plants have output caps, but all are able to provide reserves (including bus 85 and 117). Figure 8.4 shows the cost with cap control relative to the case without cap control, for various values of the cap threshold $\bar{\omega}_i$ in (8.3). Around the cap threshold values of $(-45MW, -45MW)$, the total cost is minimized, and it is significantly lower ($\sim -6\%$) than the case without caps but with reserves. The cost increases as we move away from this band.

To explain the trend in cost, we first investigate how the output cap influences the power production of the two wind power plants on bus 85 and 117. In Figure 8.5, the expected amount of wind power curtailment (left) and the standard deviation (right) for each power plant is shown as a function of the output cap. When the cap thresholds are lowered, the amount of utilized wind power drops due to higher expected curtailment. However, the standard deviation of the wind power output also drops significantly, which reduces wind power variability and thus the requirement for reserves.

To investigate how this trade-off influences the cost in more detail, we look at the total wind power used for generation, the total system reserves and the total wind energy curtailment (which includes both

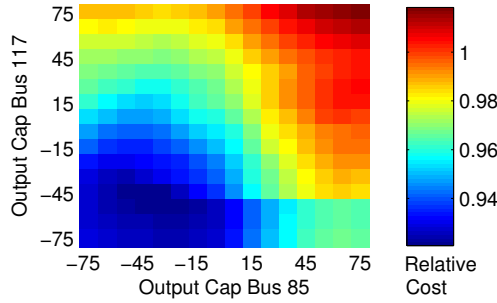


Figure 8.4: Total cost of the case with caps on the output, relative to the case without any output caps. The output caps on the two different buses are defined relative to the mean output, and are varied from -75 MW to $+75$ MW.

“wasted” energy and energy curtailed to provide reserves). These values are plotted for different pairs of output caps in Figure 8.6. We observe that, as the output caps increase, the total system reserve requirement also increases, and reaches the highest value for the case without output caps. At output caps above $(-45, -45)$, the use of wind power for nominal energy production decreases with higher reserve requirements, since more of the wind power is allocated for reserves. With low output caps $(-75, -75)$, higher total wind curtailment (due to both reserve provision and output caps) outweighs the benefits of reduced reserve requirements compared to case with the optimal output caps $(-45, -45)$. The lowest utilization of wind for energy production happens with output caps of $(+75, +75)$, which is observed to be the point with highest cost in Figure 8.4. In this case, high wind power variability and no possibility of procuring wind power reserves at buses 85 and 117 leads to an increased curtailment of the mean wind power v at those buses.

In the case with output caps $(+45, +45)$, the total wind power generation is lower than in the case without output caps. Still, the total cost of the solution is approximately 1% lower with caps $(+45, +45)$ than without caps as observed in Figure 8.4. This is because the output caps reduces variability and thus the risk of overload on some critical transmission lines, which allows for better utilization of the transmission capacity and dispatch of cheaper conventional generators.

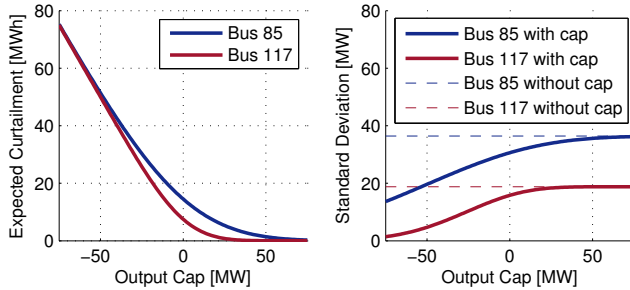


Figure 8.5: The upper and lower plots shows the expected curtailed wind energy (assuming a time step of 1 h) and the standard deviation of the wind power output at bus 117 and bus 85, respectively.

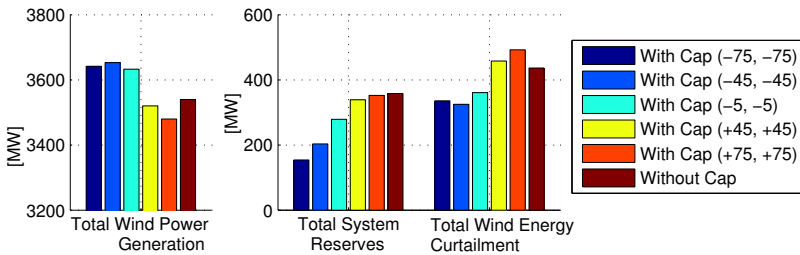


Figure 8.6: Comparison of the cases with and without caps on the wind power output at bus 117 and bus 85. From left to right, the total generation by all wind power plants, the total amount of reserves in the system and the expected total wind energy curtailment are shown.

8.8 Summary and Conclusions

In this chapter, we investigated two types of wind power control capabilities, ΔP control which reduces the mean power output by a constant value and allows the wind power plants to provide reserves, and output cap control which enforces a hard threshold on the total power output. The corresponding control and reserve models were developed and incorporated into an Optimal Power Flow formulation with Weighted Chance Constrains, which allows for controlling the risk of overloads due to wind fluctuations. Leveraging the convexity of the WCC, we solved the optimization problem using an efficient cutting-plane algorithm. Based on a case study on the IEEE 118 bus system, we observed that ΔP control with wind reserves can provide substantial cost benefits. For wind power plants with high variability, the output cap control was shown to outperform ΔP control with reserves.

Chapter 9

Conclusions and Outlook

9.1 Summary

This thesis proposes and discusses different formulations of the chance-constrained and risk-based optimal power flow, and investigates their application to selected aspects of power systems operation.

The first part is related to chance-constrained optimal power flow (CC-OPF). Chapter 2 introduces the DC security-constrained optimal power flow with chance constraints, and discusses different analytical reformulations of the problem. In Chapter 3, we investigate a scheme for integrated balancing and congestion management, where the reserves are activated not only based on the total power mismatch, but also based on the location of the fluctuations in the system. Chapter 4 proposes a framework for corrective control to handle forecast uncertainty, based on control of HVDC and PSTs. An efficient solution algorithm is developed, allowing us to demonstrate scalability on large scale test cases. Chapter 5 suggests an extension of the CC-OPF to AC power flow. We determine the nominal operating point based on the full AC power flow equations, but use a linearization around that point to account for the impact of uncertainty. This allows us to apply the analytical reformulations from Chapter 2, and we obtain an approximate, closed-form formulation of the AC CC-OPF. We further propose an iterative solution method which can be implemented as an outer loop on any AC OPF solver. This method allows us to leverage scalability and robustness of existing AC OPF implementations.

The second part of the thesis concerns risk-based optimal power flow, where we account for both the probability and severity of events. In Chapter 6, we use risk-based constraints to limit the severity of post-contingency overloads, and a joint chance constraint to limit the probability of high-risk events. The risk is defined as a product between the probability of the contingency and the severity of the resulting operating condition. The severity is modelled using a piecewise linear function, where the parameters are defined based on the cost and availability of remedial actions. Chapter 7 proposes a risk-based formulation based on weighted chance constraints (WCC), which limit the expected risk. The WCC are a generalization of the standard chance constraints, and can be used to assign higher risk to higher overloads. One of the advantages of the WCC is that they remain convex for convex weight functions and general generation control policies. This property is used in Chapter 8, where we review wind power control capabilities and investigate how they can be optimally applied in power systems operation.

9.2 Conclusions

Below, we summarize some of the main observations and conclusions in this thesis.

Solution Optimality vs. Chance Constraint Feasibility

We have proposed different methods to limit operational risk in OPF problems where the constraints are affected by uncertainty. It is usually challenging to reformulate the stochastic optimization problem into a tractable, deterministic problem. In most cases, it is not possible to find an approach that guarantees a solution which is both feasible (enforces the stochastic constraints) and optimal (achieves this at optimal cost). In particular, this becomes evident when comparing the different analytical reformulations of the chance constraints discussed in Chapter 2, where we see that more conservative solutions might easily render the reformulated CC-OPF infeasible although the true CC-OPF is actually feasible. To tackle this problem, we pursue the development of methods that are based on sound scientific and theoretical considerations, but accept simplifications and assumptions that are justifiable and necessary from a practical point of view. Two such examples are found in Chapter 5, where we discuss the AC CC-OPF. First, we use a partial linearization to obtain an analytical chance constraint reformulation,

which is an approximation, but performs well and has low computational requirements. Second, we implement a sampling-based approach where we assume that local convexity will be sufficient to ensure chance constraint feasibility. Although the locally convex region is not guaranteed to be large enough, the method performs well in simulations.

Distributional Assumptions

One assumption that is not true in general, but often practically relevant, is the assumption of a normal distribution. In this thesis, we observed that even when the uncertain power injections are not normally distributed, relevant variables such as power flows or voltage magnitudes often tend to be close to normal. This is explained by the fact that each power flow depends on a large number of random variables, and arguments similar to the central limit theorem can be invoked. For power systems with a large number of uncertainty sources, it is thus often justifiable to assume a normal distribution. This allows us to keep the nominal cost of operation lower, while accepting that the risk in some cases might be higher than expected.

Choice of Risk Measure

Another challenge in the above problems is the choice of a risk model. In this thesis, we have considered both chance constraints, which measure risk in terms of violation probability, and the more general weighted chance constraints, which limit the expected risk of overloads. A number of factors influence the choice of risk measure. It matters whether the constraint in question is physically hard, such as the PST tap limits, or soft, such as power flow limits or voltage constraints. Further, the resulting impact of a violation is also important. If a constraint violation leads to load shedding, it is much more severe than if it simply requires operational measures such as changing the tap position of a PST.

Throughout the thesis, we have discussed the physical interpretation of the introduced stochastic constraints. For example, we believe that the probability of transmission line overloads should be interpreted as the frequency with which the operator needs to implement additional remedial measures, rather than the probability of an actual overload. This is particularly true for violations of N-1 constraints, which usually do not result in physical overloads since the corresponding outage is a rare event. However, regulation typically requires the operator to implement remedial actions to ensure N-1 security.

Another observation is that the generator chance constraints depend only on the total power mismatch, which is a scalar random variable. Therefore, these generator chance constraints are actually jointly enforced. The violation probability thus has a direct interpretation as the probability of reserve deficiency, which is a commonly applied risk measure in practical system operation.

Security vs. Cost

Higher security requirements and stricter risk limits inherently increase operational cost. The acceptable level of risk is therefore not really a fixed value, but also a function of how much we are willing to pay to avoid risk exposure. There currently exist no widely accepted stochastic equivalent of the N-1 criterion, and defining the right security-cost trade-off remains a question to be addressed. In Chapter 5, we suggested a way of relating the risk limits to the N-1 criterion, which could be a way of making the choice of risk limits more tangible to system operators.

Flexibility from Corrective Control

Corrective control extends the ability to react to uncertainty in real time. This leads to less conservative solutions and lower overall cost. Throughout the thesis, we have investigated several different means of control, such as the reserve activation policies in Chapter 3, corrective control with HVDC and PSTs in Chapter 4, or the control of wind power plants in Chapter 8. One general conclusion is that with increasing number of controllable devices and more general control policies, we not only reduce cost, but also increase computational complexity. It is therefore important to find computationally efficient ways of modelling the control actions, like the (piecewise) affine control policies implemented in this thesis. However, even though we choose a certain class of policies to model control actions during operational planning, we do not need to limit the real-time control actions to the same class of policies. In real-time operation, the operator can re-optimize the set-points and obtain solutions that are specifically tailored to the current system situation. The optimized affine policies might therefore be more important as a simplified approach to ensure real-time feasibility, as compared to actual implementation in real-time operation.

Scalability to Large Systems

Real power systems are large systems, and practical applicability requires efficient solution methods. In this thesis, we have developed scal-

able solution algorithms for both the DC and AC CC-OPF, and demonstrated their applicability on large scale test instances such as the Polish Winter Peak test case with 2383 buses. Further, an outer-approximation algorithm based on cutting planes was implemented for the WCC-OPF. These implementations show that, under many practical circumstances, it is possible to solve stochastic OPF problems without losing computational tractability.

9.3 Outlook

The conclusions listed above provide not only answers, but also directions for future work.

Optimized Risk Limits

An important question raised above is how to define reasonable risk limits. One approach to achieve this is to assign a monetary value to the risk of constraint violations, and minimize risk along with generation cost. Such approaches have been investigated in risk-based OPF, both with [26] and without [106] consideration of uncertainty. In [122], co-optimization of the risk limits for WCC was implemented. It is challenging to define a reasonable cost penalty for the violations, however, preliminary results showed that a heuristic based on the cost of redispatch might be promising. Further work is needed to come up with coherent criteria for the choice of risk model and the corresponding limits.

Hierarchy of Policies

In the above implementations, we observed that more general control policies provide lower cost solutions, but also increase computational time. The application of increasingly complex control policies can be understood as a process where we gradually extend the feasible space, creating a hierarchy of solutions corresponding to different types of policies. One interesting direction for future work is to investigate if there exist optimal classes of policies for the OPF problem and how well simpler approaches, such as affine control, compare with those classes.

Comparison of Stochastic Approaches

The chance-constrained and risk-based OPF approaches presented in

this thesis only represent some possible ways of accounting for uncertainty within the OPF. There are many other approaches to stochastic and risk-based OPF, each with their own pros and cons. Taking a step back and reviewing the reasons for choosing a specific model, comparing how risk is measured within each approach and understanding the approximations made to maintain computational tractability would be useful to understand to find the best way forward.

Appendix A

Definition of AC Sensitivity Factors

To solve the approximate chance-constrained AC OPF, the sensitivity factors $\mathbf{\Gamma}_P$, $\mathbf{\Gamma}_Q$, $\mathbf{\Gamma}_V$ and $\mathbf{\Gamma}_I$ must be properly defined. In the following, we provide the mathematical derivations of those factors.

The changes in the power injections arise from the sum of uncertain fluctuations and the resulting change in generation. The change in generation can be divided into the fluctuations themselves, the direct AGC response, as well as the changes in the injections of active power δp (at the reference bus) and reactive power δq (at the PV and reference buses) which are necessary to keep the system balanced and the voltage magnitudes constant.

$$\begin{bmatrix} \Delta p \\ \Delta q \end{bmatrix} = \begin{bmatrix} \Delta p_G \\ \Delta q_G \end{bmatrix} + \begin{bmatrix} \omega \\ \gamma\omega \end{bmatrix} = \begin{bmatrix} \alpha\Omega \\ 0 \end{bmatrix} + \begin{bmatrix} \delta p_G \\ \delta q_G \end{bmatrix} + \begin{bmatrix} \omega \\ diag(\gamma)\omega \end{bmatrix} \quad (\text{A.1})$$

Rearranging the terms into the direct and indirect influence of the fluctuations ω , we obtain

$$\begin{bmatrix} \Delta p \\ \Delta q \end{bmatrix} = \begin{bmatrix} -\alpha 1_{\{m,m\}} + I \\ diag(\gamma) \end{bmatrix} \omega + \begin{bmatrix} \delta p_G \\ \delta q_G \end{bmatrix} = \mathbf{\Psi}\omega + \begin{bmatrix} \delta p_G \\ \delta q_G \end{bmatrix} \quad (\text{A.2})$$

where the matrix $\mathbf{\Psi}$ is a matrix of Generation Distribution Factors (GDFs) and describes the direct influence of ω on the bus injections.

This matrix is either known a-priori or specified through the optimization. Similarly, we can define the change in voltage angles and magnitudes as

$$\begin{bmatrix} \Delta\theta \\ \Delta v \end{bmatrix} = \begin{bmatrix} \delta\theta \\ \delta v \end{bmatrix}. \quad (\text{A.3})$$

Arranging (A.2), (A.3) into groups corresponding to the PQ, PV and reference buses, we observe that some of the changes Δp , Δq , Δv , $\Delta\theta$ are either zero, such as the change in voltage magnitude at PV buses, or directly determined by ω , γ . Other quantities, such as the change in voltage magnitude at PQ buses δv_{PQ} , are not known a-priori, but are indirectly determined as a function of the overall system changes.

$$\begin{bmatrix} \Delta p_{PQ} \\ \Delta p_{PV} \\ \Delta p_{\theta V} \\ \Delta q_{PQ} \\ \Delta q_{PV} \\ \Delta q_{\theta V} \end{bmatrix} = \Psi \omega + \begin{bmatrix} 0 \\ 0 \\ \delta p_{\theta V} \\ 0 \\ \delta q_{PV} \\ \delta q_{\theta V} \end{bmatrix}, \quad \begin{bmatrix} \Delta\theta_{PQ} \\ \Delta\theta_{PV} \\ \Delta\theta_{\theta V} \\ \Delta v_{PQ} \\ \Delta v_{PV} \\ \Delta v_{\theta V} \end{bmatrix} = \begin{bmatrix} \delta\theta_{PQ} \\ \delta\theta_{PV} \\ 0 \\ \delta v_{PQ} \\ 0 \\ 0 \end{bmatrix} \quad (\text{A.4})$$

By linearizing the system around the nominal operating point, we obtain a linear relation between the change in power injections and voltages. This relation is given by the system Jacobian $\mathbf{J}_{pow} \in \mathbb{R}^{2m \times 2m}$, evaluated at the nominal operating point (θ, v, p, q) :

$$\begin{bmatrix} \Delta p \\ \Delta q \end{bmatrix} = \left. \begin{bmatrix} \frac{\partial p}{\partial \theta} & \frac{\partial p}{\partial v} \\ \frac{\partial q}{\partial \theta} & \frac{\partial q}{\partial v} \end{bmatrix} \right|_{(\theta, v, p, q)} \begin{bmatrix} \Delta\theta \\ \Delta v \end{bmatrix} = \mathbf{J}_{pow}^{full} \begin{bmatrix} \Delta\theta \\ \Delta v \end{bmatrix} \quad (\text{A.5})$$

The expressions for the derivatives in the Jacobians $\mathbf{J}_{pow}$ can be found in, e.g., [123]. By rearranging the rows and columns of the variable vectors and the matrices $\mathbf{J}_{pow}$ and Ψ , we obtain the following set of

equations:

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ \delta p_{\theta V} \\ \delta q_{PV} \\ \delta q_{\theta V} \end{bmatrix} = \mathbf{J}_{pow} \begin{bmatrix} \delta \theta_{PQ} \\ \delta \theta_{PV} \\ \delta v_{PQ} \\ 0 \\ 0 \\ 0 \end{bmatrix} - \mathbf{\Psi} \omega = \begin{bmatrix} \mathbf{J}_{pow}^A & \mathbf{J}_{pow}^C \\ \mathbf{J}_{pow}^B & \mathbf{J}_{pow}^D \end{bmatrix} \begin{bmatrix} \delta \theta_{PQ} \\ \delta \theta_{PV} \\ \delta v_{PQ} \\ 0 \\ 0 \\ 0 \end{bmatrix} - \begin{bmatrix} \mathbf{\Psi}^A \\ \mathbf{\Psi}^B \end{bmatrix} \omega \quad (\text{A.6})$$

In the above equation, $\mathbf{J}_{pow}$ and $\mathbf{\Psi}$ were rearranged and divided into submatrices, with columns and rows corresponding to the respective groups of variables at PV, PQ and θV buses. With this, we obtain the following expressions for the changes in voltage and generation,

$$\begin{bmatrix} \delta \theta_{PQ} \\ \delta \theta_{PV} \\ \delta v_{PQ} \end{bmatrix} = (\mathbf{J}_{pow}^A)^{-1} \mathbf{\Psi}^A \omega, \quad \begin{bmatrix} \delta p_{\theta V} \\ \delta q_{PV} \\ \delta q_{\theta V} \end{bmatrix} = (\mathbf{J}_{pow}^B (\mathbf{J}_{pow}^A)^{-1} \mathbf{\Psi}^A - \mathbf{\Psi}^B) \omega,$$

which allows us to define the sensitivity factors for generation and voltages:

$$\mathbf{\Gamma}_V = (\mathbf{J}_{pow}^A)^{-1}_{PQ} \mathbf{\Psi}^A, \quad \begin{bmatrix} \mathbf{\Gamma}_P \\ \mathbf{\Gamma}_Q \end{bmatrix} = (\mathbf{J}_{pow}^B (\mathbf{J}_{pow}^A)^{-1} \mathbf{\Psi}^A - \mathbf{\Psi}^B)$$

Note that the matrix $\mathbf{J}_{pow}^A$ is square matrix and invertible under normal conditions. Normal conditions refer to conditions under which a Newton-Raphson power flow algorithm, which is based on the inversion of the same matrix $\mathbf{J}_{pow}^A$, would converge.

The approximate change in the current magnitude Δi can be computed based on the voltage changes and a Jacobian matrix $\mathbf{J}_I$ evaluated at the nominal operating point.

$$\Delta i \approx \left[\frac{\partial i}{\partial \theta} \quad \frac{\partial i}{\partial v} \right] \bigg|_{(\theta, v, p, q)} \begin{bmatrix} \Delta \theta \\ \Delta v \end{bmatrix} = \mathbf{J}_I (\mathbf{J}_{pow}^A)^{-1}_{PQ} \mathbf{\Psi}^A \omega \quad (\text{A.7})$$

From this expression, we define the sensitivity factor $\mathbf{\Gamma}_I$ as

$$\mathbf{\Gamma}_I = \mathbf{J}_I (\mathbf{J}_{pow}^A)^{-1}_{PQ} \mathbf{\Psi}^A. \quad (\text{A.8})$$

The derivatives $\frac{\partial i}{\partial \theta}$ and $\frac{\partial i}{\partial v}$ can be found in [102].

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